MEWRE RENEWABLE PROJECT MANAGER · TRIAL CRASH COURSE

Days 2 to 4 in three hours

Kuwait's power sector and its decision makers, solar, CSP and batteries, grid connection and grid strength, project delivery, contracts, money and the ministry. Written to be studied once, on paper, in about 3.25 hours.

How to study it

1. Work through one part, then take a 5 minute break.
2. Each section opens with what you will be able to do. Read the explanation, then the figure and its caption.
3. Do the worked example yourself before reading the result.
4. The dark **Remember** box is the minimum to retain. Say each line out loud.
5. Answer **Check yourself** on paper, then compare with the answers at the back.
6. Kuwait facts carry their confidence: H official, M trade press, L weak, V verify before quoting.

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PART A · TRIAL DAY 2 · ABOUT 29 MIN

Kuwait's power sector and who decides

Know the system, its numbers and the institutions that approve every project, with their Arabic names.

A1 Kuwait's power system and economics

about 12 min

After this section you can

- Explain why Kuwait is short of capacity every summer and compute the reserve margin.
- Read the summer load curve and say where the net peak sits once PV is added.
- Explain why a solar kWh is worth 26 to 37 fils to the state when the citizen pays 2 fils.
- State the targets and the pipeline with the right status, and price a day of delay.

Demand: an air-conditioning peak

Kuwait's demand is set by **air conditioning** (compressor motors driving refrigerant cycles in every building). The system's highest demand, the **peak**, occurs on July and August afternoons at 48 to 50 °C ambient. MEWRE reports summer peaks of 17,640 MW (2024), 17,610 MW (2025) and **17,540 MW (2026)** M, ministry figures reported in the press.

MEES reported a 17.71 GW record in July 2025 M. The two disagree because of the measurement basis: **gross** (generator terminals, station auxiliaries included) or **net** (sent out), and instantaneous or averaged over 15 minutes or an hour. Always quote a peak with source, year and basis.

New connections add about 1 GW of load a year M; the measured peak is flat only because of demand measures such as a shorter government working day.

Installed versus reliably available capacity

Installed capacity is the sum of nameplate ratings. **Available capacity** is what can be dispatched at the peak hour after three deductions: forced outages (unplanned trips, frequent on old steam units), planned maintenance, and **temperature derating** (a gas turbine loses about 0.5 to 0.7 % of output per °C above 15 °C). Kuwait has about 21 GW installed but only about 17 GW reliably available, of which about 14 GW is gas-fired M.

> $RM = (C_{avail} - P_{peak}) / P_{peak}$
>
> where RM is the reserve margin, C_{avail} the available capacity and P_{peak} the peak demand, both in MW

Hot-climate utilities plan for +10 to +20 %, to cover the loss of the largest unit plus forecast error. In June 2025 Kuwait had 16.98 GW of peak against 16.275 GW available: RM = −4.2 % M. A negative margin forces load shedding: cuts came in June 2024 (300 MW Az-Zour South trip), August 2024 (gas supply), April 2025 (heat during maintenance) and May 2025 (more than 1.5 GW short) M.

GCCIA imports (a 500 MW deal in 2024; KWD 161.381 m approved for June to December 2025) M only narrow the gap: a full 500 MW in June 2025 leaves RM at −1.2 %. The structural answer is 14.05 GW of new generation by 2031 M, plus renewables and storage.

Fuel mix

Generation is about 62 % gas, 36 % oil and 2 % renewable M. Kuwait's own gas is mostly associated gas (produced with crude), so the summer shortfall is met with imported **LNG** (5.2 Mt in January to August 2025) M and liquid fuels burned directly: crude, fuel oil and gasoil. The **heat rate** (HR, fuel energy in per kWh out) links electricity to fuel: η = 3,600 / HR with HR in kJ/kWh. An old steam unit at 10,500 kJ/kWh is 34 % efficient; a combined-cycle unit at 7,500 kJ/kWh is 48 %.

Load shape and net load

Figure A1.1 shows a summer day. Demand rises through the morning as buildings absorb heat, peaks around 15:00, and stays high into the evening because walls release stored heat and residents return home. The **load factor** measures how much of the year the system runs near peak:

> $LF = E / (P_{peak} \cdot 8{,}760)$
>
> where E is annual energy in MWh, P_{peak} in MW, 8,760 hours a year

With the course's illustrative 84 TWh a year V and a 17.5 GW peak, LF = 0.55: the last gigawatts of capacity run only a few hundred hours a year.

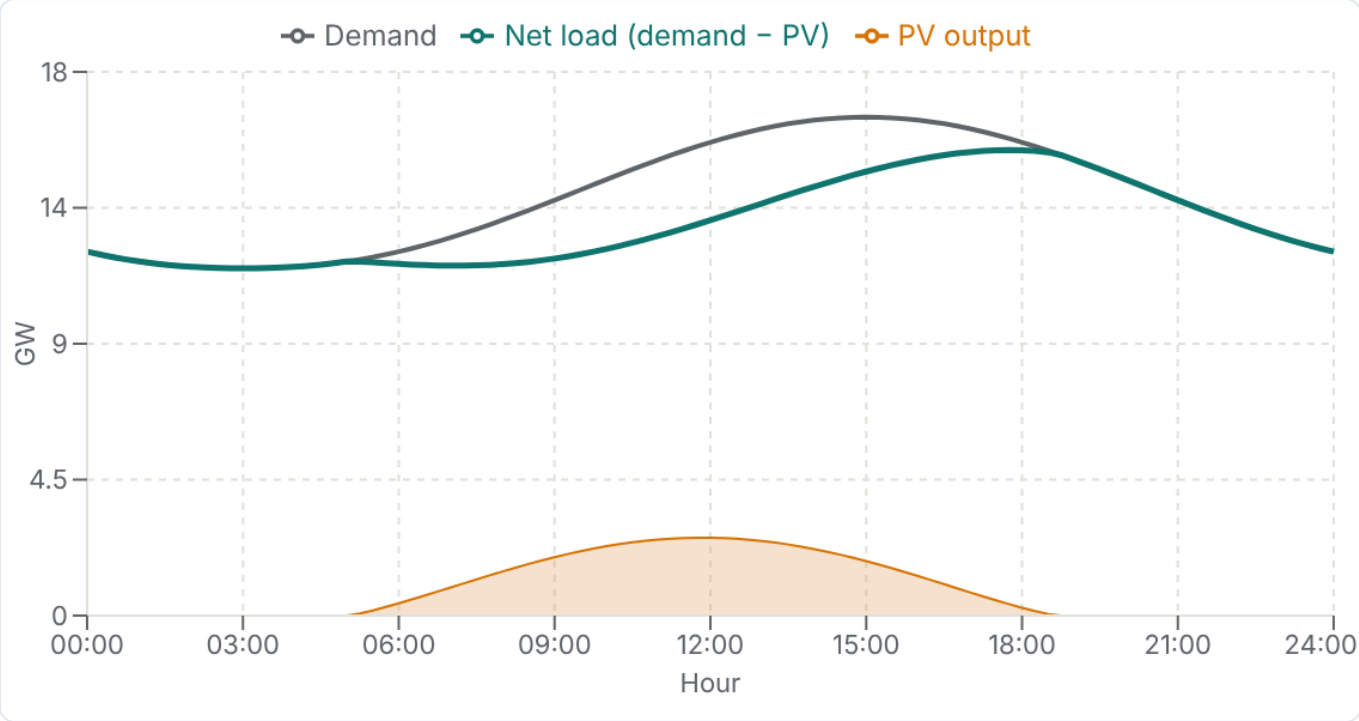


Figure A1.1 Illustrative Kuwait summer day: demand (grey) peaks near 15:00, PV (orange) peaks at noon and ends near 19:00, net load (green) peaks near 18:00. Notice that the net peak is later and only slightly lower than the demand peak.

Net load is demand minus PV output: the load the thermal fleet must still serve. PV peaks at solar noon, three hours before the demand peak, and is zero after sunset while cooling load remains high. As PV grows, the net peak moves into the evening, with a steep ramp at sunset. In the course's illustrative case, 3 GW of PV turns a 17.5 GW gross peak at 15:00 into a 16.55 GW net peak at 18:00: only 0.95 GW of thermal peak removed. Shifting midday energy into the evening is the job of the BESS programme. At the peak hour PV also delivers below nameplate (hot cells lose about 0.3 to 0.4 % per °C, plus sun angle and dust), so quote MWac at the peak hour, never MWp.

Tariffs, subsidy and the value of displaced fuel

Kuwaiti citizens in private housing pay **2 fils/kWh**, unchanged since 1966 H. Apartments pay 5 / 10 / 15, commercial 5, industrial and agricultural 3 fils/kWh H. There is no official current **cost of supply** (full cost of a delivered kWh); estimates run from 14 to 38 fils/kWh M, mainly because fuel can be valued at export price or at a low transfer price.

$$s = c - t \qquad v = HR \cdot p_{fuel}$$

s unit subsidy, c cost of supply, t tariff (fils/kWh); v value of displaced fuel, HR in GJ/kWh, p_{fuel} in currency per GJ

A citizen therefore pays 5 to 14 % of what the kWh costs. When a PV plant produces 1 kWh, the marginal thermal unit produces 1 kWh less. Crude at USD 70/bbl in a 10.5 MJ/kWh steam unit gives v ≈ 37 fils/kWh; LNG at USD 12/MMBtu in a 7.5 MJ/kWh combined-cycle unit gives v ≈ 26 fils/kWh M (illustrative prices). Analysts put solar at 7 to 8 fils/kWh L; the Zone 1 tariff is not published.

Hence the economic case: crude not burned is exported at world prices, and LNG not burned is a cargo not bought. The saving is set by the fuel's opportunity cost, not by the retail tariff.

Targets and pipeline

The official target is **15 % renewable electricity by 2030** (set 2012) H. MEWRE has stated 30 / 40 / 50 % by 2030 / 2040 / 2050 M; cabinet adoption is unconfirmed. National net zero is **2060**; 2050 applies to the oil and gas sector H. Rystad forecasts about 3.3 GW (7 %) by 2030 M.

$$P_{RE} = \sigma \cdot E_{tot} / (CF \cdot 8{,}760)$$

P_{RE} capacity in MW, σ target share, E_{tot} annual generation in MWh, CF capacity factor

15 % of 84 TWh at CF 0.25 needs about 5.75 GW of PV; Phase I plus Zones 1 and 2 (1,670 MW) give about 4 %. Status labels in the table below matter: only signed items count.

What a day of delay costs

$$C_d = \Delta t \cdot P \cdot CF \cdot 24 \cdot (v - T)$$

Δt delay in days, P plant MWac, CF capacity factor, v fuel value and T tariff in KWD/kWh

▸ **Worked example: Zone 1 slips one day**

Inputs: P = 1,100 MW, CF = 0.25, crude displaced at v = 36.95 fils/kWh (USD 70/bbl ÷ 6.12 GJ/bbl × 0.0105 GJ/kWh ÷ 3.25 USD/KWD), assumed tariff T = 8 fils/kWh.

Daily energy: 1,100 × 0.25 × 24 = 6,600 MWh = 6.6×10^6 kWh. Net value: 36.95 − 8 = 28.95 fils = 0.02895 KWD/kWh. Cost per day: $6.6 \times 10^6 \times 0.02895$ = **KWD 191,070**. A 30-day deferral costs KWD 5.73 m. With LNG as the displaced fuel (26.25 fils), it is KWD 120,450 per day.

In Kuwait: numbers to know

Item	Value	Conf.
Summer peak 2024 / 2025 / 2026	17,640 / 17,610 / 17,540 MW (MEWRE)	M
Installed / reliably available / gas	~21 / ~17 / ~14 GW	M
June 2025 reserve margin	16.275 vs 16.98 GW: −4.2 %	M
Mix; LNG Jan to Aug 2025	62 % gas, 36 % oil, 2 % RE; 5.2 Mt	M
Citizen tariff; cost of supply	2 fils/kWh; 14 to 38 fils/kWh	H / M
Targets	15 % by 2030 official; 30/40/50 % stated; net zero 2060	H / M
Shagaya Phase I	70 MW: 50 CSP + 10 PV + 10 wind, KISR; CSP data after 2019 not public	H / L
Zone 1 PV, 1,100 MW (KAPP)	Masdar + Fouad AlGhanim reported selected ~Aug 2026; tariff unknown	M
Zone 2 PV, 500 MW (KAPP)	Bids 26 Jul 2026 (two or three, sources conflict)	M
400 kV Shagaya to Al-Wafra line	Power Grid Co., KD 48.6 m, Mar 2026	M
BESS programme	1 to 1.5 GW / 4 to 6 GWh; first 500 MW for summer 2027	M / L
China G2G (NEA/SPIC)	Framework 17 Mar 2025; 4.8 GW claimed; **not contracted**	L
Al-Khairan Phase 1	1,800 MW **gas-fired** IWPP, two bids 1 Jun 2026; not solar	H
GCCIA imports	500 MW deal 2024; KWD 161.381 m approved for Jun to Dec 2025	M

What the PM does

- Labels every number with source, date, basis and confidence; reports project status in tiers (framework, bids, selected, signed, closed).
- Schedules grid outages for connection works from November to March, never in summer.
- Puts the cost of delay in front of any committee about to defer an item.

Questions to ask

- Is this peak figure gross or net, instantaneous or hourly, and whose?
- Which fuel and heat rate does the ministry use for avoided-cost calculations?
- Where is the net peak now, and where does it move after Zone 1 is online?

Remember

- Peak about 17.5 GW against about 17 GW reliably available: the reserve margin is negative.
- PV covers the afternoon; the net peak moves to the evening, so storage carries the evening.
- A solar kWh saves 26 to 37 fils of exportable or imported fuel; the 2 fils tariff is irrelevant to that saving.
- 15 % by 2030 official; net zero 2060; G2G is not contracted; Al-Khairan is gas.
- A 1.1 GW plant late by one day costs about KWD 191,000 in fuel.

Check yourself

1. Peak demand is 17,540 MW and reliably available capacity is 17,000 MW. What is the reserve margin in %?
2. Why does adding PV move the net peak later in the day?
   A. PV output is highest in the evening.
   B. PV output peaks at noon and falls to zero after sunset while cooling demand stays high.
   C. Thermal plants are derated in the evening.
   D. Demand peaks at noon.
3. Zone 2 (500 MW, CF 0.25) is delayed. With LNG displaced at 26 fils/kWh and a tariff of 8 fils/kWh, what is the cost of delay in KWD per day?
4. Which statement is correct?
   A. Al-Khairan Phase 1 is a 1,800 MW solar IPP.
   B. The China G2G track has 4.8 GW under signed contract.
   C. The Zone 1 tariff has been published.
   D. Al-Khairan Phase 1 is a gas-fired IWPP that received two bids on 1 June 2026.

A2 Institutions, approval chains and the ministry

about 13 min

After this section you can

- Name each institution in English and Arabic and say what it decides.
- Route a purchase to EPC, PPP or G2G and recite the 10-step IPP chain.
- Apply the procurement thresholds that hold during the trial week and after Law 94/2026.
- Read authority inside MEWRE and estimate an approval chain's duration.

Why the chain is long

Kuwait separates four functions on purpose: the body that needs something (MEWRE), the body that buys it (CAPT or KAPP), the body that checks legality (Fatwa and Legislation) and the body that checks financial propriety (State Audit Bureau). Each signatory is personally accountable. Officials are rarely sanctioned for delay but can be for an irregular approval, so an incomplete file is returned. The PM's lever is a complete file, not pressure.

MEWRE's four roles

MEWRE (established 1962) is at once: (1) **policy owner**, setting targets and strategy; (2) **system operator and transmission owner** of the 400/132 kV network, which connects new plants; (3) **offtaker**, the buyer under a 30-year PPA for solar IPPs and a 25-year ECWPA for gas-fired power and water plants (IWPPs); (4) **direct procurer** of grid works and EPC contracts through public tenders, such as the Shagaya to Al-Wafra 400 kV line M.

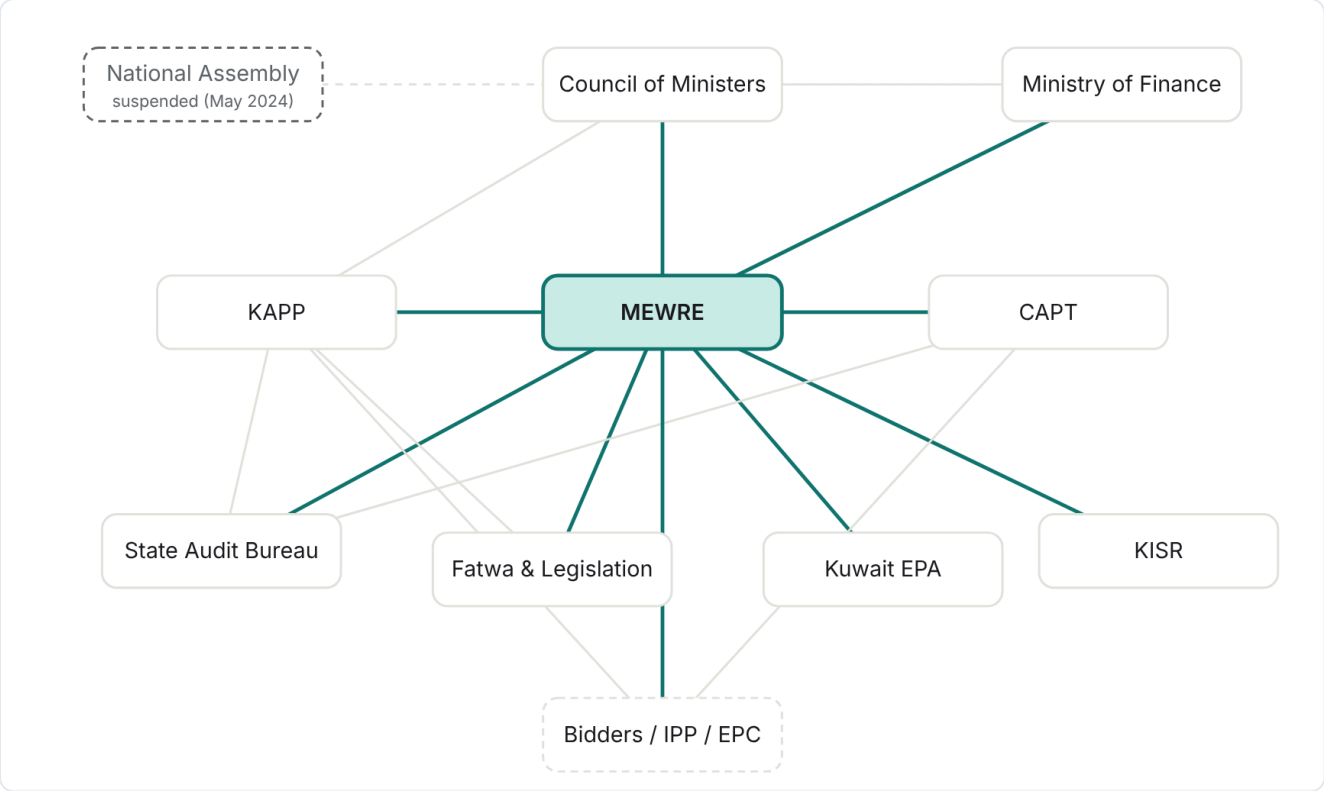


Figure A2.1 MEWRE and the institutions around it. Dark links are MEWRE's direct interfaces; grey links connect the control bodies to each other. The National Assembly is dashed because it is suspended.

The institutions

Figure A2.1 shows the map; the table gives each official Arabic name as it appears on letters and file covers.

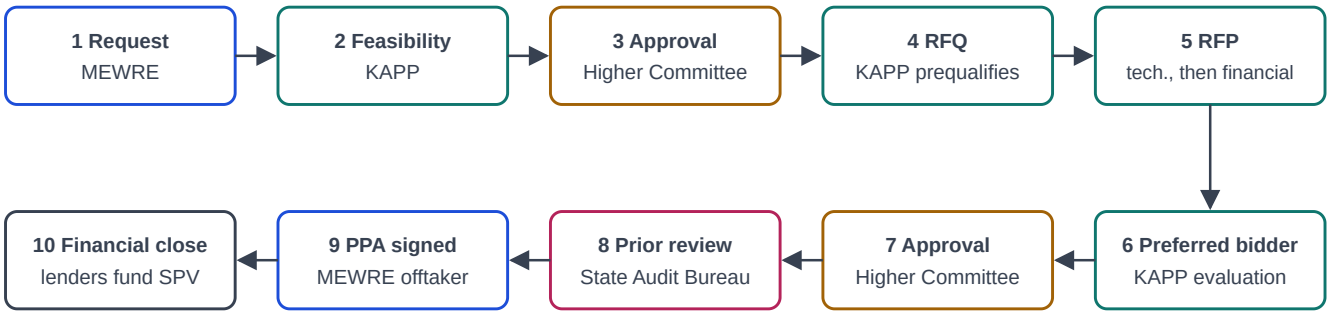
English	Arabic	Role for a renewable PM
MEWRE	وزارة الكهرباء والماء والطاقة المتجددة	Policy, grid, offtaker, direct procurer
KAPP, Kuwait Authority for Partnership Projects	هيئة مشروعات الشراكة بين القطاعين العام والخاص	Procures PPP/IPP projects (PPP Law 116/2014, Decree 78/2015); attached to the Minister of Finance; holds 60 % of IWPP companies such as Az-Zour North 2&3 H
Higher Committee for PPP	اللجنة العليا لمشروعات الشراكة بين القطاعين العام والخاص	Chaired by the Minister of Finance; approves the project, the procurement stages and the selected investor
CAPT, Central Agency for Public Tenders	الجهاز المركزي للمناقصات العامة	State-funded tenders above KD 75,000 (Law 49/2016)
State Audit Bureau	ديوان المحاسبة	Prior review before signature (Law 30/1964); approve, reject or conditional
Fatwa and Legislation Department	إدارة الفتوى والتشريع	Legal review of contract terms, PPAs included; reached through MEWRE legal affairs; its stage in the chain is unconfirmed V
Council of Ministers	مجلس الوزراء	Top executive decisions; issues decree-laws
Ministry of Finance	وزارة المالية	State budget (fiscal year 1 April to 31 March); funds ministry capex
KISR	معهد الكويت للأبحاث العلمية	Built Shagaya Phase I; solar resource stations at five inland sites
SCPD, Supreme Council for Planning and Development	المجلس الأعلى للتخطيط والتنمية	Kuwait Vision 2035 and the development plan
National Assembly	مجلس الأمة	Suspended since 10 May 2024 H: laws issued by decree-law, so the bottleneck is cabinet agendas
EPA, Environment Public Authority	الهيئة العامة للبيئة	Environmental approvals; the EIA law number for PV is not confirmed V
KPC, Kuwait Petroleum Corporation	مؤسسة البترول الكويتية	Oil sector; its solar plans compete for land and grid capacity
KDIPA, Direct Investment Promotion Authority	هيئة تشجيع الاستثمار المباشر	Licenses up to 100 % foreign ownership (FDI Law 116/2013)
GCCIA, GCC Interconnection Authority	هيئة الربط الكهربائي لدول مجلس التعاون الخليجي	400 kV regional grid; emergency imports

Three delivery routes

EPC via CAPT: the state pays capex from the MEWRE budget and buys a plant or line through a public tender. **PPP/IPP via KAPP**: a private project company finances, builds and operates the plant, and MEWRE pays a tariff per MWh under a 30-year PPA. **G2G**: a government-to-government framework, such as the MEWRE to China NEA agreement of 17 March 2025 M; the legal basis exempting G2G from the tenders law was not found, so do not assert one.

	PPP / IPP track	Conventional tender track
Procurer	KAPP	MEWRE under CAPT rules
Law	PPP Law 116/2014	Public Tenders Law 49/2016
Typical projects	Solar IPPs, IWPPs	Grid works, consultancy, O&M
Gatekeeper	Higher Committee (project, then investor)	CAPT (above KD 75,000)
Before signature	State Audit Bureau prior review on both tracks	
MEWRE's role	Offtaker	Direct buyer

The 10-step IPP approval chain



Blue MEWRE, green KAPP, amber Higher Committee, red Audit Bureau. Fatwa review: stage unconfirmed.

The Higher Committee appears twice: once to approve the project and its procurement, once to approve the investor. Zone 1 shows the pace: RFQ January 2024, bids 15 January 2026, preferred bidder reported about August 2026 M; steps 7 to 10 were not publicly confirmed. A press report of "selected" is step 6, not a signed PPA. On IWPPs the shareholding has been 40 % developer and 60 % KAPP, with 50 % of the company earmarked for a citizen IPO H.

Laws and thresholds

Instrument	What it sets	Conf.
Public Tenders Law 49/2016	CAPT tender required above **KD 75,000**	M
Law 94/2026	Gazetted 27 Sep 2026, effective about **27 Dec 2026**. Direct contracting without CAPT up to **KD 1 m per financial year** for the same items; local and commission **agents banned**; at least **30 %** local products and 30 % local subcontracting; KPC under CAPT only above KD 20 m	M
Audit Bureau, Law 30/1964	Prior review, reported from **KD 100,000**	V
PPP Law 116/2014, Decree 78/2015	KAPP and the Higher Committee	H

During the trial week (from 15 November 2026) the **old KD 75,000 rule applies**: Law 94/2026 is not yet in force. The KD 1 m limit does not remove Audit Bureau prior review or internal delegation limits.

Elapsed-time benchmarks M: Zone 1 RFQ to reported selection about 31 months; Az-Zour North 2&3 letter agreement to ECWPA about 6 months; Al-Khairan RFP to bids 9 months; 400 kV line bid to award about 11 months (not to the lowest bidder; reason not found).

Inside MEWRE: authority

Level	Arabic	What the PM needs from it
Minister	الوزير	Political head and Council of Ministers member; reached through the Undersecretary
Form of address for the Minister	معالي الوزير	Used in letters and meetings
Undersecretary	وكيل الوزارة	Runs the ministry; approves decision memos and senior external letters
Assistant Undersecretary	الوكيل المساعد	Heads a sector; your internal client and budget owner
Department Director	مدير إدارة	Technical review, first signature on memos going up
Controller	مراقب	Day-to-day supervision within a department
Section Head	رئيس قسم	Execution, data, drafting

Formal authority is written: laws, ministerial decisions and the **delegation of authority** (the decision stating who may approve what up to which value), exercised by a numbered, dated signature. **Practical authority** is held by those who shape what is signed: the drafters who frame the options, the gatekeepers (office managers, committee secretaries, legal and financial reviewers) who control timing, and trusted advisers.

Each level writes an **annotation** (tashira) on a memo, and the annotated memo is the decision record. A verbal approval is agreement in principle only. The annotation is:

تأشيرة

Annotation	Meaning
موافق	Approved
للعرض	For presentation to the next level up
لاتخاذ اللازم	Take the necessary action

Expected approval time

Each body reviews a file in time t_i and returns it with observations with probability p_i; the review then restarts. The number of cycles has mean $1/(1 - p_i)$, so:

> $E[T] = \Sigma\, t_i / (1 - p_i)$
>
> t_i nominal review time of step i, p_i probability it returns the file

▸ **Worked example: preferred bidder to PPA (illustrative)**

Higher Committee 4 weeks, p = 0.20: 4/0.8 = 5.0. Audit Bureau 6 weeks, p = 0.40: 6/0.6 = 10.0. Fatwa 4 weeks, p = 0.25: 4/0.75 = 5.33. Signing 3 weeks, p = 0: 3.0. E[T] = **23.3 weeks** against a nominal 17: returns add 6.3 weeks (37 %), four of them at the Audit Bureau. Cutting its p to 0.2 with a complete file brings that step to 7.5 weeks.

In Kuwait

- Even emergency GCCIA imports needed Audit Bureau conditional approval (2025) M.
- A course example for a KD 450,000 extension needs about 46 working days of approvals, mostly outside the ministry (illustrative).

What the PM does

- Identifies the track of every file first: it decides the gates.
- Submits complete files with compliance matrices built from past observations, to cut p.
- Goes through MEWRE legal affairs, never directly to Fatwa or the Audit Bureau, and gets every decision on paper.

Questions to ask

- Who holds the delegation of authority for this decision, and up to what value?
- Which of our tenders fall under Law 94/2026 and which under the old law?
- At which stage does Fatwa review the PPA for this project?

Remember

- For a solar IPP, KAPP procures and MEWRE is the offtaker; the Higher Committee approves twice.
- Chain: MEWRE, KAPP, Higher Committee, RFQ, RFP, preferred bidder, Higher Committee, Audit Bureau, PPA, financial close.
- Now: CAPT above KD 75,000. From about 27 Dec 2026: direct up to KD 1 m per FY, agents banned, 30 % local content.
- The annotated memo is the decision; a verbal yes is not.
- Rework, not review time, dominates: $E[T] = \Sigma\, t/(1 - p)$.

Check yourself

1. In the IPP chain, what comes immediately after the second Higher Committee approval?
   A. RFP
   B. State Audit Bureau prior review
   C. Financial close
   D. KAPP feasibility study
2. On 20 November 2026 MEWRE needs a grid consultancy worth KD 400,000. Which applies?
   A. Direct contracting under Law 94/2026
   B. KAPP PPP procurement
   C. CAPT tender, then Audit Bureau prior review
   D. No approval is needed below KD 1 m
3. A contract needs Fatwa review (5 weeks, p = 0.3) then Audit Bureau review (8 weeks, p = 0.5). What is E[T] in weeks?
4. Which annotation sends a memo up to the next level for presentation?
   A. موافق
   B. للعرض
   C. لاتخاذ اللازم

PART B · TRIAL DAY 2 · ABOUT 40 MIN

Solar PV, CSP, wind and batteries

Explain how each technology produces or stores energy, how plants are built and how their output is predicted and guaranteed.

B1 Solar resource and the PV cell

about 10 min

After this section you can

- Use irradiance, irradiation and GHI = DNI cos θ_z + DHI.
- Read an I-V curve: I_{sc}, V_{oc}, MPP, fill factor.
- Compute module output on a Kuwaiti July afternoon.

Measuring sunlight

Irradiance is solar power per area at one instant (W/m²); **irradiation** is its time integral (kWh/m² per day or year). A clear noon gives about 1,000 W/m². A pyranometer reading 800 W/m² for 3 h records 0.8 × 3 = 2.4 kWh/m².

DNI (direct normal irradiance) is the beam from the sun's disc on a surface facing the sun. **DHI** (diffuse horizontal irradiance) is light scattered by air and dust, arriving from the whole sky. **GHI** (global horizontal irradiance) is the total on a flat surface. The beam meets flat ground at the zenith angle θ_z, so only its cosine share counts:

$$GHI = DNI \cdot \cos\theta_z + DHI$$

θ_z = angle between sun and vertical; W/m²

Example: DNI 650 W/m², DHI 250 W/m², θ_z = 30°: GHI = 650 × 0.866 + 250 = 813 W/m². Dust shifts energy from DNI to DHI. PV converts both; concentrating mirrors use only DNI (B3). **POA** (plane-of-array irradiance) is what the tilted or tracking module actually receives; on single-axis trackers annual POA is about 1.22 to 1.28 × GHI.

The cell and its I-V curve

A photon above silicon's band gap frees an electron and leaves a hole (the **photovoltaic effect**). A **p-n junction** (boron-doped p layer against phosphorus-doped n layer) has a built-in field that sweeps electrons to the n side and holes to the p side before they recombine. Through a load this separated charge flows as a current proportional to irradiance, in one direction only: **DC**.

The **I-V curve** (Figure B1.1) runs from the **short-circuit current I_{sc}** at zero voltage to the **open-circuit voltage V_{oc}** at zero current. Power V × I peaks at the knee, the **maximum power point (MPP)**. The inverter's **MPPT** (maximum power point tracking) keeps adjusting the DC voltage to stay there. Irradiance mainly sets current; temperature mainly sets voltage.

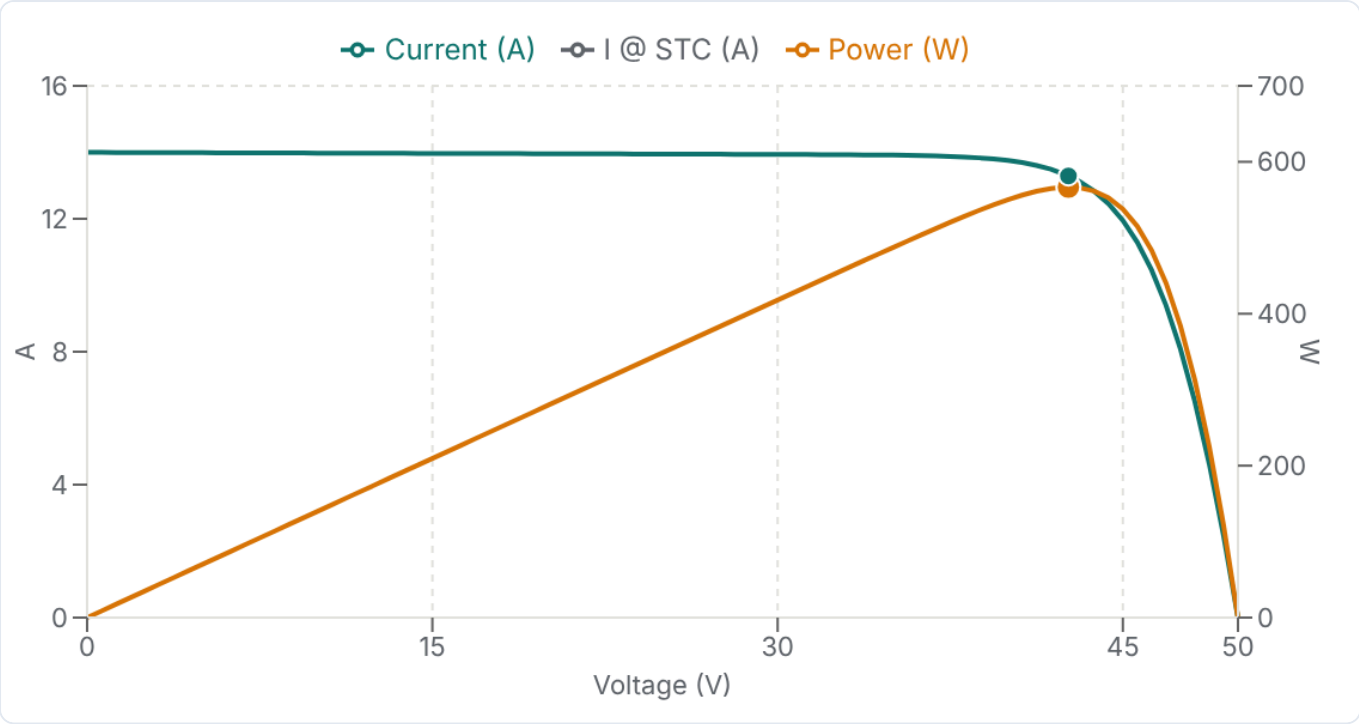


Figure B1.1 I-V curve (green) and power curve (orange) of one module. Current is nearly flat until the knee, then collapses toward V_{oc} at about 50 V. Power peaks at the MPP near 42.5 V and 13.3 A.

The **fill factor** FF = $V_{mp}I_{mp}$ / ($V_{oc}I_{sc}$) measures how square the curve is. For Figure B1.1: 42.5 × 13.3 / (50 × 14.0) = 565 / 700 = 0.81. Rising series resistance (corroded ribbons, cracked cells) lowers FF; commissioning I-V traces detect it.

Ratings and heat

STC (standard test conditions) are 1,000 W/m², 25 °C cell and the AM1.5 spectrum; the **Wp** (watt-peak) rating is MPP power at STC. In the field, output scales with irradiance and falls with cell temperature through the **temperature coefficient γ**. Cell temperature is estimated with the **NOCT** (nominal operating cell temperature, measured at 800 W/m², 20 °C ambient, 1 m/s wind; about 45 °C):

$$P = P_{STC} \cdot (G/1{,}000) \cdot [1 + \gamma(T_{cell} - 25)] \quad T_{cell} \approx T_{amb} + (NOCT - 20)/800 \cdot G$$

G in W/m²; γ per °C, negative

▸ **Worked example: 600 Wp in July**

Ambient 45 °C, G = 1,000 W/m², NOCT 45 °C: T_{cell} = 45 + 25/800 × 1,000 = 76.3 °C. TOPCon (γ = −0.30 %/°C): 1 − 0.0030 × 51.3 = 0.846, so 508 W. PERC (−0.35 %/°C) gives 492 W; HJT (−0.25 %/°C) 523 W. At 800 W/m² the cell runs at 70 °C and a PERC module loses about 16 % to heat alone.

Module technologies

Cell designs differ in how well they suppress **recombination** (carriers lost at surfaces and contacts); less recombination raises voltage and lowers γ. **PERC** (p-type, older mainstream) has γ about −0.35 %/°C, **TOPCon** (n-type, tunnel-oxide contact, now mainstream) about −0.30 %/°C, **HJT** (n-type heterojunction) about −0.25 %/°C. **Bifacial** modules also collect ground-reflected light through a glass rear; over sand (**albedo**, reflected fraction, 0.25 to 0.4) on trackers they gain about 5 to 9 %.

In Kuwait

- Shagaya at about 29.3° N: noon sun about 84° high in June, 37° in December. Record 53.9 °C, Mitribah, 21 July 2016 (WMO) H.
- Annual GHI roughly 1,900 to 2,100 kWh/m², from a secondary report; official Shagaya GHI and DNI not retrieved V.
- KISR (EuroSun 2014): satellite GHI acceptable after site adaptation, satellite DNI diverges because of aerosols M.

Questions to ask

- Is the DNI/DHI split in your hourly file measured or modelled?
- Which γ and cell-temperature model did you use at 50 °C ambient, and what measured albedo backs the bifacial gain?

Remember

- W/m² is a rate, kWh/m² a total. GHI = DNI cos θ_z + DHI; CSP uses DNI only.
- Light sets current, heat lowers voltage. A July cell at 70 to 76 °C gives about 82 to 87 % of Wp.
- Low γ (HJT, TOPCon) is worth most in Kuwait's heat.

Check yourself

1. DNI 700 W/m², DHI 200 W/m², θ_z = 60°. What is GHI?
2. What does the inverter's MPPT keep adjusting?
   A. The AC frequency
   B. The DC operating voltage
   C. The module tilt
   D. The short-circuit current
3. A 600 Wp TOPCon module (γ = −0.30 %/°C) at 1,000 W/m² has a 70 °C cell. Roughly what power?
4. Irradiance drops from 1,000 to 500 W/m² at constant cell temperature:
   A. Voltage halves, current unchanged
   B. Current roughly halves, voltage changes little
   C. Both halve
   D. Nothing changes

B2 The utility PV plant and its energy yield

about 14 min

After this section you can

- Trace a plant from module to 400 kV and explain MWp, MWac and clipping.
- Compute specific yield, CF and PR, including temperature-corrected PR.
- Find the optimal cleaning interval and convert P50 to P90.

Architecture

Modules in series form a **string** whose voltages add, up to 1,500 V DC. V_{oc} rises in the cold, so the coldest morning sets the maximum length: 49.8 V at STC with +0.26 %/°C gives 53.3 V at −2 °C, and 1,500 / 53.3 = 28.1, so 28 modules. The hottest afternoon sets the minimum, keeping string voltage inside the inverter's MPPT window.

Strings join in a **DC combiner box** (fuses, disconnect) and feed the **inverter**, which makes 50 Hz AC (Figure B2.1). **Central inverters** (3 to 5 MVA) are cheaper but each failure removes a large block; **string inverters** (250 to 350 kW) spread the risk over many units to maintain. An **MV skid** adds a step-up transformer to 33 kV and a **ring main unit** (RMU, MV switchgear), 4 to 7 MVA per block. Buried 33 kV **collector** cables return to the plant substation, whose HV transformer steps up to 400 kV.

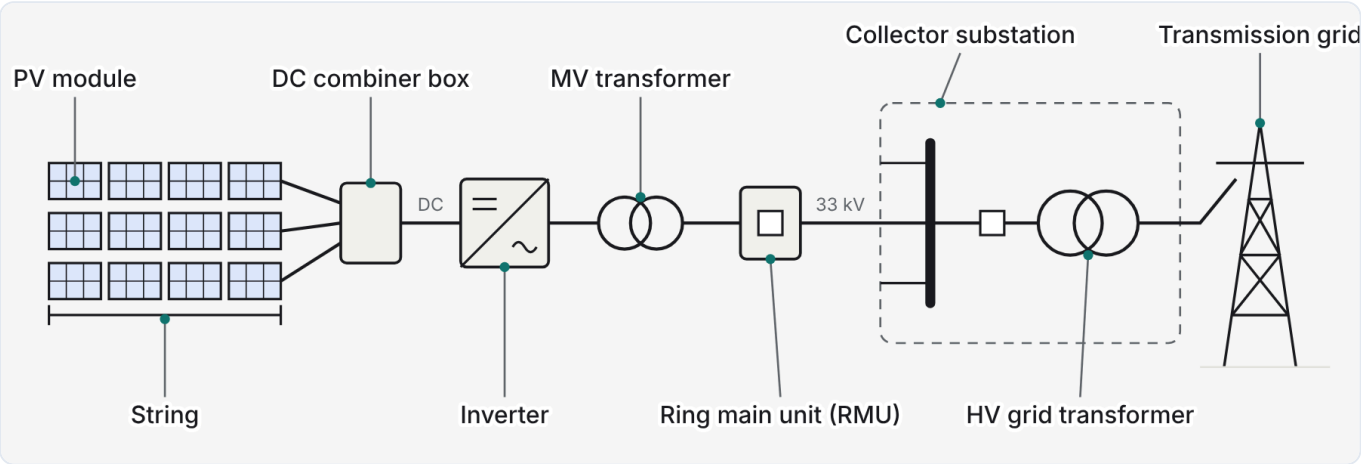


Figure B2.1 Modules, string, DC combiner, inverter (the only DC/AC boundary), MV transformer, RMU, 33 kV collector substation, HV grid transformer, transmission line.

MWp, MWac, clipping and trackers

MWp is the sum of module STC ratings (DC); **MWac** is inverter or export capacity. The **DC/AC ratio** (MWp/MWac) is typically 1.15 to 1.40. Field DC output rarely exceeds 80 to 90 % of nameplate, so extra DC fills the inverters earlier in the morning, later in the evening and on hazy days. The cost is **clipping** at clear noons (Figure B2.2): about 0.5 to 3 % of annual energy at 1.3.

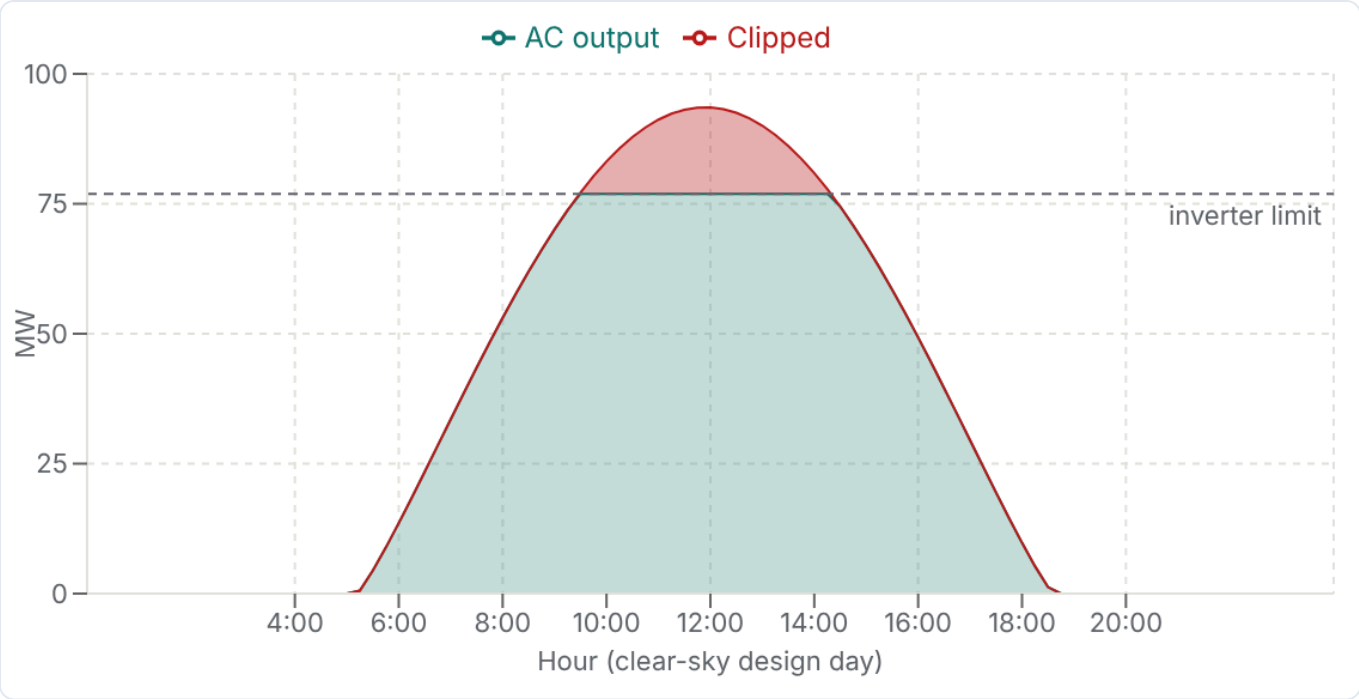


Figure B2.2 Clear-sky day: around noon the array could deliver more than the inverters allow, so output is capped at the inverter limit. The red area is clipped; the extra DC raises every hour either side of noon.

A **single-axis tracker** turns north-south rows east to west. At 29° N the summer sun rises north of east, so trackers gain 15 to 25 % over fixed tilt. At low sun they **backtrack** (rotate toward flat) to avoid row-to-row shading; in high wind they **stow**. **GCR** (ground coverage ratio, module width / row pitch) is about 0.28 to 0.45: higher saves land but adds shading.

Yield, losses and PR

Specific yield is annual kWh per kWp. **Capacity factor** = energy / (capacity × 8,760 h), stated on MWac or MWp. Check: specific yield ≈ GHI × (POA/GHI) × PR = 2,100 × 1.25 × 0.82 ≈ 2,150 kWh/kWp. Kuwaiti trackers: 2,000 to 2,300 plausible, above 2,300 optimistic.

The yield model (usually PVsyst) reports a **loss waterfall** (Figure B2.3). Typical Kuwaiti tracker losses: temperature 7 to 10 % (largest), soiling 2 to 5 %, **mismatch** (the weakest module limits string current) 0.7 to 1.5 %, DC wiring 1 to 1.5 %, inverter 1.2 to 2 %, clipping 0.5 to 3 %, AC and transformers 1 to 2 %, unavailability 0.5 to 1.5 %. Losses multiply: mid-range values of all ten in the figure give 0.79.

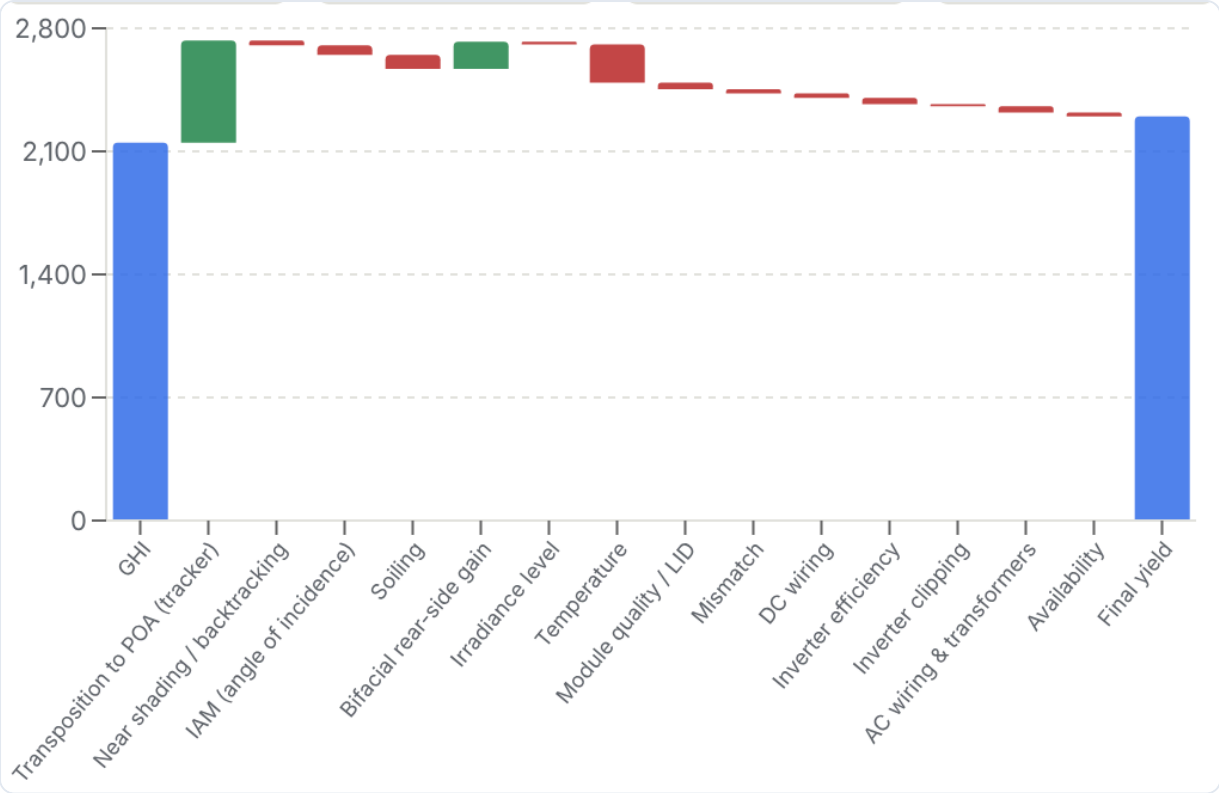


Figure B2.3 Loss waterfall in kWh/kWp. Green bars are gains (tracker transposition, bifacial rear), red bars losses; temperature is the largest red bar.

The **performance ratio** removes the weather. **Final yield** Y_f is exported kWh per kWp; **reference yield** Y_r is POA irradiation / 1 kW/m², the hours of equivalent STC sun.

$$PR = Y_f / Y_r = (E_{AC}/P_{STC}) / (H_{POA}/1\ \text{kW/m}^2)$$

E_{AC} kWh; P_{STC} kWp; H_{POA} kWh/m²

▸ **Worked example: PR test**

50 MWp, 65 kWh/m² POA over a 10-day test, 2,665 MWh exported. Y_f = 2,665/50 = 53.3 h; Y_r = 65 h; PR = 53.3/65 = 0.82. Irradiance-weighted cell temperature was 55 °C with γ = −0.30 %/°C: correction 1 − 0.0030 × 30 = 0.91, so PR corrected to 25 °C = 0.82/0.91 = 0.90.

Guarantees use **temperature-corrected PR** (IEC 61724-1) so a hot month is not read as underperformance. The contract must fix T_{ref} (25 °C, or the annual average cell temperature) and whether PR is front-side or bifacial. Raw annual PR for Kuwaiti trackers is about 0.78 to 0.84.

Soiling and cleaning

Soiling (dust on glass) grows roughly linearly at rate s per day and resets at cleaning, averaging sN/2 over an interval of N days. Minimising cleaning cost 365C/N plus lost value (sN/2)Ep gives:

$$N^* = \sqrt{(730\ C / (s\ E\ p))}$$

C USD per full clean; s per day; E kWh/yr; p USD/kWh

▸ **Worked example: module 10 base case**

C = USD 143,000, s = 0.003, E = 3.003 × 10^9 kWh, p = USD 0.018/kWh: N* = √(104,390,000 / 162,162) = √643.7 = 25.4 days. Average loss 3.8 %; cleaning USD 2.06 m/yr equals lost value USD 2.06 m/yr, total USD 4.1 m/yr. At an avoided-fuel value of USD 0.06/kWh, N* = 13.9 days: the ministry's optimum cleans more often than the IPP's.

Degradation, P50 and P90

After a first-year loss (LID, LeTID), modules **degrade** about 0.4 to 0.5 %/yr; at 0.4 %/yr year 30 gives 0.996^{29} = 0.89 of year 1. Yield is uncertain (resource 4 to 6 % without site data, 2.5 to 3.5 % with 12 months of site-adapted data, plus model, soiling and weather), combining to σ of about 6 % with good data and 7 to 8 % without. **P50** is exceeded in half of years, **P90** in 9 of 10 (Figure B2.4). Lenders size debt on P90 with a **DSCR** (debt service cover ratio) of about 1.20 to 1.30×, so lower σ means more debt and a lower tariff.

$$P90 = P50 \cdot (1 - 1.282\,\sigma)$$

normal distribution; P99 uses 2.326

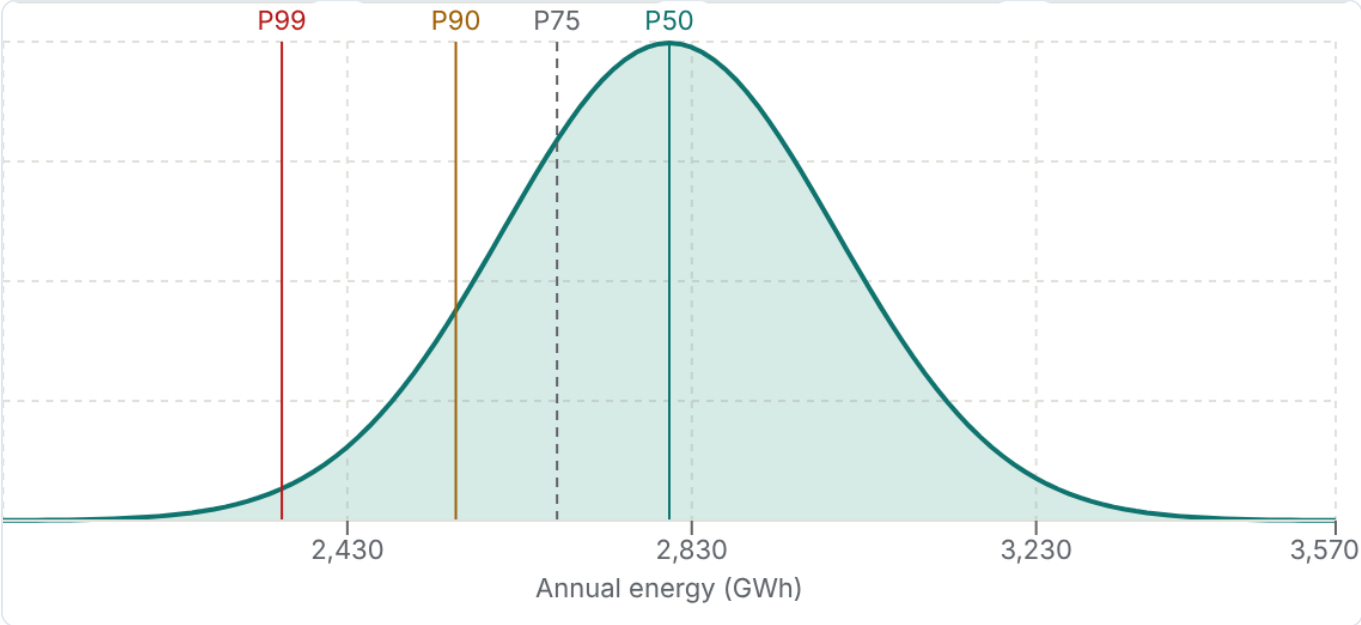


Figure B2.4 Distribution of annual energy: P50 at the peak, P75, P90 and P99 further left. A wider curve pushes P90 away from P50.

▸ **Worked example: illustrative Zone 1 base case**

1,430 MWp / 1,100 MWac (DC/AC 1.30), 2,100 kWh/kWp: 3,003 GWh in year 1. CF = 3,003,000 / (1,100 × 8,760) = 31.2 % on AC, 24.0 % on DC. P90 at σ 6 %: 3,003 × (1 − 0.0769) = 2,772 GWh; at 8 %: 2,695 GWh.

In Kuwait

- Shagaya Phase III Zone 1: 1,100 MW PV with its own 400 kV substation; Masdar with Fouad AlGhanim reported selected around August 2026; tariff not published M. The 1,430 MWp and 2,100 kWh/kWp above are illustrative, not bid data.
- Shagaya to Al-Wafra 400 kV line: separate MEWRE contract, KD 48.6 m, March 2026 M, an export interface risk.
- Kuwaiti studies: 8.7 to 15.1 % soiling loss with monthly cleaning M.

What the PM does

- Fixes the PR test method in advance: duration, PR basis, T_{ref}, data filters, sensor calibration.
- Rebuilds each bid's yield from GHI, POA/GHI and PR, and checks the soiling assumption against the paid cleaning regime.
- Witnesses flash tests, EL and thermal imaging, string I-V traces, tracker backtracking and stow tests.

Questions to ask

- What export in MWac is guaranteed at 50 °C, on which inverter derating curve?
- Show the uncertainty table: is the weather term 1-year or 10-year?

Remember

- MWp is DC, MWac is export; clipping is a designed trade-off.
- PR = Y_f/Y_r; guarantees correct it for temperature.
- N* = √(730C/(sEp)); P90 = P50(1 − 1.282σ).

Check yourself

1. A 500 MWac plant has DC/AC 1.3. How many MWp?
2. A 20 MWp plant exports 1,066 MWh in a week with 65 kWh/m² POA. What is its PR?
3. P50 = 1,000 GWh/yr, σ = 7 %. What is P90?
4. If the soiling rate doubles, the optimal cleaning interval:
   A. halves
   B. falls by about 29 %
   C. doubles
   D. does not change

B3 CSP and wind

about 9 min

After this section you can

- Explain why CSP needs DNI and size molten-salt storage.
- Use $P = \frac{1}{2}\rho A v^3 C_p$ with the Betz limit and air density.

Concentrated solar power

CSP (concentrated solar power) focuses the direct beam with tracking mirrors, heats a fluid and runs a steam turbine. Mirrors redirect only rays from one direction, so CSP uses DNI alone. In a **parabolic trough** (Figure B3.1) curved mirror rows focus onto an evacuated **receiver tube**, heating a **heat transfer fluid** (HTF), usually synthetic oil, from about 293 to 393 °C. A **tower** uses flat **heliostats** to heat molten salt to about 565 °C, giving a more efficient turbine (41 to 44 % against 37 to 38 %) and about three times the heat stored per tonne of salt.

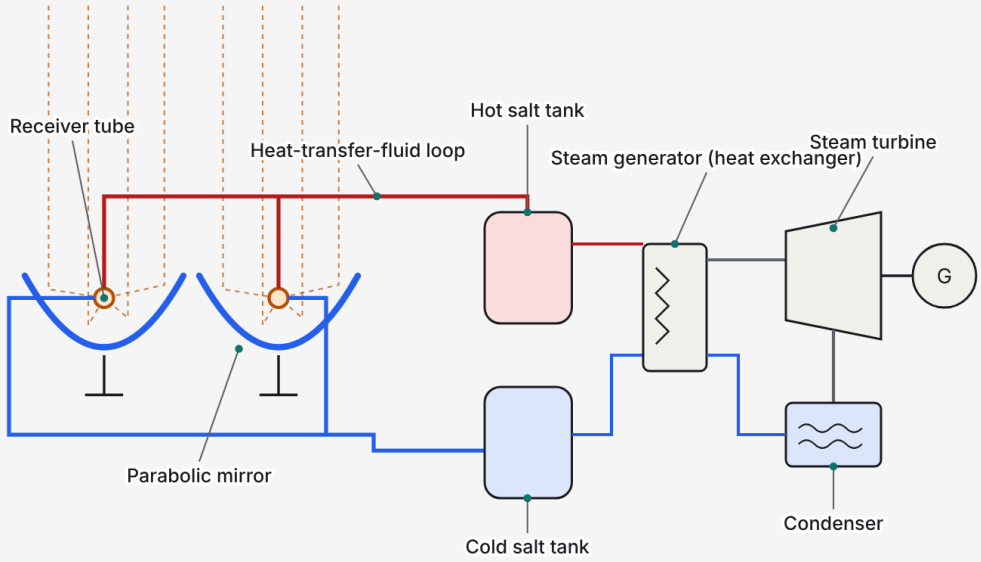


Figure B3.1 Trough CSP: mirrors, receiver tube, HTF loop to either the hot salt tank or the steam generator, then turbine and condenser. Storage sits between collecting heat and generating.

Storage uses two tanks of solar salt (60 % $NaNO_3$, 40 % KNO_3): by day salt moves from the cold tank (about 290 °C) to the hot tank, after sunset it raises steam. Salt freezes at about 238 °C, so pipes need heat tracing.

$$E_{th} = m \cdot c_p \cdot \Delta T \qquad \text{hours} = E_{th} \cdot \eta_{cycle} / P_e$$

$c_p \approx 1.5$ kJ/(kg·K); 1 MWh = 3.6×10^6 kJ

▸ **Worked example**

30,000 t, ΔT = 96 K: $3 \times 10^7 \times 1.5 \times 96 = 4.32 \times 10^9$ kJ = 1,200 MWh_{th}. At 37 %: 444 MWh_e, or 8.9 h at 50 MW.

Receiver heat losses are roughly fixed, so dust is amplified: if losses take 20 % of design heat, a 10 % DNI shortfall leaves 0.70 instead of 0.80 useful, 12.5 % less (typically 12 to 15 %). Mirrors soil by 0.2 to 1 % reflectivity per day in dusty seasons.

CSP or PV + BESS? Compare the cost of the same delivery profile (for example 200 MW from 18:00 for 6 h). Battery cost rises almost in proportion to MWh; extra salt hours are cheap. PV + BESS wins at 2 to 4 h; CSP competes from about 8 to 12 h and offers a synchronous turbine (inertia, fault current).

Wind

> $P = ½ \cdot \rho \cdot A \cdot v^3 \cdot C_p$
>
> ρ kg/m³; $A = \pi D^2/4$; v m/s; $C_{p,max} = 16/27 = 0.593$ (Betz)

The **power coefficient** C_p is the fraction of wind power captured. Extraction slows the air, and a rotor that stopped it would block the flow, so the ideal maximum (**Betz limit**) is 59.3 %; real rotors reach 0.48 to 0.50. Power goes with v^3: 10 % more speed gives 1.33 times the power. Energy therefore depends on the mean of v^3: with a **Weibull distribution** (k = 2) it is 1.91 times the cube of the mean. At 7.9 m/s and ρ = 1.10 kg/m³, $½\rho\bar{v}^3$ gives 271 W/m² but the Weibull average is 518 W/m².

Hot air is thinner: 1.225 kg/m³ at 15 °C, about 1.11 at 45 °C, about 9 % less power. A 100 m rotor at 10 m/s, C_p 0.45, gives 2.16 MW at 15 °C and 1.96 MW at 45 °C. **Capacity factor** = energy / (rating × 8,760 h). Desert design needs hot-climate packages (45 °C), leading-edge protection against sand erosion (1 to 5 % energy otherwise) and dust sealing.

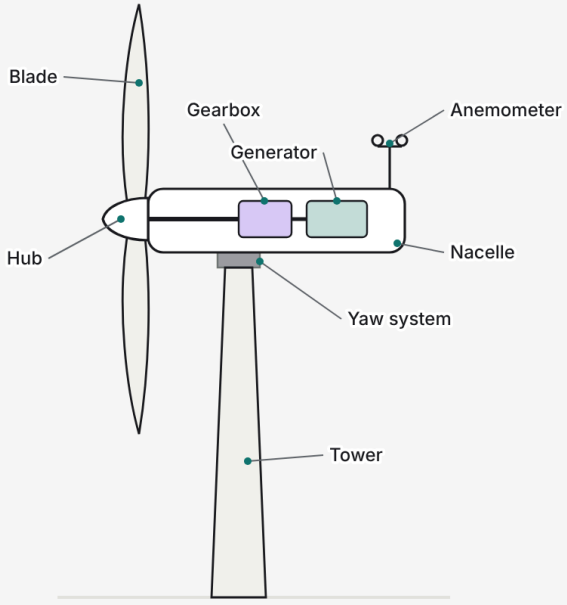


Figure B3.2 Blades and hub (with pitch control) drive the shaft; the gearbox raises speed for the generator in the nacelle; the yaw system faces the rotor into the wind measured by the anemometer.

In Kuwait

- Shagaya Phase I, 70 MW KISR pilot: 50 MW CSP trough, 10 MW PV, 10 MW wind H.
- CSP storage quoted as about 10 h, 1,200 MWh_{th}, full operation about 2019 M (our check: 8.9 h). "About 225,000 MWh/yr" means CF 51 % if CSP only, 37 % over 70 MW; performance since 2019 not public L.
- Wind: five 2 MW turbines, May 2017 H; mean about 7.9 m/s, figures conflict V; CF 45.2 % in year 1, about 70 % in summer months (peer-reviewed study) M.

Questions to ask

- Is the CSP yield on ground-measured DNI? Are storage hours gross or net, at full load?
- Who operates Shagaya CSP now, and can we have monthly generation and DNI since 2019?

Remember

- CSP: DNI only; its case is cheap long storage, not efficiency.
- $E = m\ c_p\ \Delta T$; hours = $MWh_{th} \times \eta / MW_e$.
- Wind: v^3, mean of cubes, Betz 0.593, hot air about −9 %.

Check yourself

1. Why does a 10 % DNI shortfall cost a trough plant more than 10 % of output?
   A. Salt freezes
   B. Receiver heat losses are roughly fixed
   C. The turbine trips
   D. DHI rises
2. 600 MWh_{th} of storage, 25 MW turbine at 38 %. Full-load hours?
3. Wind rises from 8 to 9 m/s. By what factor does power rise (below rated)?
4. For Kuwait's evening peak:
   A. CSP is always cheaper because it is dispatchable
   B. PV + BESS usually wins at 2 to 4 h; CSP improves from about 8 h
   C. PV LCOE alone decides
   D. Duration is irrelevant

B4 Battery energy storage

about 12 min

After this section you can

- Name the blocks of a BESS and follow energy from DC nameplate to the POI.
- Size a 500 MW / 4 h BESS and explain augmentation, warranties and fire safety.

Architecture and ratings

In a **BESS** (battery energy storage system, Figure B4.1) LFP **cells** of about 1 kWh form **modules**, then **racks** of about 1,331 V; a 20-ft container holds about 5 MWh. The **BMS** (battery management system) monitors every cell's voltage, current and temperature, estimates charge and health, balances cells and trips contactors on faults. The **PCS** (power conversion system) is a bidirectional inverter that charges and discharges and provides grid-code behaviour (reactive power, fault ride-through). A transformer steps up to 33 kV. The **EMS** (energy management system) schedules dispatch; **HVAC** holds cells near 25 °C.

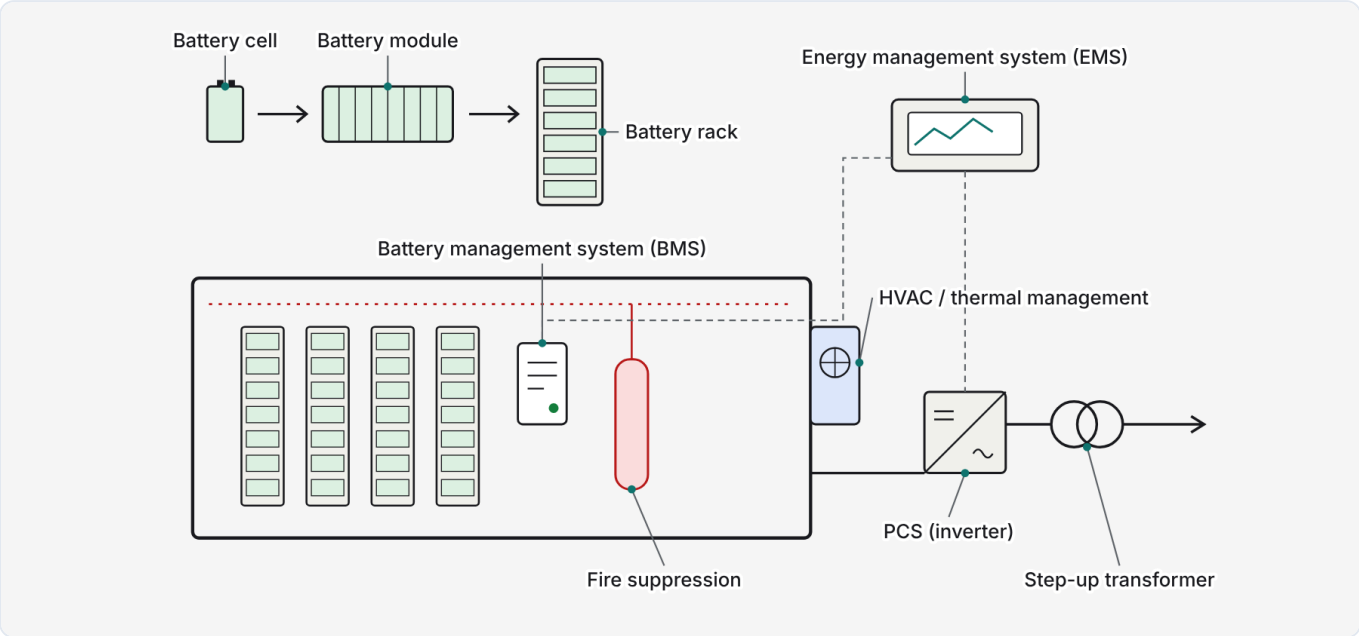


Figure B4.1 Cell, module and rack inside the container with BMS, HVAC and fire suppression; PCS and step-up transformer outside; EMS supervising.

Power (MW) is set by the PCS; energy (MWh) by the number of cells. **Duration** = MWh/MW; **C-rate** = MW/MWh, so 4 h is 0.25C. **State of charge** (SoC) is the stored fraction; the BMS holds it inside a window such as 2.5 to 97.5 %, whose width (95 %) is the usable **depth of discharge** (DoD). **State of health** (SoH) is capacity relative to new. **Round-trip efficiency** (RTE) is energy out / energy in at a stated boundary.

▸ **Worked example: "250 MW / 1,000 MWh" at the POI, 50 °C day**

200 containers × 5.016 = 1,003.2 MWh DC (0.25C). SoC window: × 0.95 = 953.0. Cell losses (DC round trip 95 %, one way 0.975): 928.9. PCS 98.5 % × MV 99.3 % × HV 99.6 % = 0.974: **904.9 MWh at the POI**, 3.62 h at 250 MW. AC-AC RTE 0.95 × 0.985^2 × 0.993^2 × 0.996^2 = 90.2 %; with auxiliaries of 12 kW per container (57.6 MWh/day), 904.9 / (1,003.7 + 57.6) = 85.3 %.

Chemistry, aging and warranties

LFP (lithium iron phosphate) gives 6,000 to 10,000+ cycles and runs away later without releasing oxygen. **NMC** (nickel manganese cobalt) stores more per kg but lasts 2,000 to 5,000 cycles and releases oxygen in runaway. Weight is irrelevant for a stationary plant, so LFP is the grid standard.

Capacity fades through **calendar aging** (time, temperature, high SoC) and **cycle aging** (throughput, DoD, C-rate): about 2 to 4 % in year 1, then 1 to 2 %/yr. Aging roughly doubles per +10 °C. Kuwait's duty (charge from PV by 15:00, sit full in the heat, discharge 18:00 to 22:00) is severe. To keep the POI MWh for 20 years the owner **overbuilds** or **augments**, adding containers every 4 to 6 years (Figure B4.2).

Augmenting is usually cheaper, provided a creditworthy party must do it and space, feeder bays and EMS capacity were reserved.

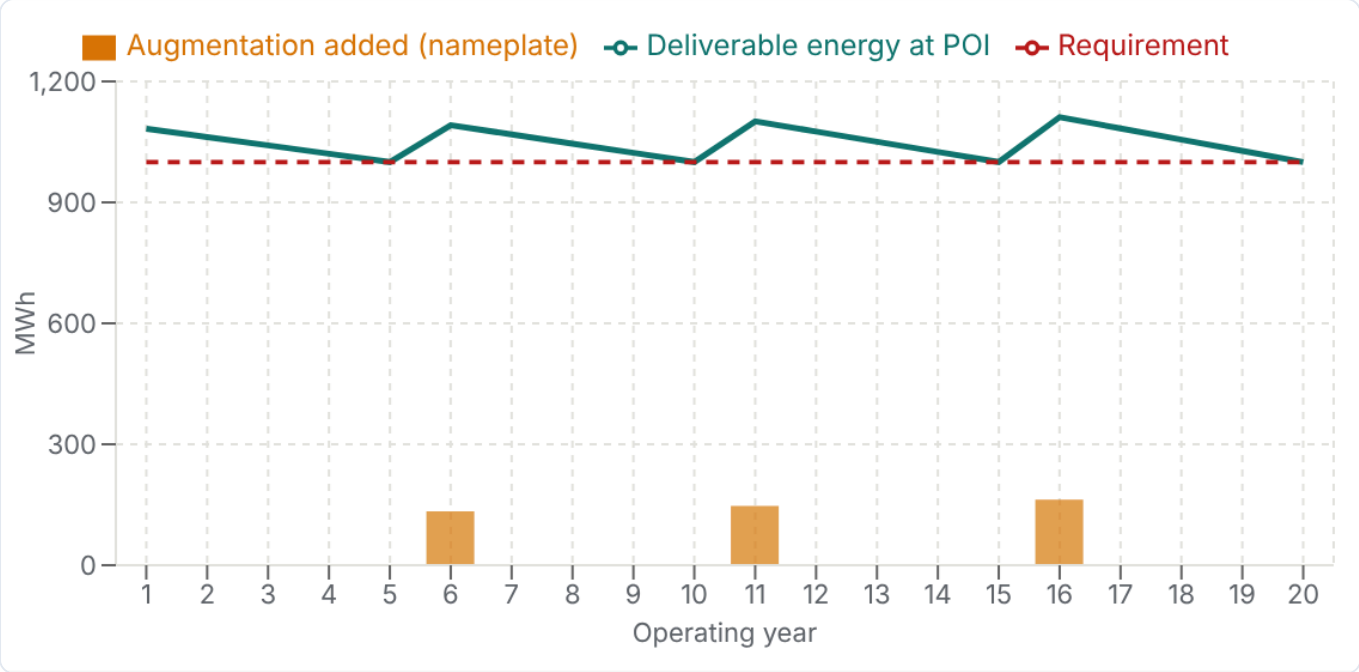


Figure B4.2 Deliverable POI energy (green) fades to the requirement (red dashed); augmentation (orange) in years 6, 11 and 16 restores the margin.

Warranties read "X years or Y equivalent full cycles (EFC), whichever first", valid only inside a window of temperature, C-rate and SoC. "20 years or 7,300 EFC" at 1.3 cycles/day ends after 7,300 / (1.3 × 365) = 15.4 years.

Fire safety

In **thermal runaway** a cell's internal layers break down (80 to 120 °C), the separator melts (130 to 165 °C), the cell shorts internally and vents flammable, toxic gas, which can accumulate and explode when a door opens. Defences: BMS and HVAC prevent; gas detection warns; spacing stops cell-to-cell propagation; explosion vents or exhaust; an emergency plan with the fire authority. Suppression knocks down flame but does not stop runaway. **UL 9540A** is the test method giving propagation data for the specific cell, module and layout; **NFPA 855** governs installation and spacing.

Services

Energy shifting moves midday PV into the evening peak (Kuwait's main need, 2 to 6 h). **Fast frequency response** injects or absorbs power within 0.5 to 1 s and needs MW headroom rather than MWh. The PCS supplies **reactive power** for voltage support, so it is rated in MVA. A **grid-forming** PCS sets its own voltage and frequency, steadying weak grids and enabling black start. Stacked services add cycles.

▸ **Worked example: 500 MW × 4 h at the POI**

2,000 MWh at the POI; auxiliaries 10 MW in discharge; augment at SoH 0.90. MV bus 2,000/0.996 + 40 = 2,048 MWh; PCS AC 2,048/0.993 = 2,062 MWh; DC 2,062/0.985 = 2,094 MWh; cells 2,094/0.9747 = 2,148 MWh; SoC window 2,148/0.95 = 2,261 MWh; SoH floor 2,261/0.90 = **2,513 MWh DC** (501 containers), 1.26 × the POI figure. PCS: (500/0.996 + 10)/0.993 = 515.6 MW, 542.8 MVA at 0.95 power factor.

In Kuwait

- MEWRE BESS plan: 1 to 1.5 GW / 4 to 6 GWh (4 h), first 500 MW for summer 2027 M-L; counterparty and chemistry unverified V.
- No Kuwaiti BESS code found: NFPA 855, UL 9540/9540A and IEC 62619 are adopted by reference.
- Chiller COP falls from about 4 at 35 °C to 2.5 at 50 °C: specify 50 °C design, operation to 55 °C, N+1.

What the PM does

- Writes every MWh guarantee at the POI meter, at 50 °C, with auxiliaries and SoC window defined.
- Tests the degradation model against the real duty and assigns augmentation with security.
- Requires UL 9540A data for the exact design and a drilled emergency plan before energisation.

Questions to ask

- Is the PCS rated for full MW plus reactive power at 50 °C, and is it grid-forming capable?
- What cycle cap and operating window does the warranty set?

Remember

- Duration = MWh/MW; 0.25C = 4 h. 1,003 MWh DC delivers about 905 MWh at the POI.
- DC nameplate ≈ 1.26 × POI MWh; RTE about 90 % (85 % with summer auxiliaries).
- LFP; aging ×2 per +10 °C; warranties end at the first of years or cycles.

Check yourself

1. A BESS is 300 MW / 1,200 MWh at the POI. C-rate and duration?
2. 1,000 MWh nameplate, 95 % DoD, SoH 0.97. Usable DC energy?
3. "15 years or 6,000 EFC" at 1.2 cycles/day. When does it end?
4. What does a UL 9540A report provide?
   A. A pass/fail certificate for any battery
   B. Propagation test data for the specific cell, module and layout
   C. Installation spacing rules
   D. The RTE guarantee

PART C · TRIAL DAY 3 · ABOUT 56 MIN

Grid connection

Read the Kuwaiti grid, the grid code, grid strength (SCR), inverter behaviour in faults, and the studies (including EMT) a project must pass.

C1 Transmission, substations and the Kuwaiti grid

about 12 min

After this section you can

- explain why bulk power moves at 400 kV and quantify the loss saving;
- read the Kuwaiti single-line diagram from generator to socket;
- name every device in a high-voltage bay and say what it does;
- quote typical ratings for each voltage level and the transformer data a PM must check.

Why transmission runs at high voltage

For a given power, three-phase line current is $I = P / (\sqrt{3} \cdot V \cdot \cos\varphi)$. Conductor heating and energy loss are $3I^2R$. Raising the voltage lowers the current in proportion, so losses fall with the square of the voltage. Higher voltage also raises the power a line can transfer before stability or voltage drop limits it, because transfer capacity scales with V^2/X.

$$P_{loss} = 3\, I^2\, R \propto P^2\, R / V^2$$

for fixed P, power factor and conductor resistance R

▸ **Worked example**

Moving 1,100 MW at power factor 0.95: at 400 kV, I = 1,100 / (1.732 × 400 × 0.95) = 1,671 A. At 132 kV, I = 5,064 A, about 3.03 times more. Losses scale with I^2, so $(400/132)^2 = 9.2$: the 132 kV option loses about nine times more energy in the same conductor and would need several more circuits to carry the current at all.

The Kuwaiti voltage ladder

Figure C1.1 shows the ladder. Power stations generate at 11 to 21 kV and step up to 400 kV. The 400 kV network moves bulk power between stations and to grid substations, which step down to 132 kV. Primary substations take 132 kV down to 33 kV (large consumers, desalination, districts) and 11 kV (street cable networks). Distribution transformers then supply 415 V three-phase and 240 V single-phase. A large PV plant collects its inverter output at 33 kV and steps up to join the 400 kV level directly.

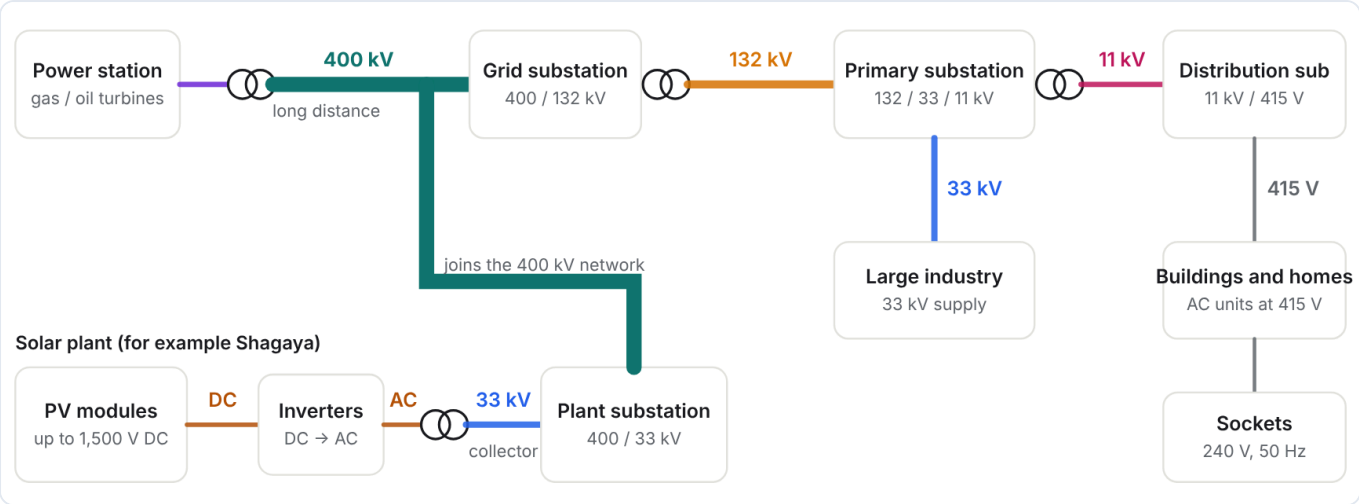


Figure C1.1 The voltage ladder. Each step down is a substation with transformers; notice that the PV plant skips the lower levels and connects at 400 kV through its own 33 kV collector system.

In Kuwait

- 400 kV is confirmed for new renewable connections: the Shagaya to Al-Wafra 400 kV line (awarded to Power Grid Company, KD 48.6 m, March 2026) and the 400 kV substation in Zone 1 (1,100 MW) scope M
- An older EHV level is quoted as 300 kV or 275 kV; sources differ and it is unverified V
- 132, 33 and 11 kV and 415/240 V are commonly cited but not verified in the course sources M
- System frequency 50 Hz H. The GCC interconnector links at 400 kV, with a back-to-back HVDC converter to Saudi Arabia's 60 Hz system; Kuwait imported 500 MW through it in summer 2024 M

Reading the single-line diagram

A **single-line diagram (SLD)** draws a three-phase system with one line per circuit and standard symbols for each device. Figure C1.2 is a simplified Kuwaiti SLD; read it top to bottom.

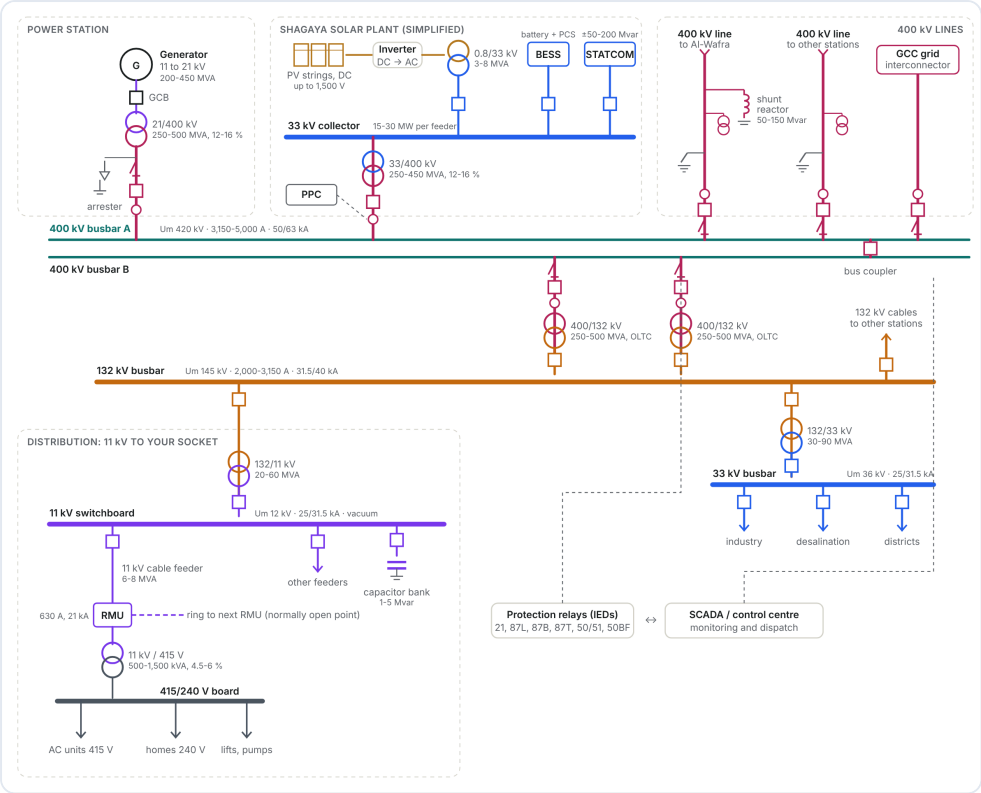


Figure C1.2 Simplified Kuwaiti SLD with typical ratings (teaching drawing, not an as-built). Notice the two 400 kV busbars joined by a bus coupler, two 400/132 kV transformers for N-1, and the PPC measuring at the plant's 400 kV connection.

Top left: the generator (11 to 21 kV, 200 to 450 MVA) feeds a **generator circuit breaker (GCB)** and a 21/400 kV **generator step-up transformer (GSU)**. **Top centre:** the PV plant: DC strings up to 1,500 V, inverters to 0.8 kV AC, 0.8/33 kV transformers of 3 to 8 MVA, a 33 kV collector bus carrying 15 to 30 MW per feeder, BESS and a STATCOM on the same bus, then a 33/400 kV plant transformer. The **point of interconnection (POI)** is where the plant meets the grid and where all grid code requirements are measured. **Top right:** 400 kV lines with a **shunt reactor** (an inductor that absorbs the reactive power long lines generate at light load) and the GCC interconnector.

Middle: two 400 kV busbars with a coupler, then 400/132 kV transformers with **on-load tap changers** into the 132 kV busbar. **Bottom:** 132/33 kV to industry and desalination; 132/11 kV to an 11 kV switchboard with a capacitor bank; 11 kV feeders through **ring main units (RMUs)**, compact switchgear on cable rings run with a normally open point so a faulted section can be isolated and supply restored from the other side; finally 11 kV/415 V transformers to air-conditioning, homes and pumps. Protection relays and SCADA tie every bay to the control centre.

Inside a substation bay

A **bay** is the set of equipment that connects one circuit (a line or transformer) to the busbar. Figure C1.3 shows one, from the line inwards.

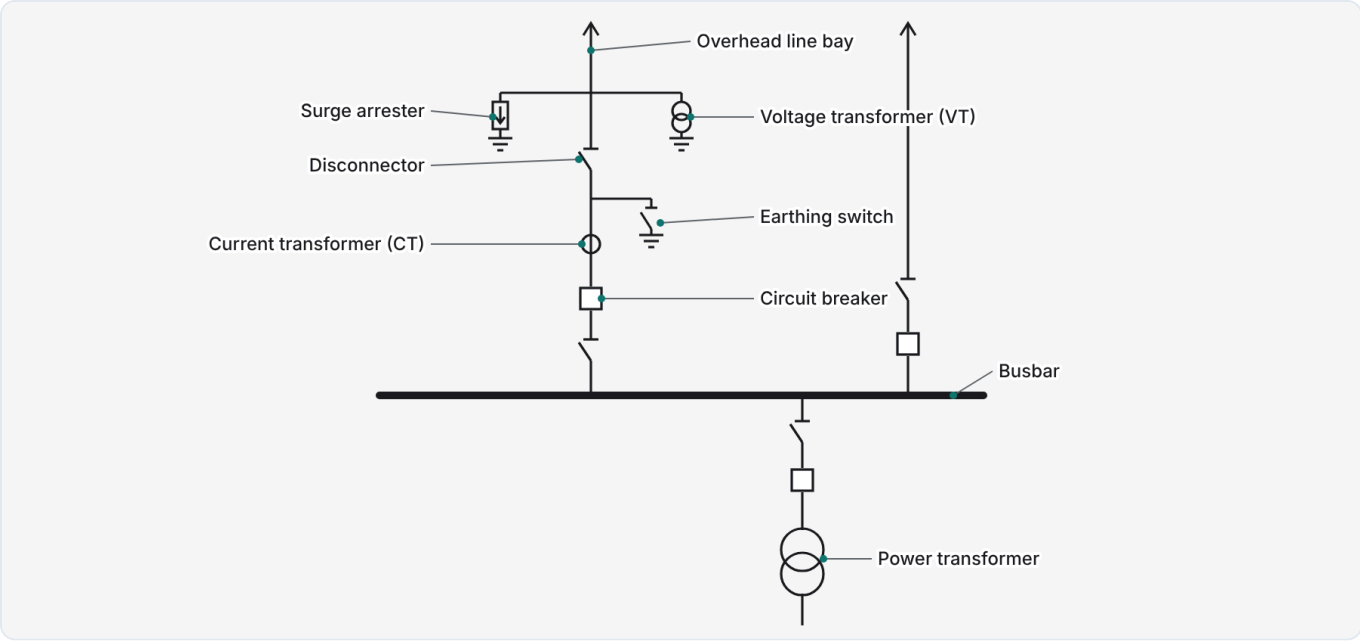


Figure C1.3 One HV bay. Notice that the breaker sits between disconnectors, so it can be isolated on both sides for maintenance.

- **Circuit breaker:** interrupts load and fault current (a 400 kV breaker clears up to 50 or 63 kA in about 40 to 60 ms after the trip command).
- **Disconnector:** gives a visible open gap for safe isolation; it is operated only with no current flowing.
- **Earthing switch:** connects an isolated section to earth so people can work on it.
- **Current transformer (CT):** scales primary current to 1 A for relays and meters.
- **Voltage transformer (VT) or capacitor voltage transformer (CVT):** scales voltage to 110 V for relays and meters; the CVT is the cheaper form at 400 kV.
- **Surge arrester:** a metal-oxide device that conducts to earth above a set voltage, limiting lightning and switching overvoltages.
- **Busbar:** the common conductor all bays connect to.

Bus schemes decide what a single failure costs. With **double bus and coupler**, each circuit can be switched to either busbar, so a bus fault or maintenance loses a bus, not the station. In **breaker-and-a-half**, three breakers serve two circuits between two buses; any one breaker can be removed without losing a circuit, which is why it is the EHV standard. **Air-insulated switchgear (AIS)** uses air spacing: cheap but large and exposed to dust and salt flashover. **Gas-insulated switchgear (GIS)** encloses everything in SF_6: compact and sealed against Kuwaiti dust, but needs SF_6 leak management and ties you to one manufacturer for spares.

Transformers: the data that matters

The voltage ratio equals the turns ratio, and current scales inversely. **Percentage impedance (z)** is the voltage needed to drive rated current through a short-circuited winding, as a percentage of rated voltage; it sets both the voltage drop on load and the fault current let through ($I_{sc} \approx I_{rated} / z$). A 14 % transformer passes about 7 times rated current into a fault on its terminals, before source impedance is added. The **OLTC** changes the ratio in steps of about 1.25 % under load to hold voltage. Cooling class **ONAN** (oil natural, air natural) gives the base rating; **ONAF** (fans) adds roughly 25 to 35 %. Ratings are quoted at a design ambient; at 50 °C expect roughly 1 % derating per °C above it, to be confirmed by the manufacturer.

System	U_m	BIL	Rated current	Breaker	Transformers
400 kV	420 kV	1,425 kV	3,150 to 5,000 A	50 / 63 kA	GSU, plant, 400/132 kV: 250 to 500 MVA, z 12 to 16 %
132 kV	145 kV	650 kV	2,000 to 3,150 A	31.5 / 40 kA	132/33 kV 30 to 90 MVA; 132/11 kV 20 to 60 MVA
33 kV	36 kV	170 kV	1,250 to 2,500 A	25 / 31.5 kA	inverter transformers 3 to 8 MVA
11 kV	12 kV	75 kV	630 to 2,500 A	25 / 31.5 kA	distribution 500 to 1,500 kVA, z 4.5 to 6 %
415/240 V	< 1 kV	n/a	boards to 4,000 A	25 to 50 kA	1,000 kVA gives about 1,390 A on LV

U_m is the highest voltage the equipment is designed for; **BIL** (basic insulation level) is the lightning impulse it withstands. These are IEC values and common practice, not MEWRE specifications.

What the PM does

- Checks that the SLD, bus scheme and breaker ratings match the fault study and MEWRE's standards.
- Confirms transformer MVA, z and cooling are stated at 50 °C ambient.
- Tracks the interface between the IPP's 400 kV substation and MEWRE's separately procured line.

Questions to ask

- What bus scheme does MEWRE require at a 400 kV renewable connection: breaker-and-a-half or double bus?
- AIS or GIS, and what is the SF_6 handling plan?
- At what ambient are the transformer ratings guaranteed?

Remember

- Losses $\propto 1/V^2$: 400 kV instead of 132 kV cuts current by 3 and losses by about 9.
- Only 400 kV is confirmed in the sources; other Kuwaiti levels are commonly cited.
- Bay: disconnector, breaker, CT, VT, earthing switch, arrester. Disconnectors never break current.
- Transformer z sets voltage drop and fault current; rate everything at 50 °C.

Check yourself

1. For the same power and conductor, by what factor do I^2R losses fall when voltage rises from 132 kV to 400 kV?
   A. 3.0
   B. 9.2
   C. 27.5
   D. 1.7
2. Which device provides visible isolation but must never open under load?
   A. Circuit breaker
   B. Surge arrester
   C. Disconnector
   D. CVT
3. Why is breaker-and-a-half preferred at EHV?
4. What full-load current does a 1,000 kVA, 11 kV/415 V transformer deliver on the LV side (A)?

C2 Grid code requirements and plant control

about 13 min

After this section you can

- list what a grid code demands of an inverter-based plant and why each item exists;
- calculate reactive capability at the POI, droop response and RoCoF;
- explain how the power plant controller, SCADA and communications deliver compliance.

What a grid code is

A **grid code** is the technical rulebook every plant must meet to connect: how it behaves in normal operation, during disturbances, and how compliance is proven. System security depends on every plant behaving predictably at the same instant. Synchronous generators provide inertia, fault current and voltage support through their physics; inverters do only what their software tells them, so the code specifies those behaviours.

In Kuwait

- No published Kuwaiti grid code for renewable or inverter-based plants was found in the course sources. Requirements are likely in the RFP technical schedules, PPA and connection agreement. Obtain the current text from MEWRE Transmission before relying on any value below V
- Values in this section are international comparators: ENTSO-E RfG, IEEE 2800, Saudi and Abu Dhabi codes, GCCIA.

Operating ranges

The plant must stay connected across a defined frequency and voltage range, for example 49.0 to 51.0 Hz continuously and 0.90 to 1.05 pu at 400 kV, with wider bands for limited times. It must also withstand a **rate of change of frequency (RoCoF)**, the slope of frequency in Hz/s just after a disturbance, typically 1 to 2 Hz/s in modern codes.

Fault ride-through

A transmission fault pulls voltage down across a wide area until protection clears it in 80 to 120 ms. If every inverter in that area tripped, a cleared line fault would become the loss of hundreds of megawatts. **Low-voltage ride-through (LVRT)** requires the plant to stay connected while the POI voltage stays above an envelope; **high-voltage ride-through (HVRT)** does the same for swells, typically to 1.2 to 1.3 pu. Figure C2.1 shows a comparator envelope.

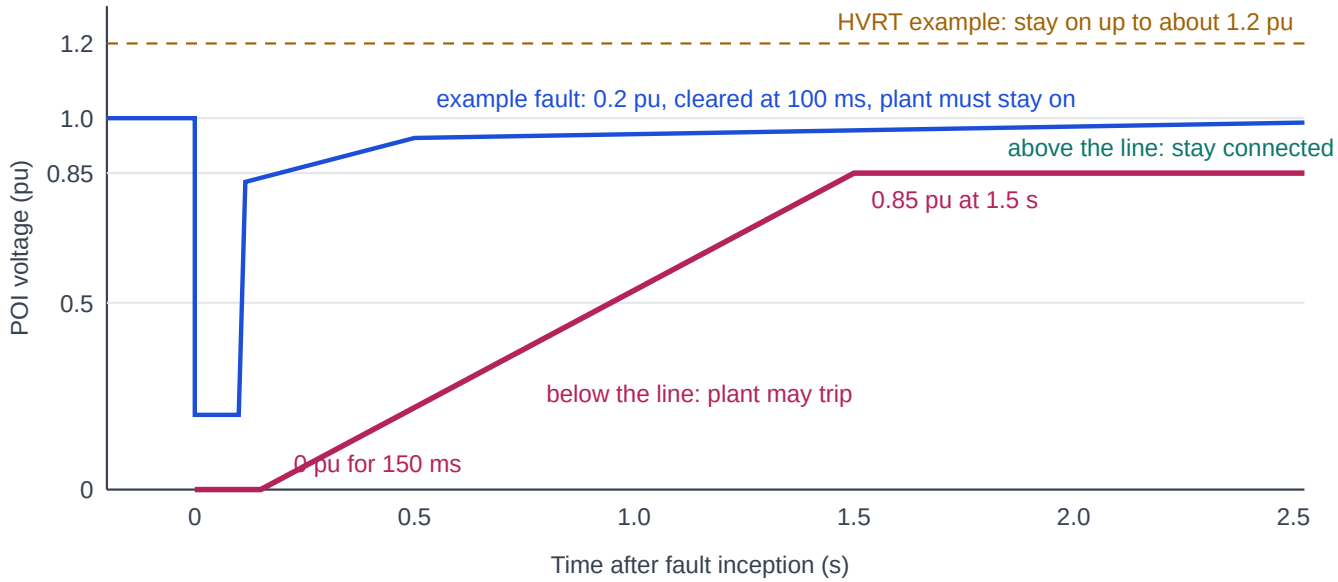


Figure C2.1 Ride-through envelope (red), an international comparator only: 0 pu for 150 ms, then a ramp to 0.85 pu by 1.5 s. The blue trace stays above it, so the plant must ride through and inject reactive current. Kuwait's values must be obtained from MEWRE Transmission.

During the dip the plant must inject **reactive current** proportional to the voltage drop, to hold voltage up near the fault. After clearance it must restore active power within a set time (typically under 1 s): too fast causes voltage overshoot, too slow deepens the frequency dip.

> $\Delta I_q = k \cdot \Delta V$
>
> in pu; k typically about 2 (codes range 2 to 6). A dip to 0.7 pu with k = 2 asks for 0.6 pu extra reactive current.

Reactive capability and voltage control

Voltage across a line changes by approximately $\Delta V \approx (R \cdot P + X \cdot Q) / V$. On transmission lines X is 10 or more times R, so reactive flow Q dominates voltage, while active flow P mainly sets the phase angle between buses. Supplying Q raises local voltage; absorbing it lowers voltage. Figure C2.2 shows this on a line with R = 0.01 and X = 0.10 pu: 0.5 pu of Q flow costs 0.05 pu of voltage, ten times what the same P would cost.

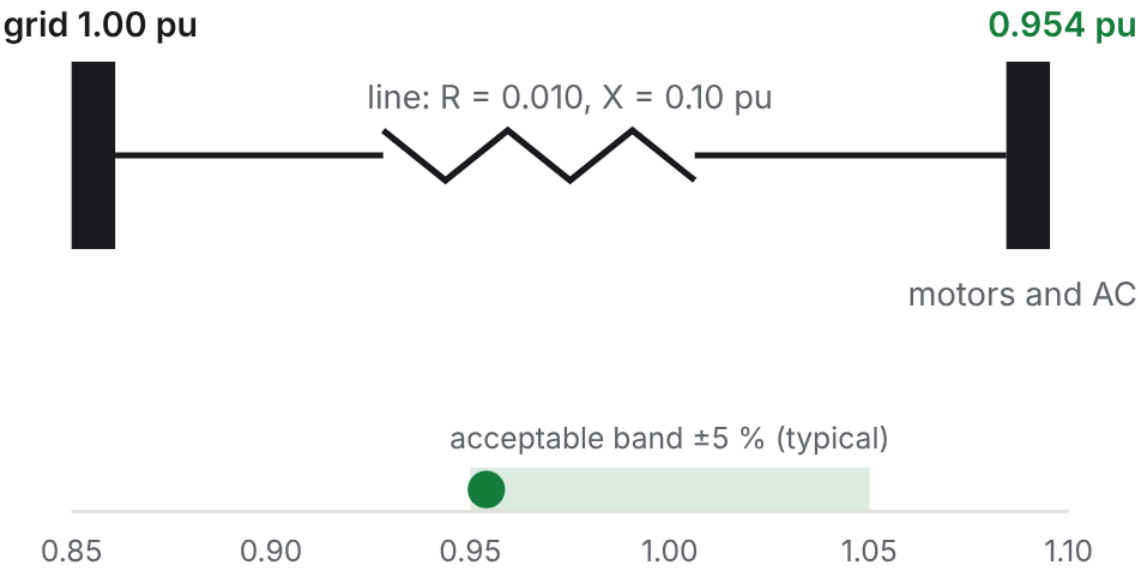


Figure C2.2 Receiving-end voltage set by flow through R and X. Notice the 0.954 pu result sits just inside a typical ±5 % band; the X term does most of the damage.

Codes therefore require reactive capability at the POI, typically 0.95 power factor lagging and leading at full output, across 0.90 to 1.10 pu voltage. In Kuwait it must hold at 50 °C. Figure C2.3 shows the triangle: $S^2 = P^2 + Q^2$, so reactive power competes with active power for the same inverter MVA.

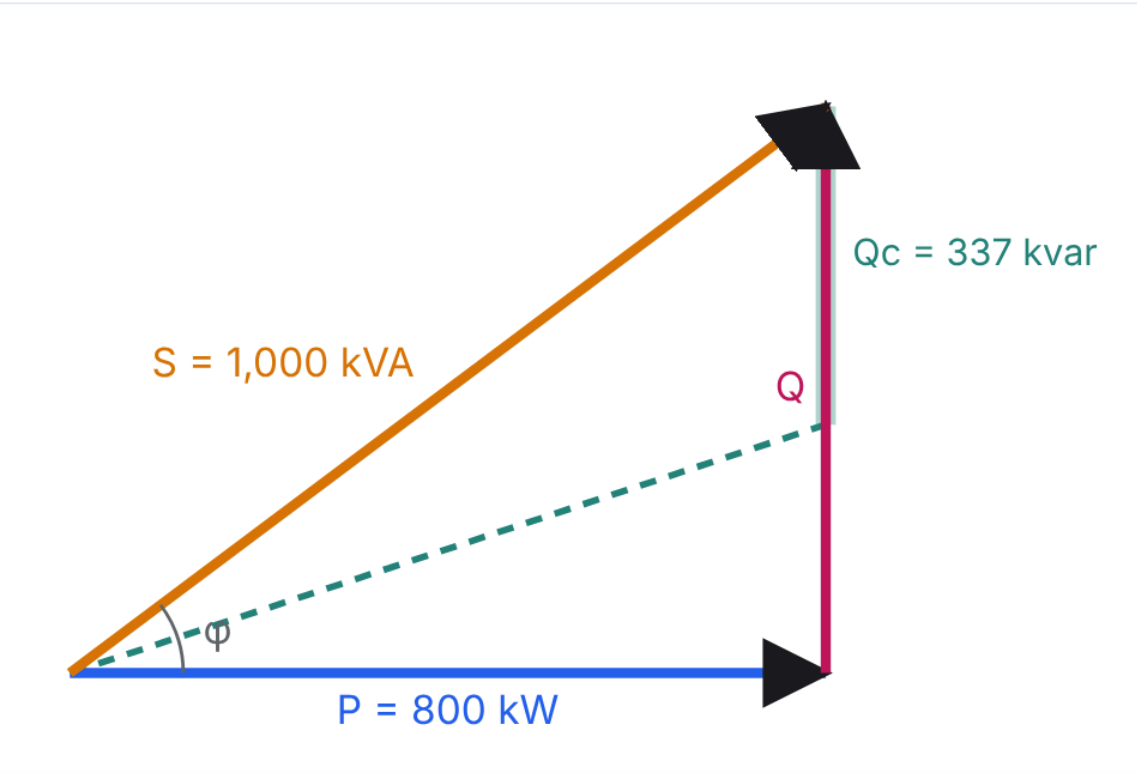


Figure C2.3 P, Q and S. Notice that at fixed S, every extra Mvar reduces the MW the inverter can carry.

▸ **Worked example: 1,100 MW at 0.95 PF at the POI**

Q = 1,100 × tan(arccos 0.95) = 1,100 × 0.329 = 361.5 Mvar each direction. Between inverters and POI, four 300 MVA, 14 % HV transformers absorb 156.4 Mvar and the MV transformers (1,250 MVA, 8 %) absorb 85.8 Mvar; cables supply about 20 Mvar. The inverters must produce about 584 Mvar while delivering 1,116.5 MW (P before losses), so S = √(1,116.5² + 584²) ≈ 1,260 MVA, 14.5 % above the plant MW rating. If 1,250 MVA of inverters derate 8 % at 50 °C to 1,150 MVA, available Q is √(1,150² − 1,116.5²) = 275.5 Mvar: about 308 Mvar short. Fix with more inverter MVA, a STATCOM, or a reduced-power capability curve at peak temperature.

Grid operators prefer **voltage control with slope**: the plant changes Q in proportion to the voltage error (for example a 2 to 5 % slope, meaning full rated Q for that percentage of voltage error), so several plants share the work without fighting each other.

Active power and frequency response

The plant follows the control centre's set-point and limits its **ramp rate**, typically to about 10 % of P_{max} per minute (110 MW/min for 1,100 MW). Only storage can limit a downward ramp from a dust front.

Frequency is held at 50 Hz only while generation equals demand. After a trip, the initial RoCoF depends on system inertia H (stored rotating energy, in seconds of rated power):

$$\mathrm{RoCoF} = \Delta P \cdot f_0 / (2\,H\,S_{sys})$$

300 MW trip, H = 4 s, S_{sys} = 15,000 MVA: 300 × 50 / (2 × 4 × 15,000) = 0.125 Hz/s

Losing 1,100 MW with H = 3 s on 12,000 MVA gives 0.76 Hz/s: more PV displacing turbines lowers H and raises RoCoF. Three modes answer frequency: **LFSM-O** (limited frequency sensitive mode, over-frequency) cuts output above a threshold such as 50.2 Hz; **LFSM-U** raises output below a threshold, which needs headroom; **FSM** (frequency sensitive mode) responds continuously around 50 Hz. Response follows a **droop** (Figure C2.4): the frequency change, as a percentage of 50 Hz, that moves output by 100 % of rating.

$$\Delta P = -(P_n / R) \cdot (\Delta f_{eff} / f_n)$$

R = droop (0.04 for 4 %), Δf_{eff} = deviation beyond the deadband

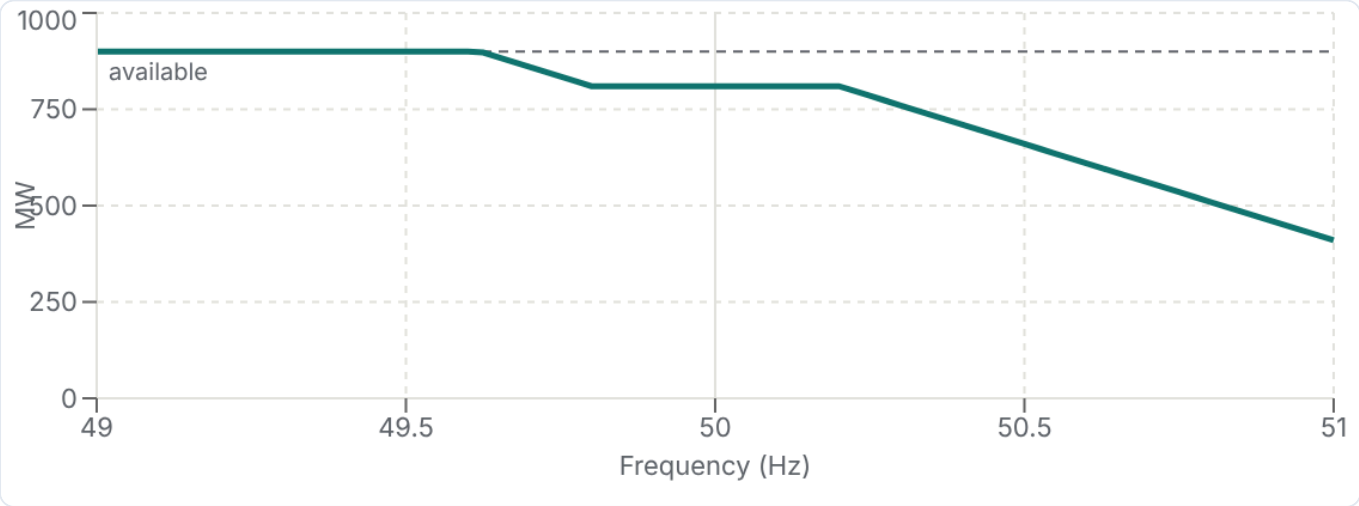


Figure C2.4 Droop characteristic. Notice the flat deadband around 50 Hz, output cut along the slope above it, and the rise below it stopping where available power (dashed) is reached.

▸ **Worked example: 500 MW BESS, 4 % droop**

Frequency falls to 49.6 Hz with a ±0.2 Hz deadband, so Δf_{eff} = 0.2 Hz. $\Delta P = (500 / 0.04) \times (0.2 / 50)$ = 50 MW. Without the deadband, Δf = 0.4 Hz gives 100 MW. A battery delivers this from any state; a PV plant only if it is already curtailed below available power.

Power quality

Inverters emit **harmonics**, currents at integer multiples of 50 Hz. **Total harmonic distortion** THD = $\sqrt{(\Sigma V_h^2)} / V_1$. IEEE 519 limits voltage THD to 1.5 % above 161 kV, with 1.0 % for any single harmonic. A study giving V_5 1.2 %, V_7 0.9 %, V_{11} 0.5 %, V_{13} 0.4 % yields THD 1.63 %: non-compliant, needing a filter or resonance mitigation. Codes also limit **flicker** (fast voltage fluctuation, indices P_{st} and P_{lt}) and **unbalance** (negative-sequence voltage, typically 1 to 2 %).

The power plant controller, SCADA and curtailment

The **power plant controller (PPC)** measures P, Q, V and f at the POI, compares them with control-centre set-points, and sends new set-points to every inverter and PCS every 100 to 200 ms. Its modes: P set-point with ramp limit, Q, power factor, voltage with slope, and frequency response. Response times form a ladder: inverters ride through faults alone in milliseconds; the PPC loop cycles in 100 to 200 ms; voltage control reaches 90 % of a step in 1 to 5 s; set-point changes settle in about 1 minute.

Inside the substation, **IEC 61850** links relays and controllers; its **GOOSE** messages carry trip signals in about 4 ms. The plant gateway speaks **IEC 60870-5-104** to the control centre over **OPGW** (optical fibre in the line's earth wire). **IEC 62443** sets cyber security: zones and conduits, a DMZ, vendor access only via a jump host with multi-factor login.

When the grid cannot take the output, the control centre curtails. Curtailment for grid constraint, security, oversupply or late grid is usually compensated as **deemed energy**, $\Sigma(P_{avail} - P_{actual})\Delta t$ above any annual allowance; plant-caused curtailment is not. Capping 950 MW available to 600 MW for 5 h deems 1,750 MWh.

What the PM does

- Obtains MEWRE's requirement text and writes it, with values, into the RFP and connection agreement.
- Checks that reactive capability is demonstrated at the POI, at 50 °C, across 0.90 to 1.10 pu.
- Agrees the signal list, protocol profile and deemed-energy estimation method early.

Questions to ask

- Which grid code version or comparator applies, and where is it written?
- Is the 0.95 PF met by inverters alone at 50 °C or with a STATCOM?
- What frequency response must the BESS give, with what droop and deadband?

Remember

- Kuwait's grid code text was not found: ask MEWRE Transmission; use comparators meanwhile.
- FRT keeps one fault from becoming a generation loss; reactive current $\Delta I_q = k \cdot \Delta V$.
- 0.95 PF = 0.329 P; 1,100 MW needs 361.5 Mvar at the POI and about 584 Mvar at the inverters.
- Q moves voltage because X >> R; droop $\Delta P = (P_n/R)(\Delta f/f_n)$.
- The PPC works at the POI in 100 to 200 ms loops; FRT is handled by each inverter.

Check yourself

1. How many Mvar must a 400 MW plant supply at 0.95 PF at the POI?
2. A 200 MW BESS, 5 % droop, ±0.2 Hz deadband, sees 49.7 Hz. How many MW does it inject?
3. Why does Q, not P, dominate voltage on a 400 kV line?
   A. Because P flows only on the DC side
   B. Because X is much larger than R
   C. Because voltage depends only on frequency
   D. Because Q is measured in Mvar
4. Which device handles fault ride-through?
   A. The control centre
   B. The PPC
   C. Each inverter, on its own
   D. The SCADA historian

C3 Grid strength and inverter behaviour

about 14 min

After this section you can

- calculate fault level and SCR, WSCR and ESCR, and classify a connection point;
- explain grid-following and grid-forming control and when each is needed;
- explain inverter current limiting and its consequences for protection, control and studies;
- check a switchgear rating against a fault calculation.

Fault level: the measure of grid strength

The **fault level** (short-circuit power) at a bus is the three-phase power that would flow into a bolted fault there. It equals the system voltage divided by the Thévenin impedance behind the bus, so it is set by the generators, lines and transformers electrically close to it. Many synchronous machines and short, parallel lines mean low impedance and a high fault level; a remote node at the end of a long line with few machines nearby has a low one.

$$S_{sc} = \sqrt{3} \cdot V \cdot I_{sc}$$

15,000 MVA at 400 kV corresponds to I_{sc} = 15,000 / (1.732 × 400) = 21.7 kA

Fault level matters in two directions. The maximum sets breaker and busbar ratings. The minimum (N-1 outage, minimum generation, typically winter night) sets voltage stiffness, which is what inverter controls

depend on. The **X/R ratio** of the source impedance sets the DC offset in fault current: transmission X/R of 10 to 20 or more means the offset decays with a time constant of X/(ωR), about 48 ms at X/R = 15, adding duty on breakers.

The SCR family

> $SCR = S_{sc,POI} / P_{plant}$ $\quad WSCR = S_{sc} / \Sigma P_{IBR}$ $\quad ESCR = (S_{sc} - Q_c) / P$
>
> S_{sc} in MVA without the plant; P in MW; Q_c = shunt capacitors and filters at the POI

The **short-circuit ratio (SCR)** compares grid strength with the plant that will push power into it. A low SCR means the plant's own current changes the POI voltage noticeably, so its control loop is acting on a voltage it is itself moving. **Weighted SCR (WSCR)** divides by the sum of all **inverter-based resources (IBRs)** on the same electrical area, because they interact. **Effective SCR (ESCR)** subtracts capacitors and filters, which resonate with grid inductance and make the node behave as weaker. **Composite SCR (CSCR)** reduces a whole area of IBRs to one equivalent.

SCR	Meaning (rule of thumb)
> 5	Strong: RMS studies usually sufficient
3 to 5	Moderate: EMT studies required (course rule: below about 5)
< 3	Weak: grid-following inverters risk control instability and voltage oscillation
< 2	Very weak: needs grid-forming inverters, synchronous condensers or reinforcement

▸ **Worked examples**

1,100 MW at a 15,000 MVA 400 kV bus: SCR = 13.6, strong. The same plant at 3,000 MVA: SCR = 2.7, weak. 500 MW at a 2,000 MVA bus: SCR = 4.0, moderate, EMT required. If a 300 MW plant already sits on that bus, WSCR = 2,000 / 800 = 2.5: weak. If the 1,100 MW plant at 3,000 MVA also has 150 Mvar of filters, ESCR = 2,850 / 1,100 = 2.6.

Grid-following and grid-forming

A **grid-following (GFL)** inverter is a controlled current source. A **phase-locked loop (PLL)** measures the angle and frequency of the POI voltage, and the current controller injects current at a commanded angle relative to it. This works when the voltage is stiff. On a weak grid the injected current shifts the voltage angle, the PLL tracks the shift, the current reference moves again, and the loop can oscillate at a few hertz to tens of hertz or lose lock.

A **grid-forming (GFM)** inverter is a controlled voltage source behind an impedance. It sets its own voltage magnitude and angle and lets current flow according to the network, as a synchronous machine does. It provides **synthetic inertia** (power released in proportion to RoCoF within milliseconds), holds up weak grids and can **black start** a dead network. It needs energy headroom to deliver these on demand, which is why it is specified mainly for BESS. The course procurement rule: require GFM-capable BESS power conversion systems even if operated grid-following at first.

Inverter current limiting (clamping)

An inverter's IGBTs (insulated-gate bipolar transistors) have a junction temperature limit reached within microseconds to milliseconds of overcurrent; they have almost no thermal mass. The control therefore **clamps** output current at about 1.0 to 1.2 pu of rating, in every operating state including a short circuit. A

synchronous generator's fault current is limited only by its subtransient reactance (around 0.15 to 0.25 pu), giving 4 to 7 pu for the first cycles (Figure C3.1).

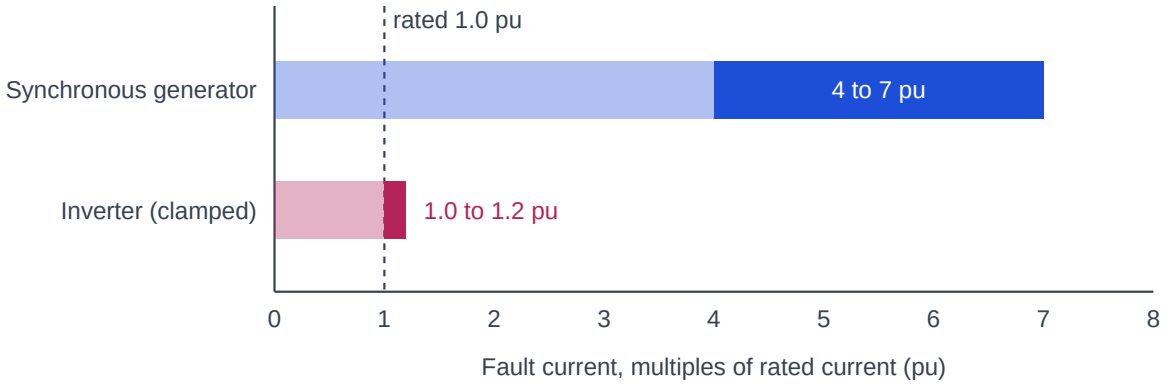


Figure C3.1 Typical fault current in the first cycles. Notice the inverter gives roughly one fifth of what a generator of the same rating gives; protection designed around generator fault current loses margin as IBRs displace machines.

The consequences are what you must recognise in studies and design reviews:

- **Protection sensitivity and directionality.** Overcurrent and distance relays detect faults by large current and its angle. With little and controlled fault current from the plant side, relays may not pick up or may misjudge direction. Hence line differential (87L) protection over fibre on export lines, and protection studies using vendor fault-current models.
- **Q-priority or P-priority.** With total current fixed at about 1.1 pu, the inverter must divide it. Q-priority (normal grid-code requirement) serves reactive current first: a dip to 0.6 pu with k = 2 demands 0.8 pu reactive current, leaving $\sqrt{(1.1^2 - 0.8^2)} = 0.76$ pu active current, so active power falls to about 0.45 pu.
- **Controller saturation and anti-windup.** While clamped, outer loops keep integrating an error the inverter cannot correct. Without **anti-windup** logic (stopping the integrators at the limit) the stored error releases at fault clearance as overshoot and oscillation.
- **Negative sequence.** In unbalanced faults the inverter's negative-sequence current depends entirely on vendor control (suppressed, or injected as IEEE 2800 requires). Relays that use negative sequence for direction depend on it.
- **GFM loss of synchronism.** At the current limit a grid-forming inverter stops acting as a voltage source and behaves as a current source; if its angle control is not designed for this, it can slip out of step. The vendor's current-limit design is a key study item.

Related terms: the **DC-link chopper** is a braking resistor that dissipates power the inverter cannot export during a dip, keeping DC voltage safe (a PV inverter can instead move off its maximum power point). **Clipping** is not a fault behaviour: output flat-topped at the AC rating when DC/AC ratio exceeds 1. **Momentary cessation** is an inverter stopping current injection during a dip; after the Blue Cut (2016) and Odessa (2021) events in the US, codes restrict it.

▸ **Worked example: can the 33 kV switchgear survive?**

A 400/33 kV, 300 MVA, 14 % transformer fed from a 15,000 MVA 400 kV bus. On a 300 MVA base, z_s = 300 / 15,000 = 0.02 pu; total 0.16 pu. S_{sc} = 300 / 0.16 = 1,875 MVA; I_{sc} = 1,875 / (1.732 × 33) = 32.8 kA, above 31.5 kA switchgear: under-rated. Two such transformers in parallel on one bus: z = 0.02 + 0.07 = 0.09 pu, 3,333 MVA, 58.3 kA. Keep the 33 kV bus coupler normally open, buy 40 kA gear, or raise transformer impedance.

In Kuwait

- Fault levels and SCR at Shagaya's 400 kV connection are not in the sources; the examples are illustrative. Ask MEWRE Transmission for minimum and maximum fault levels V
- BESS programme reported at 1 to 1.5 GW / 4 to 6 GWh, first 500 MW for summer 2027: a natural place to specify grid-forming capability L

Questions to ask

- What is the minimum fault level at the POI (N-1, minimum generation) and the resulting SCR and WSCR?
- Down to what SCR does the vendor guarantee stable operation?
- What is the inverter's maximum fault current, for how long, with Q-priority? Is that behaviour in the EMT model the real firmware?

Remember

- SCR = S_{SC}/P: above 5 strong, 3 to 5 moderate (EMT), below 3 weak, below 2 very weak.
- GFL follows the voltage with a PLL; GFM sets the voltage and can supply inertia and black start.
- Inverters clamp at 1.0 to 1.2 pu; generators give 4 to 7 pu. Protection, priority, anti-windup, negative sequence and GFM stability all follow.
- Always compare fault current against switchgear kA; parallel transformers nearly double it.

Check yourself

1. A 600 MW plant connects at a 2,400 MVA bus where 200 MW of PV already operates. What are its SCR and the WSCR?
2. Why do inverters clamp current at about 1.1 pu?
   A. To meet harmonic limits
   B. Because IGBTs overheat within microseconds to milliseconds
   C. To save transformer losses
   D. Because the PLL requires it
3. What goes wrong at fault clearance if a clamped inverter has no anti-windup?
4. Fault level at a 400 kV bus is 10,000 MVA. What is the three-phase fault current (kA)?

C4 Power system studies and how a PM manages them

about 15 min

After this section you can

- state what question each study answers and which cases it must cover;
- explain the difference between RMS and EMT simulation and when EMT is required;
- run the study process with MEWRE Transmission, the vendor and the owner's engineer;
- write model requirements into contracts and manage energisation hold points.

What each study answers

Load flow solves steady-state voltages and branch flows. It asks: do lines and transformers stay within thermal ratings, and voltages within limits, with all elements in service (N) and with any one out (N-1)? Kuwaiti cases must include summer peak, winter midday (low load, strong sun: overvoltage and reverse flow) and evening BESS discharge. Example: one double-circuit 400 kV line, 1,732 MVA per circuit. Exporting 1,600 MW at 0.95 PF is 1,684 MVA: 49 % per circuit under N, 97 % on the surviving circuit under N-1, which passes with 3 % margin. Adding 1,700 MW fails N-1 and needs a new line or a runback scheme.

Short circuit (IEC 60909) calculates maximum fault currents (voltage factor c = 1.1 at HV) for breaker, busbar and earthing ratings, and minimum fault levels for SCR and protection sensitivity. **Harmonic** studies combine frequency scans of network impedance with inverter emissions and background distortion to check THD and resonance. **Protection coordination** sets relays so the nearest breaker trips first, checked with low IBR fault current. **Insulation coordination** matches BIL and switching withstand to lightning, switching and temporary overvoltages and places surge arresters.

RMS (phasor) stability simulation represents voltages and currents as positive-sequence 50 Hz phasors, steps in 1 to 10 ms, and covers the whole national system for seconds to minutes. It answers: does the system stay in synchronism after a fault, does voltage recover, where does frequency go after a trip, do plants ride through at system level? Tools: PSS/E, PowerFactory RMS, PSLF.

EMT (electromagnetic transient) simulation solves instantaneous three-phase voltages and currents from the circuit differential equations, with 1 to 50 µs steps (400 steps per 50 Hz cycle at 50 µs), and includes the inverter's real control code, switching, saturation and the PLL. It is slow, so it models a local area for a few seconds. Tools: PSCAD/EMTDC, EMTP, PowerFactory EMT, RTDS for real-time hardware-in-the-loop. EMT is required when:

- SCR is below about 5: GFL control and PLL stability cannot be judged with phasors.
- IBRs, STATCOMs or HVDC are electrically close and their fast controls may interact (including sub-synchronous control interaction).
- FRT must be verified with real current limiting and unbalanced faults.
- Grid-forming behaviour and current-limit transitions must be proven.
- Energisation: transformer inrush and sympathetic inrush, cable and capacitor switching.
- Temporary overvoltage after load rejection or fault clearing; ferroresonance; harmonic resonance; protection performance with IBR fault current.

	RMS (phasor)	EMT
Quantities	50 Hz magnitude and angle, positive sequence	Instantaneous waveforms, all three phases
Time step	1 to 10 ms	1 to 50 µs
Extent	Whole system, seconds to minutes	Local area, typically seconds
Inverter model	Generic or vendor averaged model	Vendor real control code (often black-box)
Misses	PLL instability, unbalance, switching transients, harmonics	Long-term and system-wide dynamics (too slow to run)
Tools	PSS/E, PowerFactory, PSLF	PSCAD, EMTP, PowerFactory EMT, RTDS

Managing the studies in practice

Studies sit on the critical path because approval, settings freeze and equipment orders depend on them, and because the model must be the equipment actually supplied. Figure C4.1 shows the sequence.

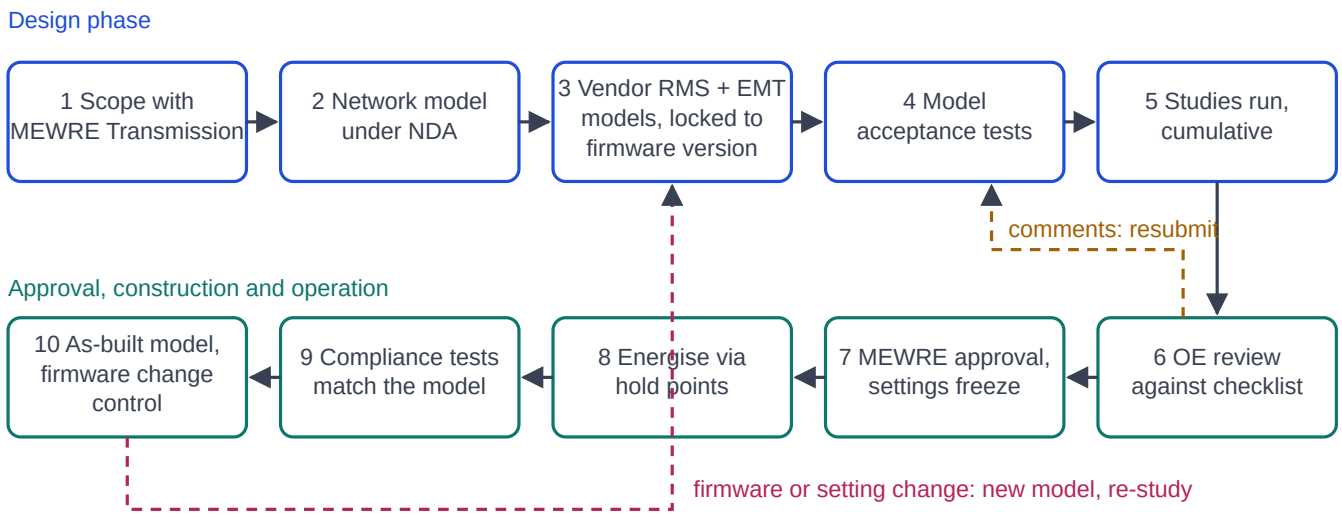


Figure C4.1 Study management workflow. Notice the two loops: review comments send the study back, and any firmware change after approval sends the plant back to new models and re-study.

1. **Scope** agreed in writing with MEWRE Transmission: study list, cases (summer peak, winter midday, evening BESS, N-1), minimum and maximum fault levels, future plants included, criteria, deliverables, and who runs which study.
2. **Network model:** MEWRE issues its PSS/E or PowerFactory model under an NDA; agree the base year.
3. **Plant models:** vendor RMS and EMT models of inverter, PPC, BESS PCS and STATCOM, locked to the firmware version ordered, with parameter sheets and a validation report.
4. **Model acceptance tests:** the model initialises cleanly, holds a flat run without drift, gives correct step responses and FRT, and RMS and EMT results agree on a benchmark case.
5. **Studies run** by the consultant or EPC, cumulative with neighbouring plants.
6. **OE review** against a checklist; comments and resubmission.
7. **Approval**, then settings and design frozen and equipment ordered.
8. **Compliance tests** on site reproduce the study results: the model must match the field.
9. **As-built model** handed over; any later firmware or setting change triggers notification, updated models and possibly re-study.

The OE review checklist: correct network base case and year; future plants included; both minimum and maximum fault levels used; worst cases (winter midday, N-1, low inertia) present; plant model equals the inverter, PPC and firmware ordered; EMT used where SCR is low or other IBRs are near; reactive capability at 50 °C and 0.90 to 1.10 pu; FRT, reactive current injection and recovery shown against the code; harmonics at the POI with background distortion; protection consistent with low IBR fault current; assumptions and sensitivities stated.

Model requirements to write into contracts

- Site-specific RMS and EMT models of inverter, PPC, BESS and STATCOM, delivered by a dated milestone and updated as design changes.
- EMT model built from the real control code, not a generic approximation; black-box (compiled) form acceptable under NDA.

- Validation against type tests, factory tests or hardware-in-the-loop, then against site tests.
- Documentation: parameters, valid range (SCR, X/R), required time step.
- Firmware change control: no change without notice, updated models and MEWRE agreement.

Energisation hold points and compliance testing

A **hold point** is a step that cannot proceed until signed off. In order: protection settings approved by MEWRE and tested; SCADA point-to-point checks; telecoms and teleprotection working; revenue metering sealed; switching programme approved; HV back-fed from the grid and soaked (often 24 h); MV feeders energised progressively; inverters and PPC tested at low power; witnessed compliance tests; trial (reliability) run; COD.

Grid compliance testing proves each requirement on site, witnessed by MEWRE: reactive capability across the P range, voltage control step response, P set-point and ramp, frequency response by injecting a simulated frequency signal into the PPC, and power quality measurement. FRT is usually proven by inverter type test certificates plus the validated model, because a real 400 kV fault cannot be staged. Each result is compared with the model prediction; a mismatch means the model or the settings are wrong.

In Kuwait

- The division of study work between developer and MEWRE Transmission is not publicly documented: agree it in writing at project start V
- Kuwait's grid code and compliance test procedure were not found in the sources; until obtained, use comparators and label them as such V
- Zone 1's 400 kV substation is IPP scope while the Shagaya to Al-Wafra line is a separate MEWRE award, so remote-end protection and OPGW sit on both critical paths M

What the PM does

- Puts study milestones, model delivery and MEWRE review periods on the programme as critical activities.
- Refuses settings freeze or equipment orders before study approval.
- Holds a firmware register and enforces change control through operation.

Questions to ask

- Has the EMT study used the vendor's real control code for the firmware version being shipped?
- Do the studies include winter midday, minimum fault level and the neighbouring plants?
- Have MEWRE approved the protection settings, or only received them?

Remember

- Load flow: overloads and voltages, N and N-1. Short circuit: ratings and minimum strength.
- RMS: phasors, ms steps, whole system. EMT: waveforms, µs steps, real control code, local area.
- EMT when SCR < ~5, IBRs near each other, real FRT and current limits, GFM, switching and resonance.
- The model must be the firmware shipped, and site tests must match the model.

Check yourself

1. Which study finds PLL instability at a weak bus?
   A. Load flow
   B. IEC 60909 short circuit
   C. RMS stability
   D. EMT
2. A double-circuit line has 1,732 MVA per circuit. What is N-1 loading when exporting 1,500 MW at 0.95 PF (%)?
3. The vendor updates inverter firmware six months after COD. What must happen?
4. Name the first hold point before back-feeding the HV substation.

PART D · TRIAL DAY 3 · ABOUT 31 MIN

Delivering and controlling a project

Walk a project from feasibility to handover and control it with a schedule, earned value and a risk register.

D1 From feasibility to handover

about 14 min

After this section you can

- name the gates of a PV or BESS project, the four gate decisions and the estimate accuracy at each;
- explain the design stages, the review codes and who reviews design for MEWRE;
- follow the testing chain to COD and calculate a temperature-corrected PR;
- state what the Taking-Over Certificate switches over and what the handover package contains.

Stages, gates and funding

A **stage** is a period of work that produces defined deliverables; a **gate** is the decision point where an accountable committee reviews them against criteria written beforehand. A project is cheapest to change at concept; development (about 1 to 3 % of capex) buys information before the large spend.

A gate has four outcomes: **go** (release budget for the next stage), **go with conditions** (proceed, with named gaps closed by a date, typically a pending geotechnical report), **recycle** (redo or extend the stage) and **stop**. Money is released one stage at a time, only on go or go with conditions.

Gate	Question	Stage before it
G0 Need	Mandated need?	1 to 3 months
G1 Pre-feasibility	Which options survive?	2 to 4 months
G2 Feasibility	Which option, site, route, cost?	6 to 12 months
G3 Investment	Commit to tender?	3 to 6 months
G4 Award / financial close	Sign?	6 to 18 months
G5 Notice to proceed	Safe to build?	1 to 3 months
G6 COD / taking over	Meets the contract?	1 to 3 months
G7 Close-out	Finished, lessons recorded?	End of defects period

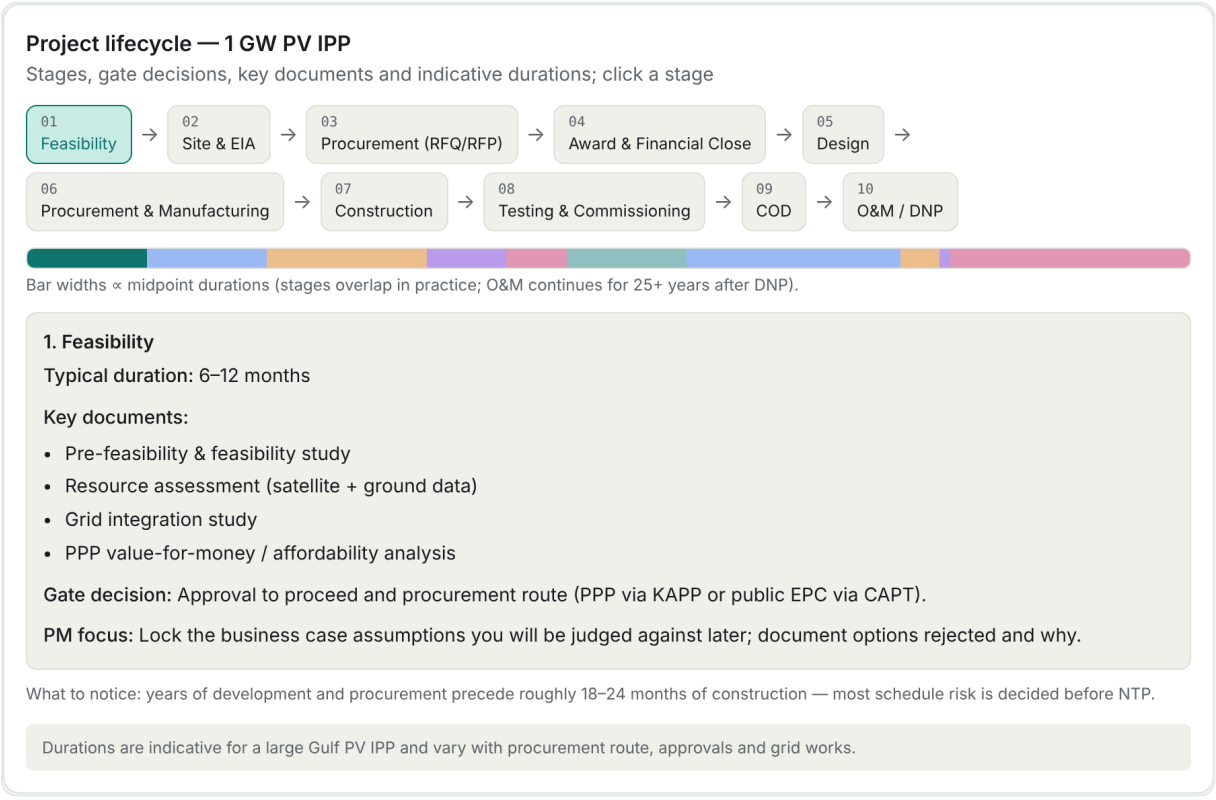


Figure D1.1 Lifecycle of a 1 GW PV IPP. The coloured bar is proportional to stage duration: feasibility, site and EIA, procurement and financial close together take as long as construction, so most schedule risk is fixed before notice to proceed.

Every estimate carries an **AACE estimate class**, a grade tying its accuracy range to design maturity.

Class	Purpose (gate)	Accuracy, approx.
5	Screening (G1)	−50 % / +100 %
4	Feasibility (G2)	−30 % / +50 %
3	Budget authorisation (G3)	−20 % / +30 %
2	Control or bid check (G4)	−15 % / +20 %
1	Check estimate	−10 % / +15 %

▸ **Worked example: a Class 5 number**

Pre-feasibility gives USD 600 m (Class 5): plausible range 600 × 0.5 = 300 to 600 × 2 = 1,200 m. At Class 3 it would be 480 to 780 m.

Site, permits and EIA

Before design freezes, the site needs land allocation for plant and corridors, **no-objection letters** (written consents from bodies whose assets or airspace are affected), topographic and geotechnical surveys, flood hydrology, an unexploded-ordnance check and the **environmental impact assessment (EIA)**, the study the environmental regulator must accept before construction.

Design stages and design review

Concept fixes capacity, technology and connection voltage. **Basic design** or **FEED** (front-end engineering design) freezes the **design basis**: site conditions, standards, equipment ratings, block size, protection

philosophy. **Detailed design** by the EPC contractor produces drawings, cable schedules and settings, each released **IFC** (issued for construction) only after review. After the freeze a change is a **variation** (a paid change of contract scope), costing more the more documents rest on it.

Submittals return with a code: **A** no objection, proceed; **B** proceed, incorporate comments, resubmit; **C** rejected, do not proceed; **D** information only. Reviewers state non-compliance with the Employer's Requirements rather than dictating a fix, which could become a variation; a Code A leaves design responsibility with the contractor. Under FIDIC 2017 Yellow the review period defaults to 21 days, and a missed deadline risks a **deemed no-objection**.

	Owner's engineer (OE)	The Engineer (FIDIC Red/Yellow)	Lenders' technical adviser
Works for	MEWRE; not neutral	Named by the Employer; neutral when determining	The banks; not neutral
Job	Reviews, witnesses tests, recalculates data, recommends	Instructions, certificates, determinations	Due diligence, drawdown and completion sign-off

The OE advises. The PM signs the review code and owns the review deadlines, priorities, MEWRE's interfaces (grid, control centre, approvals), budget and escalation.

Testing and commissioning to COD

Each test proves what the next relies on:

1. **FAT** (factory acceptance test): equipment meets specification before shipment.
2. **SAT** (site acceptance test): no transport damage, works as installed; then mechanical completion.
3. **Cold commissioning**, de-energised: insulation resistance, polarity, string V_{oc} and I_{sc}, earthing, relay secondary injection, SCADA points.
4. **Energisation** under an approved switching programme and permit to work; the site is now live.
5. **Hot commissioning**: systems running energised.
6. **Grid compliance tests** at the point of interconnection, witnessed by MEWRE: setpoints, ramps, frequency response, reactive power and voltage control.
7. **Performance tests**: PR (7 to 15 days), capacity (pass at about 97 % of guarantee), battery capacity and efficiency.
8. **Reliability run** (for example 7 to 30 days without contractor-caused trips), then **COD**.

$$PR = (E / P_0) / (H / G_{STC})$$

E energy delivered (kWh); P_0 rated DC power (kWp); H plane-of-array irradiation (kWh/m²); G_{STC} = 1 kW/m²

The **performance ratio (PR)** is delivered energy over what the DC rating would give with no losses. Module power falls about 0.3 to 0.4 % per °C of cell temperature, so the test corrects each interval's expected energy to a contractual **reference temperature** T_{ref}; otherwise the result depends on the season.

$$PR_{corr} = E / \Sigma\, P_0 \cdot (G/1000) \cdot [1 + \gamma(T_c - T_{ref})] \cdot \Delta t$$

G irradiance (W/m²); γ power temperature coefficient (negative, per °C); T_c cell temperature

▸ **Worked example: temperature-corrected PR**

100 MWp, γ = −0.30 %/°C, four hourly intervals (G W/m², T_c °C, E MWh): (600, 45, 50.0), (900, 60, 70.5), (1,000, 65, 77.0), (700, 55, 56.0). Measured 253.5 MWh; uncorrected expected 100 × 3.2 = 320 MWh; raw PR 0.792.

To 25 °C: 60 × 0.94 = 56.40; 90 × 0.895 = 80.55; 100 × 0.88 = 88.00; 70 × 0.91 = 63.70; sum 288.65; PR = 253.5 / 288.65 = **0.878**. To 45 °C: 60.00 + 85.95 + 94.00 + 67.90 = 307.85; PR = **0.823**. Same plant, same data: always read a PR guarantee with its T_{ref} (typically 0.85 to 0.88 at 25 °C, 0.80 to 0.83 at the annual average cell temperature, about 42 to 47 °C in Kuwait).

Taking over, defects period and handover

Punch list items still open at completion are graded. **Category A** (safety, reliability, performance, compliance, for example a faulty protection relay) must close before COD; **Category B** (a missing label, an as-built update) closes during the defects period against a retained sum. COD also needs all tests passed and MEWRE's grid approval.

The **Taking-Over Certificate (TOC)** (FIDIC 2017 Clause 10.1) passes care and risk to the owner, stops delay damages, starts the **defects notification period (DNP)** and releases the first half of retention, less outstanding work. The DNP is commonly 12 to 24 months (FIDIC default 365 days), extendable for a defect that prevents use by up to 2 years (Clause 11.3). The **Performance Certificate**, issued within 28 days after the last DNP ends (Clause 11.9), is the only acceptance of the works and frees the second half of retention. Retention and the performance security are still at risk during the DNP, so inspect the whole plant in its first 3 months and notify defects before expiry.

▸ **Worked example: retention at taking over**

USD 600 m contract, retention 5 %: 30 m held. First half 15 m, less 1.2 m of Category B work: **USD 13.8 m** released at the TOC.

The **handover package**: as-builts (native CAD and PDF), O&M manuals loaded into the **CMMS** (computerised maintenance management system), test baselines (IV curves, thermography, relay settings as left), assessed training, spares, SCADA and controller configuration with credentials, registered warranties and an asset register feeding the CMMS.

On a KAPP IPP, **financial close** (financing signed, conditions met) lets the project company commit capital and issue notice to proceed; **COD** under the PPA starts MEWRE's energy payments, with MEWRE's own grid works on that critical path.

In Kuwait

- IPP sequence: entity request, KAPP feasibility, Higher Committee, RFQ, RFP, preferred bidder, Higher Committee, Audit Bureau, PPA signing, financial close. M
- Shagaya Phase III zones: KAPP procurement, 30-year PPAs with MEWRE. M
- EIA under Environment Protection Law No. 42 of 2014 with the Environment Public Authority, as commonly cited; steps and durations are unverified, so confirm with the EPA. V
- No-objection letters from oil and gas, defence, civil aviation, telecoms and utilities; 1990 to 91 unexploded ordnance in parts of the desert. L
- Record 53.9 °C (Mitribah, 2016, WMO): check equipment ratings at 50 °C and above. H

What the PM does

- States the estimate class with every cost and writes gate criteria in advance.
- Approves test procedures (filters, T_{ref}, soiling, curtailment) before testing.
- Refuses COD while a Category A item or grid approval is open.

Questions to ask

- Which submittals are near their review deadline?
- Which T_{ref} does the PR guarantee use?
- When does each section's DNP end?

Remember

- **Go, go with conditions, recycle, stop**; funding stage by stage.
- Class 5 −50/+100 %, 4 −30/+50 %, 3 −20/+30 %.
- FEED freezes the design basis; codes A to D; 21 days or deemed no-objection.
- TOC: risk to owner, delay damages stop, DNP starts, half retention.

Check yourself

1. At a feasibility gate everything is complete except a geotechnical report due in 6 weeks, with no fatal flaw in the desk study. Which decision fits?
   A. Go
   B. Go with conditions
   C. Recycle
   D. Stop
2. A 50 MWp plant receives 65 kWh/m² in the plane of array during a 10-day test and delivers 2,665 MWh. What is the PR?
3. Which of these does **not** happen at the Taking-Over Certificate?
   A. Care and risk pass to the owner
   B. Delay damages stop
   C. The Performance Certificate is issued
   D. The DNP starts
4. For one 1-hour interval of a 100 MWp test, G = 800 W/m², T_c = 55 °C, γ = −0.30 %/°C. What is the expected energy corrected to T_{ref} = 25 °C, in MWh?

D2 Planning, control and reporting

about 16 min

After this section you can

- build and check a WBS for a large PV IPP and spot scope creep;
- run a forward and backward pass and turn a slip into a completion delay;
- calculate and interpret every earned value measure;
- quantify risk (EMV, PERT, Monte Carlo), run change control and write an honest monthly report.

Scope and the WBS

Scope is the plant and its performance plus the work to deliver it; under design-build it is written in the **Employer's Requirements**, and the **battery limit** states where each party's scope ends (the 400 kV gantry, the metering point). A **work breakdown structure (WBS)** splits scope into coded deliverables (nouns, not activities) down to **work packages** of about 1 to 4 % of value or one physical block. The **100 % rule**: children sum exactly to their parent in scope and money, with no overlaps.

Code	WBS element, 1 GW PV + battery EPC
1.1	Engineering
1.2	PV field: blocks 1.2.01 to 1.2.40 of about 25 MWac (40 × 25 = 1,000 MWac)
1.3 / 1.4	Battery / MV collection (33 kV cables, ring main units)
1.5	400 kV substation
1.6	Plant control and SCADA
1.7	Infrastructure and civil
1.8 / 1.9	Testing and commissioning / project management and documents

Each package has a **WBS dictionary** entry: scope, boundary, exclusions, acceptance criteria, owner, interfaces, budget. MEWRE keeps its own WBS (400 kV line, remote-end bay, SCADA integration, metering, land, test witnessing); at the point of interconnection most **interfaces** are MEWRE "gives" (line ready at the gantry, points list, outage window), tracked in an **interface register** with give and need dates and one owner. **Scope creep** is growth through informal requests; it bypasses change control.

▸ **Worked example: scope creep**

Nine informal extras at KWD 48 k, a spare 33 kV feeder bay (0.35 m) and a weather station (0.22 m): 9 × 0.048 + 0.35 + 0.22 = **KWD 1.002 m**, or 125 % of a KWD 0.80 m change allowance, none assessed for time.

Scheduling and the critical path

A schedule links **activities** (work with a duration) by **logic**, mostly finish-to-start, sometimes with a **lag** (waiting time). The **forward pass** gives earliest start and finish (ES, EF): an activity starts when its last predecessor allows, so take the maximum. The **backward pass** from the end date gives latest dates (LS, LF): it must finish before its earliest-needed successor, so take the minimum.

$ES = \max(EF_{pred})$, $EF = ES + D$; $LF = \min(LS_{succ})$, $LS = LF - D$; $TF = LS - ES$; $FF = \min(ES_{succ}) - EF$

TF total float (slip without moving completion); FF free float (slip without delaying a successor)

▸ **Worked example: the two passes (weeks)**

A design 10; B grading 6; C tracker procurement 20 after A; E piling 12 after A and B; F tracker installation 14 after C and E. Forward: C 10 to 30; E max(10, 6) = 10 to 22; F max(30, 22) = 30 to 44. Backward from 44: LS of C 10, E 18, B 12, A 0. E has TF 8; B has TF 12 but FF 10 − 6 = 4. A, C, F have zero float: the **critical path**, the longest chain, which sets completion.

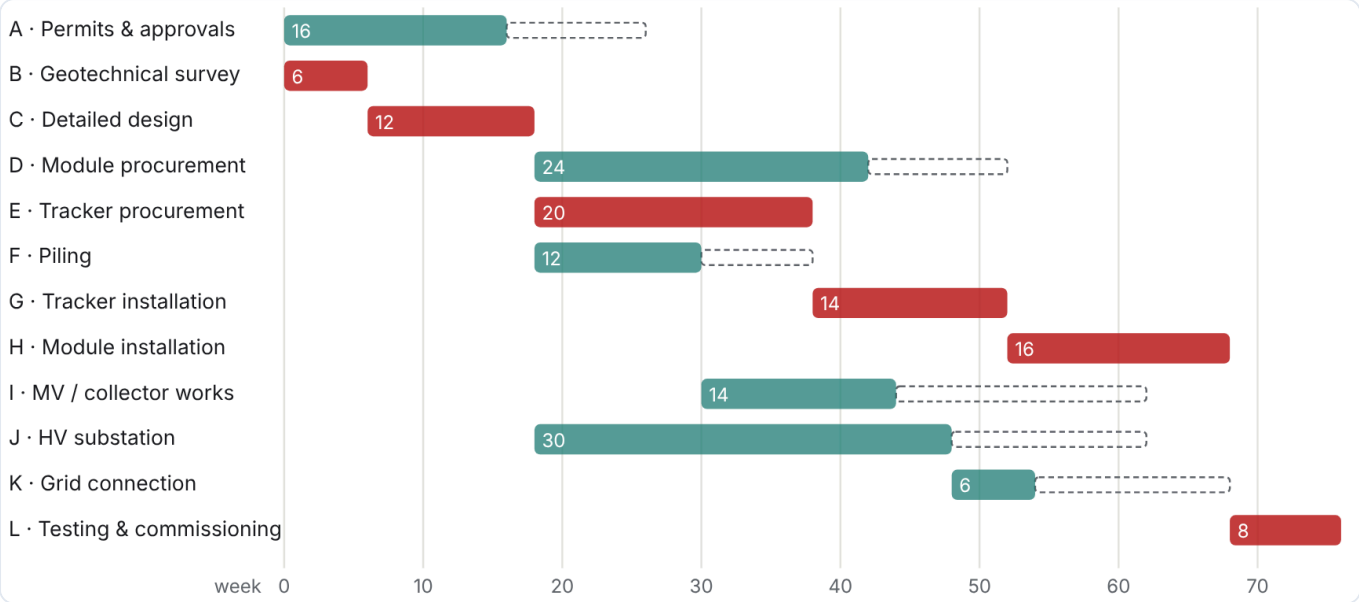


Figure D2.1 PV project network. Red bars are critical: B, C, E, G, H, L = 6 + 12 + 20 + 14 + 16 + 8 = 76 weeks. Dashed boxes are total float; the HV substation (J) and grid connection (K) each have 14 weeks.

$$\text{Completion delay} = \max(0, \text{slip} - TF)$$

Float belongs to the path: used upstream, it is gone downstream. In Figure D2.1 a 15-week slip of the HV substation (TF 14) delays completion by 15 − 14 = **1 week**, and the critical path moves to the substation and grid connection.

The approved schedule is frozen as the **baseline**; monthly updates are stated at the **data date** (actual before it, forecast after). Primavera **P6** is the standard tool; ask for the native XER file. Under FIDIC 2017 Sub-Clause 8.3 the programme is due within 28 days of the Commencement Date notice and the Engineer has 21 days to object: use them to insert MEWRE's approvals, grid works and outages. **Merge bias**: two independent paths each 50 % likely by week 40 give 0.5 × 0.5 = 25 % that both are.

Earned value

Against the **budget at completion (BAC)**, **planned value (PV)** is the budget of work scheduled to date, **earned value (EV)** the budget of work done and **actual cost (AC)** its cost. Judge cost by EV against AC and schedule by EV against PV (Figure D2.2); low spend against PV may only mean little work.

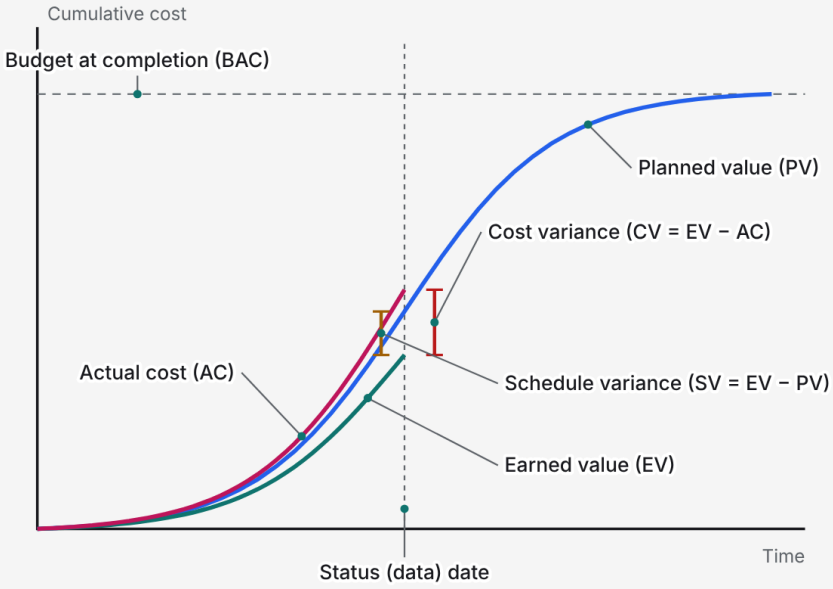


Figure D2.2 PV, EV and AC at the status date. EV lies below both: behind schedule (SV negative) and over cost (CV negative).

Measure	Formula	BAC 100, PV 40, EV 36, AC 45 (KWD m)
CV / SV	EV − AC / EV − PV	−9 / −4
CPI / SPI	EV / AC / EV / PV	0.80 / 0.90
EAC	BAC / CPI, or AC + (BAC − EV)	125, or 109
ETC / VAC	EAC − AC / BAC − EAC	80 or 64 / −25 or −9
TCPI	(BAC − EV) / (BAC − AC)	64 / 55 = 1.16

Each dinar buys 0.80 dinar of work; the project is 10 % behind. EAC = BAC/CPI assumes performance continues, AC + (BAC − EV) that the rest goes to budget: report 109 to 125. TCPI 1.16 against CPI 0.80 is not credible (the test: more than about 0.05 to 0.10 above CPI).

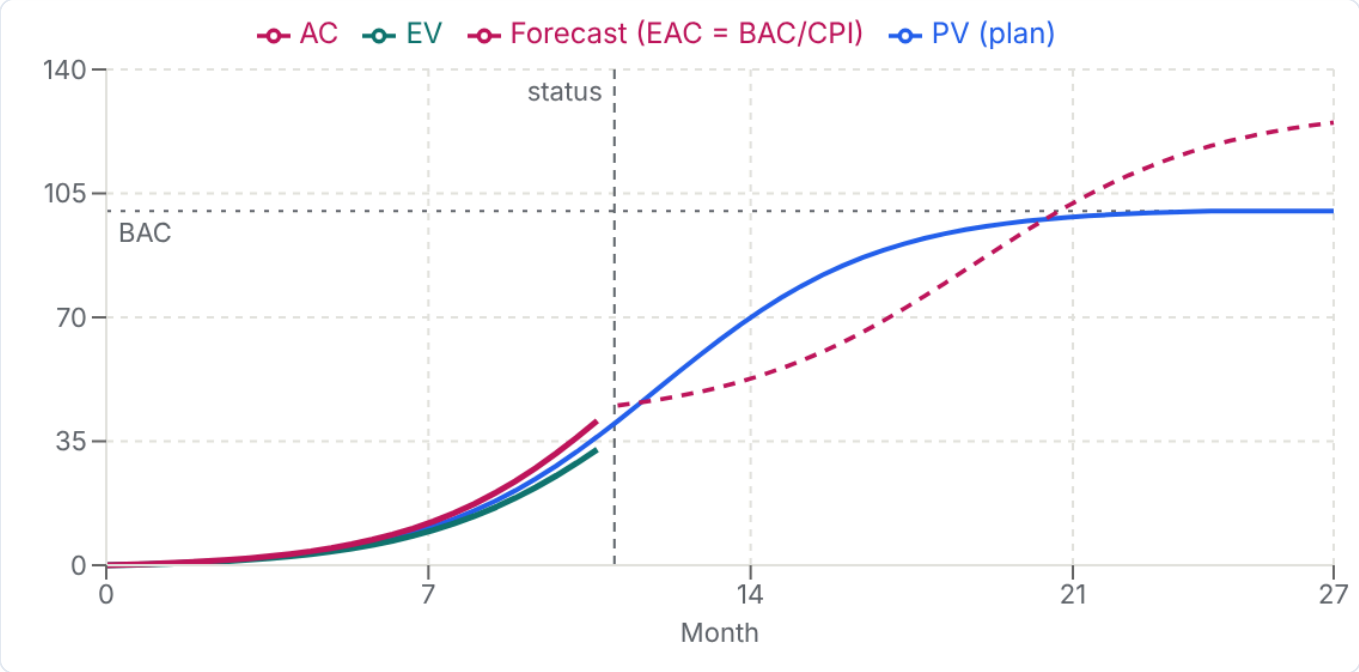


Figure D2.3 Earned value with a forecast. Notice the dashed forecast (EAC = BAC / CPI) ending above the budget line: with a CPI below 1, the project is heading for an overrun unless performance improves.

EV is only as honest as the **rules of credit**: count units installed and inspected, weight supply milestones (for modules: approval 5 %, order 5 %, FAT 15 %, shipped 15 %, delivered and inspected 60 %). Credit at purchase order or for untested work inflates progress.

Risk

The **risk register** states each risk as cause, event and effect, with one owner, a response (avoid, transfer, mitigate, accept, escalate) and a score. Probability and impact are rated 1 to 5 and multiplied: 15 to 25 red, 8 to 12 amber, 1 to 6 green; an HSE impact of 5 is red regardless. Scores rank risks and are never added. Money uses **expected monetary value**: EMV = $\Sigma\, p_i \cdot C_i$. A 40 % chance of a late 400 kV line costing USD 6.0 m gives 0.40 × 6.0 = **USD 2.4 m**, so mitigation costing well under that is good value.

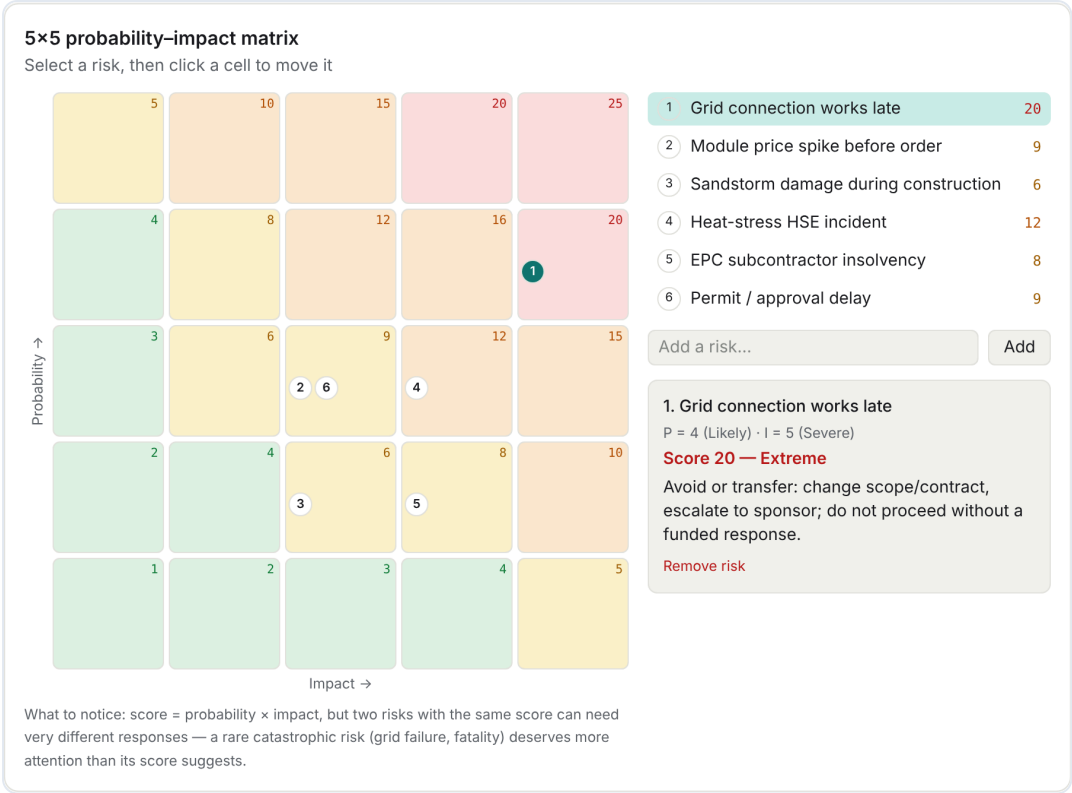


Figure D2.4 5 × 5 matrix. "Grid connection works late" at P 4 × I 5 = 20 is red, and it is a MEWRE-side risk.

$t_e = (a + 4m + b) / 6$; $\sigma = (b - a) / 6$; $P80 = \Sigma t_e + 0.842 \cdot \sqrt{(\Sigma \sigma^2)}$

a optimistic, m most likely, b pessimistic; along a path means and variances add; 0.842 is the normal z-value for 80 %

▸ **Worked example: P80 of a chain (weeks)**

Tracker supply (16, 20, 30), tracker installation (12, 14, 20), module installation (10, 12, 18). Means 21.0 + 14.67 + 12.67 = 48.33. Variances $2.33^2 + 1.33^2 + 1.33^2 = 9.0$, so $\sigma = 3.0$. P80 = 48.33 + 0.842 × 3.0 = **50.86**. The most-likely plan, 46 weeks, has z = −0.78, about a 22 % chance; contingency 50.86 − 46 = 4.9 weeks.

Monte Carlo analysis samples every duration and risk event 1,000 to 10,000 times through the full network, capturing merge bias that PERT misses. Owners hold **contingency** = P80 − base inside the baseline; **management reserve** sits outside it.

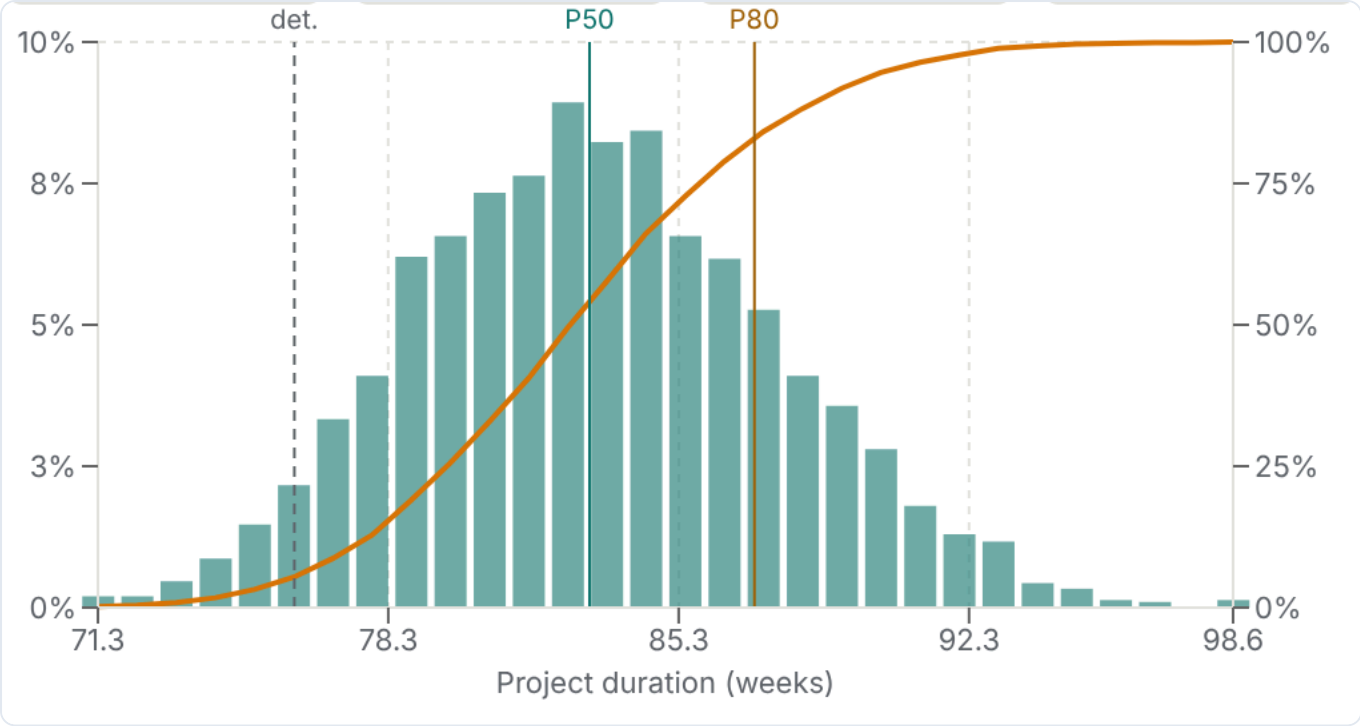


Figure D2.5 Schedule Monte Carlo of a PV programme (one simulation run). Notice that the deterministic date sits far to the left of the distribution: the single-point plan is unlikely to be met, and the P50 and P80 dates lie weeks later.

Change control, the monthly report, quality and safety

Every change goes: identify, log, assess (cost, time, risk, quality, HSE), decide at the delegated authority, instruct formally, implement and re-baseline, close. FIDIC 2017 Clause 13 covers variations; Clause 20 claims need notice within 28 days and full detail within 84. An RFI clarifies, never changes scope.

The monthly report (2 to 4 pages) opens with decisions needed, then key dates against baseline, critical path, cost and EAC range, HSE and quality, top 5 risks with owners, MEWRE interfaces. Each area gets a **RAG** (red, amber, green) status against criteria set in advance, plus trend and path to green; red asks for a decision.

Area	Green	Amber	Red
Schedule	On time	1 to 4 weeks late	Over 4 weeks or key date at risk
Cost	CPI ≥ 0.97	CPI 0.92 to 0.97	CPI < 0.92 or EAC above budget plus contingency

The **inspection and test plan (ITP)** marks each check H (hold: work stops until released), W (witness), R (review) or S (surveillance). A **non-conformance report (NCR)** records work outside specification, resolved by rework, repair, use-as-is or rejection. HSE essentials: permit to work, heat-stress control, and PV strings are live in daylight.

In Kuwait

- MEWRE procures grid works separately: Shagaya to Wafra 400 kV line, reportedly awarded March 2026 for about KD 48.6 m. M
- Approvals (KAPP, Higher Committee, Audit Bureau) belong in P6 as ranged activities. M Audit Bureau prior review is reported from KD 100,000. V
- Summer midday outdoor work ban, reported as about 11:00 to 16:00, June to August: confirm before setting calendars. V

What the PM does

- Owns MEWRE's interface dates as P6 milestones and protects their float.
- Reports EAC as a range and funds contingency at P80.
- Routes every request, however senior, through change control.

Questions to ask

- How has the longest path moved over three updates?
- Which rules of credit apply, and when were they agreed?
- Has this change been assessed for time?

Remember

- WBS: deliverables, 100 % rule, dictionary with exclusions.
- **TF = LS − ES**; delay = max(0, slip − TF); two P50 paths merge at 25 %.
- CPI = EV/AC, SPI = EV/PV, EAC = BAC/CPI or AC + (BAC − EV).
- Contingency = P80 − base; FIDIC 8.3: 28 and 21 days.

Check yourself

1. A KWD 200 m contract has EV = 60 and AC = 75 (KWD m). What is EAC if current cost performance continues?
2. At month 6, PV = 50, EV = 40, AC = 38. Which is right?
   A. On schedule and under budget, because AC is below PV
   B. Behind schedule (SPI 0.80) but under cost for the work done (CPI about 1.05)
   C. Over budget and behind schedule
   D. Ahead of schedule (SPI 1.25)
3. MEWRE's 400 kV line has 16 weeks of total float and slips 20 weeks. By how many weeks does completion move?
4. An activity is estimated at a = 6, m = 8, b = 16 weeks. What are its PERT mean and σ?

PART E · TRIAL DAY 4 · ABOUT 40 MIN

Contracts, money, procurement and the ministry

Handle FIDIC contracts, IPP and PPA structures, project finance, Kuwaiti procurement and ministry writing.

E1 Construction contracts (FIDIC 2017)

about 13 min

After this section you can

- say who designs and who carries ground risk under the Red, Yellow and Silver Books, and why;
- run a claim through its time limits, compute an EOT and value a variation;
- choose the right security or LD remedy and compute capped delay damages.

The parties

FIDIC (the International Federation of Consulting Engineers) publishes standard contracts; the 2017 editions are current. Each has **General Conditions** and **Particular Conditions** (Part A contract data; Part B the Employer's amendments, which override). Read Part B before quoting any clause.

Figure E1.1 shows the contracts. The **Employer** is the owner who pays (MEWRE on its own substations; the SPV on an IPP's EPC). The **Contractor** builds and, on design-build books, designs. The **Engineer**, appointed and paid by the Employer, instructs, certifies payments and, under Sub-Clause 3.7, determines claims **neutrally**. The **DAAB** (Dispute Avoidance/Adjudication Board) is appointed jointly.

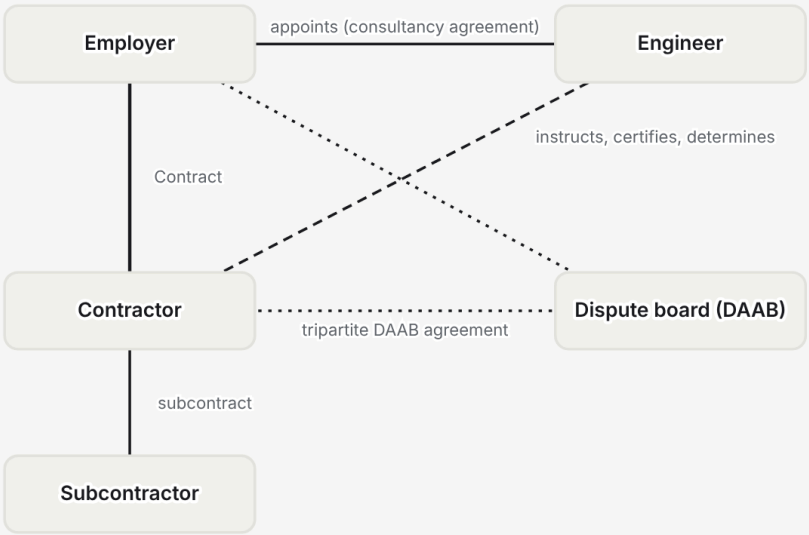


Figure E1.1 Solid lines are contracts. The Engineer has no contract with the Contractor but instructs, certifies and determines; the DAAB sits under a tripartite agreement with both Parties.

The books and who carries the risk

Red (Construction): the Employer designs and the Contractor is paid on a remeasured Bill of Quantities. **Yellow** (Plant and Design-Build): the Contractor designs to the Employer's Requirements for a lump sum. **Silver** (EPC/Turnkey): the Contractor designs, procures and builds for a fixed lump sum and carries **Unforeseeable Difficulties**, including ground (Sub-Clause 4.12); an **Employer's Representative** replaces the Engineer. **Gold** adds operation (design-build-operate) and **Emerald** covers underground works with a shared geotechnical baseline. Most IPP EPC contracts are Silver-based.

The allocation rule: give each risk to the party that **controls** it, otherwise to the party that can bear it most cheaply. A risk pushed onto a party that can neither control nor absorb it returns as a **risk premium** in the price.

> $RP = k \cdot EV,\ EV = p \cdot I$
>
> RP risk premium; EV expected value; p probability; I impact; k > 1 (about 1.5 for well-understood risks, 3 or more for risks the contractor cannot control)

A 20 % chance of a KWD 5 m loss has EV = KWD 1 m. Kept by MEWRE, it costs KWD 1 m on average; transferred, it adds KWD 1.5 to 3 m to the price. Figure E1.2 shows the default allocation by book.

Contract risk allocation matrix

Who bears each risk under FIDIC 2017 books and a typical IPP/PPA structure — click a cell

Risk	FIDIC Red 2017 Employer designs; Contractor builds (re-measured)	FIDIC Yellow 2017 Contractor designs & builds (lump sum)	FIDIC Silver 2017 EPC/Turnkey — single-point, higher price certainty	IPP / PPA Offtaker ↔ SPV (project company) ↔ EPC
Unforeseeable ground conditions	Employer Sub-Clause 4.12	Employer Sub-Clause 4.12	Contractor Sub-Clause 4.12	SPV → EPC
Design errors	Employer	Contractor Sub-Clauses 4.1 & 5.1	Contractor Sub-Clauses 4.1 & 5.1	SPV → EPC
Errors in Employer's Requirements / specification	Employer	Shared Sub-Clause 1.9	Contractor	SPV / EPC
FX and inflation	Contractor (unless indexed) Sub-Clause 13.7	Contractor (unless indexed) Sub-Clause 13.7	Contractor	Offtaker ↔ SPV
Change in law	Employer Sub-Clause 13.6	Employer Sub-Clause 13.6	Employer Sub-Clause 13.6	Offtaker (mostly)
Grid connection delay	Employer Sub-Clause 8.5	Employer Sub-Clause 8.5	Employer (if its scope)	Offtaker
Force majeure / exceptional events	Shared Clause 18	Shared Clause 18	Shared Clause 18	Shared
Performance shortfall (output / PR)	Shared	Contractor Sub-Clause 4.1, Clauses 9 & 12	Contractor Sub-Clause 4.1, Clauses 9 & 12	SPV → EPC/O&M
Permits and approvals	Shared Sub-Clauses 1.13 & 2.2	Shared Sub-Clauses 1.13 & 2.2	Shared Sub-Clauses 1.13 & 2.2	Shared
Interface risk (multiple packages)	Employer Sub-Clause 4.6	Shared	Contractor	SPV

Click a cell to see who bears the risk, why, and the governing clause.

Allocation under the General Conditions only — Particular Conditions frequently amend them, and Gulf procurers often shift risk to the contractor. Always read the actual contract. IPP column describes common market practice, not a specific Kuwait PPA.

Figure E1.2 Read across a row: design and ground move from Employer to Contractor from Red to Silver, while change in law and grid connection delay stay with the Employer or offtaker in every column. General Conditions only; Part B often changes this.

Time limits

Missing a FIDIC clock can extinguish a right, so every date below belongs in the notices register.

Item (Sub-Clause)	Limit
Notice of Claim after awareness (20.2.1)	**28 days**, time-barred if late (applies to the Employer's claims too)
Fully detailed Claim with legal basis (20.2.4)	**84 days**, else the Notice lapses
Performance security after Letter of Acceptance (4.2)	28 days
Initial programme / Engineer's review (8.3)	28 / 21 days
Agreement, then determination (3.7)	42 + 42 days
Notice of Dissatisfaction (NOD) on a determination or DAAB decision	28 days
DAAB decision	84 days, binding at once
Interim Payment Certificate (IPC) issue / payment (14.6, 14.7)	28 / 56 days after the Statement
Taking-Over Certificate (TOC) (10.1)	28 days after application
Performance Certificate (11.9)	28 days after the last Defects Notification Period (DNP)

Variations, claims and extensions of time

A **Variation** is an instructed change to the Works. It is valued in a fixed order: contract rates, then rates derived from them, then Cost Plus Profit (default profit 5 %). Omissions are credited to the Employer. A **Claim** is a request for time or money under a named clause; it needs the Notice and the detailed Claim in time.

An **extension of time (EOT)** moves the completion date when an Employer-risk event delays completion. Only delay beyond the activity's **total float** (slack before completion moves) does so. The preferred method is **time impact analysis (TIA)**: insert the event into the updated programme current at the time and measure the shift in completion. When Employer and Contractor delays overlap (**concurrent delay**), the SCL Protocol gives time but not money. Climate events give time only, and nothing on Silver.

> $\mathrm{EOT} = \max(0,\ d - \mathrm{TF})$
>
> d delay to the activity (days); TF its total float on the updated programme

Disputes

The path is: Engineer's determination, NOD within 28 days, referral to the DAAB within 42 days, DAAB decision in 84 days (comply at once), NOD on that decision within 28 days, 28 days of amicable settlement, then arbitration (ICC by default). The NOD preserves the right to arbitrate; it does not suspend payment.

Securities

Security	Typical size	Released
Bid bond	1 to 5 %	Signature, or award to another bidder
Performance security	10 %	Performance Certificate (often reduced to 5 % at TOC)
Advance payment guarantee (APG)	= advance	Reduces as the advance is recovered
Retention	10 % per IPC up to 5 %	Half at TOC, half after the DNP
Warranty bond	2.5 to 5 %	End of the DNP
Parent company guarantee (PCG)	Full liability	When all obligations end

Advance recovery rule: recovery starts once certified IPCs exceed 10 % of the contract price, at 25 % of each IPC until repaid. Gulf bank guarantees are usually **on-demand** (the bank pays a compliant demand), but the Employer may claim only what it is entitled to. Figure E1.3 shows the cash effect.

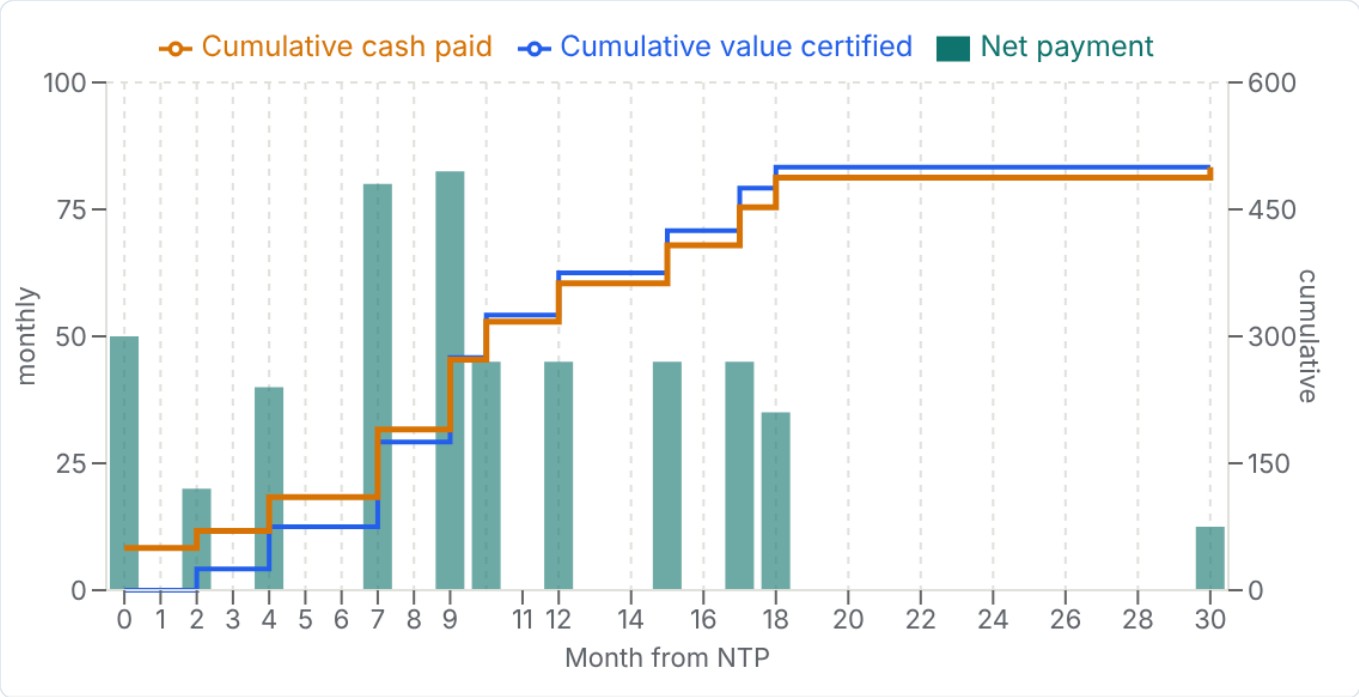


Figure E1.3 Cash paid (orange) runs ahead of value certified (blue) after the month-0 advance, then falls below it as advance recovery and retention are deducted; the last bar is the retention release.

▸ **Worked example: one IPC**

KWD 20 m contract, 10 % advance (KWD 2 m) already past the 10 % trigger. Gross IPC KWD 1.5 m: retention 10 % = 0.150, advance recovery 25 % = 0.375, net = 1.5 − 0.150 − 0.375 = **KWD 0.975 m**. The advance is repaid after 2 / 0.375 = 5.3 such IPCs.

Liquidated damages

Liquidated damages (LDs) are pre-agreed sums for a defined breach, payable without proving actual loss. **Delay LDs** accrue daily after the Time for Completion as extended by any EOT, until Taking-Over; a

sane rate is 0.03 to 0.1 % of the price per day. **Performance LDs** buy down a capacity or performance-ratio (PR) shortfall at the tests. Below a **minimum performance level**, money is not enough: the Contractor must remedy, or the Employer may reject. Caps: delay 10 to 20 %, performance 10 to 15 %, aggregate 20 to 30 %; reaching the delay cap is usually a termination trigger.

$$LD_{delay} = \min(LD_d \times \text{days late after EOT}, c \times CP) \quad \text{PR buy-down} = (PR_g - PR_m)/PR_g \cdot E_{yr} \cdot T \cdot AF$$

LD_d daily rate; c cap fraction; CP contract price; PR_g, PR_m guaranteed and measured PR; E_{yr} annual energy; T tariff; AF = $[1 - (1+i)^{-n}]/i$ (12.41 at 7 %, 30 years)

▸ **Worked example: delay LDs on a KWD 20 m EPC**

Rate 0.1 %/day = KWD 20,000/day; cap 10 % = KWD 2 m. Taken over 45 days after the extended date: 45 × 20,000 = **KWD 900,000**. The cap is reached at 2,000,000 / 20,000 = 100 days; after that, further delay costs the Contractor nothing in LDs.

System	Rule
English common law (Cavendish v Makdessi, 2015)	A clause is a penalty only if its detriment is out of all proportion to the innocent party's legitimate interest.
Gulf civil law: الشرط الجزائي	Valid, but the court may reduce it if grossly excessive. Kuwaiti Civil Code articles: not verified V

In Kuwait

- MEWRE's own contracts are often FIDIC-based with Kuwaiti Particular Conditions M
- Dispute forum (DAAB kept or not, arbitration or Kuwaiti courts) and arbitration law: not confirmed; ask legal affairs V
- Public tenders: final guarantee at least 10 % of contract value (Law 49/2016 Art. 65, single academic source) L

What the PM does

- Keeps a **notices register**: each Notice, sub-clause, date received, response deadline and owner.
- Serves MEWRE's own 28-day Notice of Claim before deducting delay damages.
- Tracks bond expiries 60 days ahead and computes entitlement before any call.

Remember

- Red: Employer designs; Yellow: Contractor designs; Silver: Contractor designs and carries ground. Part B can change all of it.
- 28 days to notify, 84 to detail; late notice can lose the claim.
- EOT = max(0, d − TF); concurrency gives time, not money.
- LDs: rate × days after EOT, capped; below minimum performance, remedy or reject.

Check yourself

1. Under the Silver Book General Conditions, who carries unforeseeable ground conditions?
   A. Employer, time and cost
   B. Employer, time only
   C. Contractor
   D. Shared 50/50
2. A late MEWRE approval delays an activity by 30 days; it had 12 days of total float. How many days of EOT are due?
3. KWD 20 m EPC, delay LDs 0.1 %/day, cap 10 %. Taken over 130 days after the extended completion date. Delay LDs payable (KWD m)?
4. An Employer delay and a Contractor delay to completion overlap for 20 days. The usual outcome under the SCL Protocol is:
   A. No EOT
   B. EOT and full prolongation cost
   C. EOT without prolongation cost
   D. LDs for the full 20 days

E2 IPPs, PPAs and project finance

about 12 min

After this section you can

- name every party in a Kuwaiti solar IPP and what MEWRE signs;
- compute a deemed-energy payment and a DSCR, and read a cash waterfall;
- explain why MEWRE's grid works are on the IPP's critical path.

The structure

An **independent power producer (IPP)** is a privately financed company that builds and runs a plant and sells its output to a public buyer. The **SPV** (special purpose vehicle) is a company created only for this plant; it owns the plant and signs every contract. **Sponsors** (developers) provide equity. **Lenders** provide most of the capital as **non-recourse** debt, repaid only from SPV revenue. The SPV hires an **EPC contractor** to build and an **O&M contractor** to operate, on terms back-to-back with the **PPA** (power purchase agreement), the sales contract with the **offtaker**, MEWRE. Figure E2.1 shows the links.

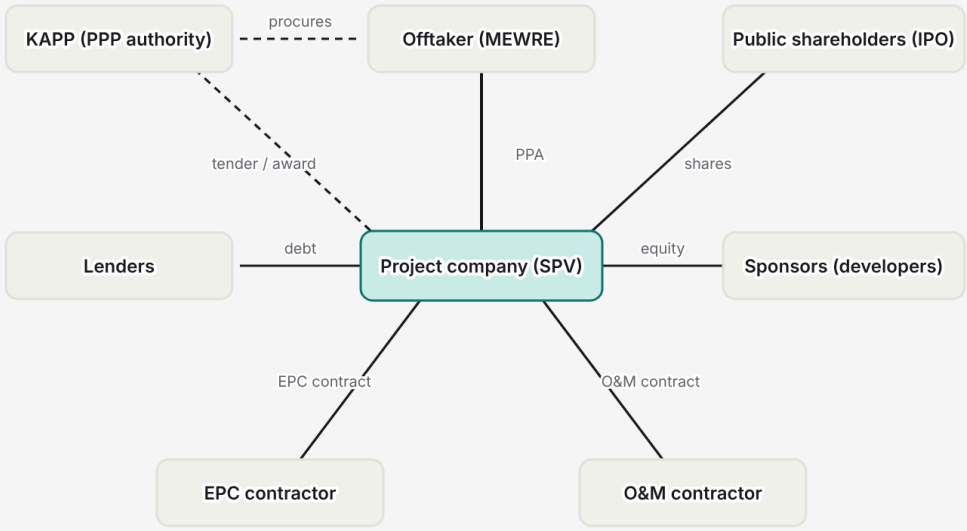


Figure E2.1 Every contract runs through the SPV. MEWRE's only contract is the PPA (plus any direct agreement with lenders); KAPP procures (dashed lines) and holds the state's shares. MEWRE cannot instruct the EPC contractor.

Tariffs and deemed energy

An **energy-only tariff** pays one price per MWh delivered; the SPV carries irradiance, soiling and plant performance. It is the norm for PV. A **two-part tariff** pays a **capacity charge** for being available (recovering capex and debt) plus an **energy charge** for output; it suits thermal plants and BESS, where the offtaker controls dispatch.

$$R = T \cdot (E_{del} + E_{deemed}) \quad R_m = CC \cdot P_{dep} \cdot \min(1, A_m/A_{ref}) + EC \cdot E_m$$

T tariff; E_{del} energy delivered; CC capacity charge per kW-month; P_{dep} contracted capacity; A_m measured and A_{ref} reference availability; EC energy charge; E_m energy in the month

Deemed energy is energy the plant could have exported but did not because of the offtaker: grid outage, curtailment beyond the annual allowance, or late grid connection. MEWRE pays for it as if delivered, because MEWRE controls those causes; if the SPV carried them, bidders would price them into 30 years of tariff. It is measured from on-site irradiance through an agreed plant model, capped at available capacity, minus an uncompensated allowance.

$$E_{deemed} = \Sigma\, P_{exp,t} \cdot \Delta t - E_{allow}$$

$P_{exp,t}$ expected output in interval t (MW); Δt interval (h); E_{allow} allowance (MWh)

▸ **Worked example**

MEWRE-side 400 kV outages total 24 hours while the model expects an average 750 MW; the allowance is already used. E_{deemed} = 750 × 24 = 18,000 MWh. At an **assumed** 8 fils/kWh: 18,000 × 1,000 × 0.008 = **KWD 144,000**, for energy the grid never received.

Termination and lenders' protection

Termination payments rise with the offtaker's responsibility: **SPV default** (at most the debt; equity loses) < **natural force majeure** (debt plus equity invested) < **offtaker default or political force majeure** (debt, breakage costs and equity with a return). A **direct agreement** between MEWRE, the lenders and the SPV

obliges MEWRE to notify lenders of an SPV default and give a cure period, during which lenders may **step in** through a nominee or **novate** the PPA to a new owner.

Project finance

Lenders size debt from **CFADS** (cash flow available for debt service: revenue minus opex, tax, working-capital changes and reserves). The **DSCR** (debt service cover ratio) is CFADS divided by interest plus principal in each period. Lenders set a sizing DSCR of about 1.2 to 1.3×, a dividend **lock-up** at 1.10 to 1.20× and a default level near 1.05×. **Debt sculpting** shapes repayments so the DSCR is constant each year; debt is then the lower of that capacity and the **gearing** cap (debt share of capex, 75 to 85 % for Gulf solar).

> $DSCR_t = CFADS_t / (I_t + P_t)$ $D_{max} = \Sigma\,(CFADS_t/DSCR_{target})/(1 + k_d)^t$ $LLCR = PV(CFADS, \text{loan life})/D$
>
> I, P interest and principal; k_d debt rate; LLCR loan-life cover ratio; PLCR uses the project life, so it includes the tail after the loan

Cash leaves the SPV in a fixed **waterfall**: opex and tax, senior debt service, the **DSRA** (debt service reserve account, about 6 months), the maintenance reserve, the lock-up test, then dividends.

▸ **Base case (illustrative, not Kuwait bid data)**

1,100 MWac plant, capex USD 560 m, flat 1.80 US¢/kWh, 25-year debt at 5.5 % sculpted at 1.25×. Sculpted capacity USD 461.4 m exceeds the 80 % gearing cap, so debt = 0.8 × 560 = **USD 448 m**, equity **112 m**. Year-1 CFADS = 3,003 GWh × 0.018 − 7.5 = USD 46.55 m against debt service 36.16 m: DSCR = **1.287×** (LLCR also 1.287×; PLCR 1.377× with a 5-year tail). Project IRR **6.45 %**, equity IRR **8.97 %**. A P90 year (7 % less energy) gives CFADS 42.77 m and DSCR **1.183×**: below a 1.20× lock-up, above default.

In Kuwait

- KAPP procures PPPs under PPP Law 116/2014 and Decree 78/2015; MEWRE is offtaker on a 30-year solar PPA M
- PPP companies above KD 60 m: 50 % IPO to citizens, 6 to 24 % public entities, at least 26 % private (one legal guide; sources conflict) V
- IWPP practice: developer 40 %, state through KAPP 60 %; KAPP holds the IPO shares until operation; Az-Zour North 1 IPO (2019) oversubscribed M
- Not confirmed: PPA currency, government support, termination regime, deemed-energy terms V
- Shagaya Zone 1: developer builds the 400 kV substation; the line to Al-Wafra is MEWRE's contract (KD 48.6 m) M

What the PM does

- Treats MEWRE's grid works as the IPP's critical path: a late line means deemed energy or a later COD, paid from the ministry's budget.
- Logs every PPA allowance and deemed-energy trigger, and the direct-agreement cure periods.

Remember

- MEWRE signs the PPA, not the EPC: its levers are the PPA and the grid.
- Deemed energy = expected energy during offtaker events minus the allowance.
- DSCR = CFADS/(I + P); debt = min(sculpted capacity, gearing cap).
- Base case: 448/112, DSCR 1.287×, IRRs 6.45 % and 8.97 %, P90 1.183×.

Check yourself

1. CFADS USD 50 m, debt service USD 40 m. DSCR?
2. A MEWRE outage outside the allowance prevents 25,000 MWh. At an assumed 8 fils/kWh, what does MEWRE pay (KWD)?
3. Which termination cause gives the SPV the largest payment?
   A. SPV default
   B. Natural force majeure
   C. Offtaker default
   D. Expiry of the PPA term
4. In a P90 year the DSCR is 1.183×, with lock-up at 1.20× and default at 1.05×. What happens?
   A. Event of default
   B. Dividends are blocked; no default
   C. Nothing
   D. MEWRE must top up the tariff

E3 Time value of money, LCOE and comparing bids

about 8 min

After this section you can

- discount a cash flow and compute NPV, IRR, WACC and a quick LCOE;
- levelise escalating tariff bids and spot an abnormally low one.

Discounting

Money received later is worth less today because capital has a cost. The **discount rate** r expresses that cost; for a project it is usually the **WACC** (weighted average cost of capital). **NPV** (net present value) sums every cash flow multiplied by its discount factor; NPV > 0 means the project earns more than r. **IRR** (internal rate of return) is the r at which NPV = 0. Decide with NPV and communicate with IRR, because IRR ignores scale, assumes reinvestment at the IRR, and can have several values when cash flows change sign more than once. **Payback** (years until cumulative cash turns positive) measures liquidity only.

$$PV = CF_t/(1+r)^t \quad NPV = \Sigma\, CF_t/(1+r)^t \quad WACC = (E/V)k_e + (D/V)k_d(1 - T)$$

E, D, V equity, debt and total capital; k_e, k_d their costs; T tax rate

Base case: WACC = 0.2 × 9 % + 0.8 × 5.5 % = **6.2 %** (4.12 % real at 2 % inflation). Discount factors at 6.2 %: year 10 = 0.548, year 20 = 0.300, year 30 = 0.165. The base-case project pays back in year 13 in plain cash but only in year 28 discounted, ending at NPV ≈ +USD 14 m.

LCOE

The **LCOE** (levelised cost of electricity) is the constant price per unit of energy that recovers all lifetime costs at the discount rate. Both costs and energy are discounted; discounting only costs makes LCOE far too low. The **CRF** (capital recovery factor) converts a present sum into an equal annual payment and gives a screening estimate.

$$\mathrm{LCOE} = \Sigma(I_t + O_t)(1+r)^{-t} / \Sigma\, E_t(1+r)^{-t} \quad \mathrm{CRF} = r(1+r)^n/[(1+r)^n - 1] \quad \mathrm{LCOE} \approx (I_0 \cdot \mathrm{CRF} + O)/E$$

I capex; O opex; E energy; n life in years

▸ **Worked example: base case (illustrative)**

1,100 MWac / 1,430 MWp, 3,003 GWh/yr (CF 31.2 %), capex USD 560 m, opex USD 7.5 m/yr (+2 %/yr), degradation 0.4 %/yr, 30 years, r = 6.2 %. Full LCOE = **1.764 US¢/kWh** (5.43 fils/kWh). Quick check: CRF(6.2 %, 30) = 0.0742; (560 × 0.0742 + 7.5)/3,003 = 0.01634 USD/kWh = 1.634 US¢, 7 % low because it ignores degradation and opex growth.

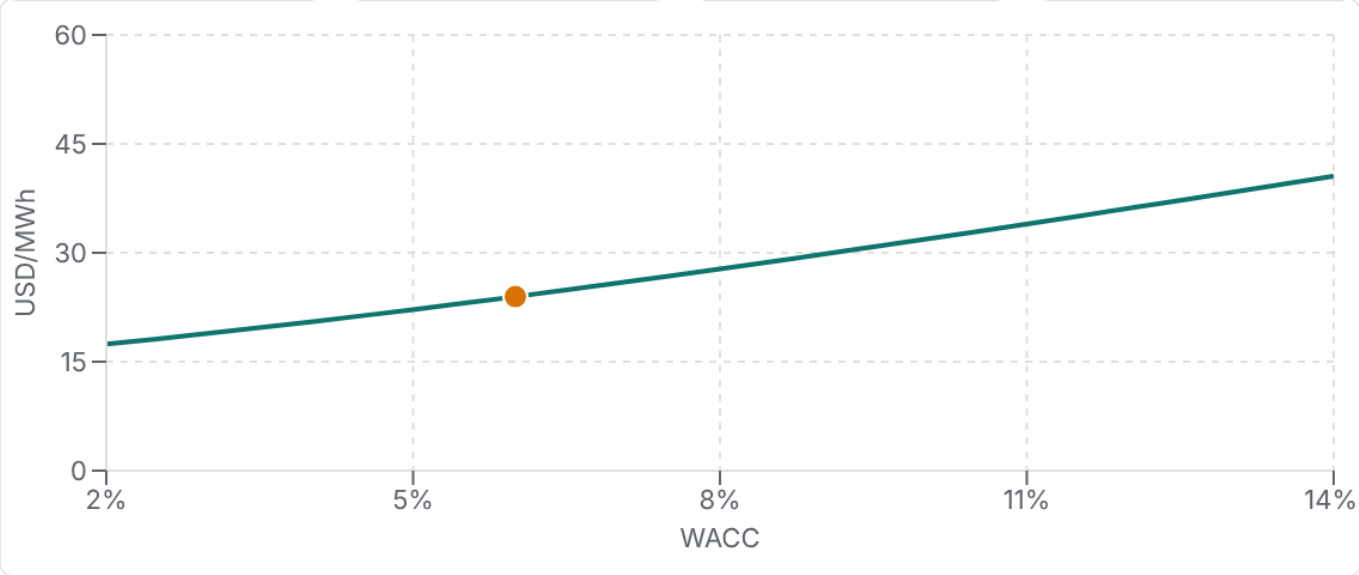


Figure E3.1 LCOE versus WACC for an illustrative PV plant (generic inputs, not the base case): LCOE more than doubles from 2 % to 14 %. PV has no fuel, so the cost of capital drives its price.

The base-case **tornado** ranks drivers by LCOE swing: yield ±10 % (0.356 US¢), capex ±10 % and WACC ±1 point (0.288 each), opex ±20 % (0.129), degradation (0.080). Reported Gulf records (verify): MBR 5 ≈ 1.69 (2019), Al Dhafra ≈ 1.35 (2020), Sudair ≈ 1.24 and Al Shuaibah ≈ 1.04 US¢/kWh (2021).

Comparing tariff bids

Escalating bids cannot be ranked by year-1 price. The **levelised tariff** is the flat tariff with the same present value of payments for the same energy.

$$\mathrm{LT} = \Sigma\, T_t E_t(1+d)^{-t} / \Sigma\, E_t(1+d)^{-t}, \quad T_t = T_1(1+g)^{t-1}$$

d evaluation discount rate; g escalation

▸ Worked example

30 years, constant energy, d = 7 %. Bid A: flat 4.20 fils/kWh. Bid B: 3.90 fils +1.5 %/yr, passing A in year 6 and reaching 6.01 by year 30. LT(B) = **4.54** fils/kWh: the lowest headline is the dearest bid.

A higher d shrinks late payments, so it favours back-loaded bids (LT(B) = 4.43 at 10 %, 4.67 at 4 %). Publish d, the energy profile and index paths in the RFP **before** bids. Test **abnormally low bids**: a bid 20.5 % below the median levelised tariff gave a 1.70 % equity IRR at base costs, signalling failed financial close, delay, quality cuts or claims.

In Kuwait

- The Zone 1 tariff is not published M
- Units used here: 1 KWD ≈ 3.25 USD (assumed); 1 US¢/kWh = 3.08 fils/kWh V

Remember

- Decide on NPV; IRR is the rate where NPV = 0.
- LCOE discounts both costs and energy; base case 1.764 US¢/kWh at 6.2 %.
- Yield, capex and WACC dominate PV LCOE.
- Rank bids by levelised tariff at a rate fixed before opening.

Check yourself

1. Present value (KWD thousand) of KWD 1 m received in year 10 at 6.2 %?
2. Capex USD 100 m, CRF 0.08, opex USD 2 m/yr, 500 GWh/yr. Quick LCOE in US¢/kWh?
3. Raising the evaluation discount rate makes an escalating bid look:
   A. Dearer
   B. Cheaper
   C. Unchanged
   D. Non-compliant
4. Convert 1.764 US¢/kWh to fils/kWh at 1 KWD = 3.25 USD.

E4 Procurement in Kuwait and working in the ministry

about 10 min

After this section you can

- route a purchase (CAPT, direct, KAPP) and apply Law 94/2026;
- score bids with QCBS and estimate an approval chain's P80;
- write a decision memo, an Arabic letter and a valid notice, and negotiate on entitlement.

Methods and rules

Method	Use
Public tender	Default; open to all registered bidders; award to the lowest technically compliant price
Limited tender	Invited, prequalified bidders
Practice: ممارسة	Negotiation with a few suppliers
Direct order	Below the direct-contracting limit, or a justified exception

CAPT (Central Agency for Public Tenders) runs tenders above its threshold; MEWRE writes the specification and runs the technical evaluation. Clarifications go in writing to all bidders. IPPs go through **KAPP** with a **two-envelope** evaluation: financial envelopes are opened only for technically passing bids. Consultancy uses **QCBS** (quality- and cost-based selection), with weights fixed before opening.

$$S_F = 100 \cdot P_{min}/P_i \quad S = w_T S_T + w_F S_F \quad P_{nat} \le (1+\pi)P_{min} \quad LC = V_{local}/V_{base} \ge 30\ \%$$

S_T technical score; P_i bid price; π national-product preference rate; LC local content (define V_{base} in the contract)

▸ **Worked example: weights decide**

Owner's Engineer bids: X (88, KWD 1.20 m), Y (75, 0.95 m), Z (92, 1.40 m). S_F: 79.17, 100, 67.86. At 70/30: X 85.35, Y 82.50, Z 84.76; X wins. At 80/20: X 86.23, Y 80.00, Z 87.17; Z wins.

Approvals and delay

The Audit Bureau approves, rejects or approves **conditionally** (conditions bind). Never start work before approval. Estimate each approval step with three points and attack the largest variance.

$$t_e = (a + 4m + b)/6, \quad \sigma = (b - a)/6, \quad P80 \approx \Sigma t_e + 0.84\sqrt{(\Sigma\sigma^2)}$$

a, m, b optimistic, most likely and pessimistic durations

▸ **Worked example (illustrative, weeks)**

Higher Committee (2, 4, 10), Audit Bureau (4, 6, 16), Fatwa (2, 4, 8), signing (1, 2, 4): mean 18.50, σ = 2.651, P80 = **20.73**. The Audit Bureau carries 57 % of the variance. Pre-clearing its file to (3, 5, 8) gives P80 = 17.96, 2.8 weeks earlier.

Lawful accelerators: a mapped committee calendar, complete pre-cleared files, the next gate prepared in parallel, standard documents, bankability settled before the RFP, managed bid validity, and escalation with a KWD-per-day delay cost.

Writing for the ministry

A **decision memo** is one page, about 450 words (2 minutes at 200 words/min), written **BLUF** (bottom line up front): the first paragraph states the decision, deadline and recommendation. Eight parts: decision required; issue; background; options including do-nothing; recommendation; risks; decision line; attachments with the next letter ready to sign. The written annotation is the decision; a verbal yes is not.

The Arabic official letter follows a fixed frame, top to bottom:

Part	Arabic
Reference number / date	الرقم / التاريخ
Addressee with title	السيد/ ... المحترم
Subject	الموضوع
Greeting	تحية طيبة وبعد،،،
Reference to prior letter	بالإشارة إلى كتابكم رقم ... المؤرخ في ... بشأن ...
Numbered points	نفيدكم بما يلي
Request	يرجى التكرم بـ ...
Closing	وتفضلوا بقبول فائق الاحترام والتقدير،،،
Signature, attachments, copies	المرفقات / نسخة إلى

A contractual **notice** is valid only if labelled as a notice, citing the exact sub-clause, sent by the right party to the contractual address, within the time limit, stating facts and reserving rights. **Minutes** are evidence, not notices: rebut disputed statements on the record and still serve the notice.

Notice element	Arabic
Subject label	إشعار
Reservation of rights	تحتفظ الوزارة بكافة حقوقها بموجب العقد

The **escalation ladder**: discussion, minuted action, letter, notice or non-conformance report, contractual remedies, senior escalation, termination.

Presenting and negotiating

Give the answer first, then three supporting points. Deliver bad news early: cause, impact, plan, ask. Never guess a number; commit to a dated follow-up.

On public money, settle on **entitlement and substantiated quantum**, never by splitting the difference. Your **BATNA** (best alternative to a negotiated agreement) is the dispute route; it sets your walk-away price. The **ZOPA** (zone of possible agreement) is $[EV - C_C, EV + C_E]$: EV the expected award, C each side's dispute cost.

AI tools: never paste ministry or bidder data into public tools. Models invent article numbers, laws and figures; verify each against the primary source.

In Kuwait

- CAPT oversees tenders above KD 75,000 (Law 49/2016 Art. 19) M
- Law 94/2026 (gazetted 27 Sep 2026, effective about 27 Dec 2026): direct contracting up to KD 1 m per financial year for the same items, agents banned, at least 30 % local products and 30 % local subcontracting, splitting still a violation M
- Audit Bureau prior review above KD 100,000 (reported by a minister) V; 28 contracts worth KD 63 m signed without approval; 2025 conditional approval of KWD 161.381 m GCCIA purchases M
- National-product preference rate (10, 15 or 20 %) not confirmed V
- Delay evidence: Zone 1 bid deadline moved Sep 2025 to Jan 2026; 400 kV line about 11 months from bids to award, to a higher bidder (KD 48.6 m vs lowest KD 41.45 m) M

Remember

- Fix weights, discount rate and pass marks before opening envelopes.
- Plan approvals at P80 and attack the highest-variance step.
- Memo: BLUF, one page, eight parts, letter attached.
- A notice needs the label, sub-clause, sender, address, time limit and facts; minutes are not notices.
- Settle on entitlement; never paste ministry or bidder data into public AI tools.

Check yourself

1. After Law 94/2026 takes effect, MEWRE has placed direct orders of KD 300,000 and KD 450,000 this financial year for the same spares. Largest further direct order (KD)?
2. An approval step has a = 3, m = 5, b = 13 weeks. PERT mean (weeks)?
3. In approved minutes the contractor states that all delay is MEWRE's fault. Best response?
   A. Ignore it; minutes are informal
   B. Rebut it on the record and serve any contractual notice separately
   C. Agree to keep the relationship
   D. Call the performance bond
4. A colleague wants a public chatbot to rank three bidders' financial offers. You:
   A. Paste the offers with names removed
   B. Use only an approved tool, or none, and let the committee score
   C. Paste them; the output is only advisory
   D. Ask the bidders for consent

ANSWERS

Check yourself: answers

Mark each one you got wrong and re-read that section's Remember box.

A1 · Kuwait's power system and economics

1. **−3.1 %.** RM = (17,000 − 17,540) / 17,540 = −0.0308: demand exceeds reliably available capacity by 540 MW.
2. **B.** PV peaks at solar noon and stops after sunset, while cooling demand stays high into the evening, so the highest demand minus PV falls later.
3. **KWD 54,000 per day.** 500 × 0.25 × 24 = 3,000 MWh = 3×10^6 kWh; (26 − 8) = 18 fils = 0.018 KWD; $3 \times 10^6 \times 0.018 = 54{,}000$.
4. **D.** Al-Khairan Phase 1 is a gas-fired IWPP with two bids on 1 June 2026; no 1,800 MW solar project was found, G2G is only a framework, and the Zone 1 tariff is not published.

A2 · Institutions, approval chains and the ministry

1. **B.** After the Higher Committee approves the selected investor, the State Audit Bureau carries out prior review before the PPA is signed.
2. **C.** Law 94/2026 takes effect only around 27 December 2026, so the KD 75,000 CAPT threshold applies; at KD 400,000 the reported KD 100,000 Audit Bureau threshold also applies.
3. **23.1 weeks.** 5 / 0.7 = 7.14 plus 8 / 0.5 = 16.0 gives 23.14 weeks, against a nominal 13 weeks.
4. **B.** للعرض means "for presentation" to the next level; the first option means approved and the third means take the necessary action.

B1 · Solar resource and the PV cell

1. **550 W/m².** 700 × cos 60° + 200 = 700 × 0.5 + 200 = 550 W/m².
2. **B.** MPPT moves the DC operating voltage to stay on the peak of the power curve as irradiance and temperature change.
3. **About 519 W.** 600 × [1 − 0.0030 × (70 − 25)] = 600 × 0.865 = 519 W.
4. **B.** Photocurrent is proportional to irradiance; voltage depends mainly on temperature and falls only slightly (logarithmically) with light.

B2 · The utility PV plant and its energy yield

1. **650 MWp.** 1.3 × 500 = 650 MWp; the extra 150 MWp fills the inverters outside clear noons.
2. **0.82 (82 %).** Y_f = 1,066 / 20 = 53.3 h; Y_r = 65 h; PR = 53.3 / 65 = 0.82.
3. **910 GWh/yr.** 1,000 × (1 − 1.282 × 0.07) = 1,000 × 0.910 = 910.3 GWh.
4. **B.** N* is proportional to 1/√s, and 1/√2 = 0.707, so the interval shortens by about 29 %.

B3 · CSP and wind

1. **B.** The receiver loses roughly the same heat whatever the DNI, so a smaller input leaves a proportionally larger cut in useful heat (12 to 15 % per 10 % DNI).
2. **About 9.1 h.** 600 × 0.38 = 228 MWh_e; 228 / 25 = 9.12 h.
3. **About 1.42.** $(9/8)^3$ = 1.424: 12.5 % more speed gives about 42 % more power.
4. **B.** Battery cost scales with MWh while salt storage hours are cheap, so the crossover sits around 8 to 12 h; compare the same delivery profile, not LCOE.

B4 · Battery energy storage

1. **0.25C, 4 h.** C-rate = 300 / 1,200 = 0.25; duration = 1,200 / 300 = 4 h.
2. **921.5 MWh.** 1,000 × 0.95 × 0.97 = 921.5 MWh, before conversion losses to the POI.
3. **After about 13.7 years.** 6,000 / (1.2 × 365) = 13.7 years, so the cycle cap ends cover before the 15-year calendar limit.
4. **B.** UL 9540A is a test method that reports how runaway propagates in that cell, module and layout; it is not a pass/fail certificate, and spacing rules come from NFPA 855.

C1 · Transmission, substations and the Kuwaiti grid

1. **B.** Losses $\propto 1/V^2$: $(400/132)^2$ = 9.2.
2. **C.** A disconnector gives a visible isolating gap and has no current-breaking capability; the breaker interrupts current.
3. Any one breaker can be taken out for maintenance or fail without losing a circuit, and a busbar fault does not disconnect any circuit, at a cost of 1.5 breakers per circuit.
4. **About 1,391 A.** I = 1,000 kVA / (1.732 × 0.415 kV) = 1,391 A.

C2 · Grid code requirements and plant control

1. **131.5 Mvar.** Q = 400 × tan(arccos 0.95) = 400 × 0.329; the inverters must supply more to cover transformer reactive losses, at 50 °C.
2. **8 MW.** Δf_{eff} = 0.3 − 0.2 = 0.1 Hz; ΔP = (200 / 0.05) × (0.1 / 50) = 8 MW.
3. **B.** $\Delta V \approx (RP + XQ)/V$ and X is 10 or more times R on HV lines, so the Q term dominates.
4. **C.** Ride-through acts in milliseconds inside each inverter's control; the PPC loop (100 to 200 ms) is too slow.

C3 · Grid strength and inverter behaviour

1. **SCR 4.0, WSCR 3.0.** 2,400 / 600 = 4.0; 2,400 / (600 + 200) = 3.0: moderate on its own, borderline weak with its neighbour, so EMT is required.
2. **B.** IGBTs have very little thermal mass and exceed their junction temperature within microseconds to milliseconds of overcurrent.
3. Integrators keep accumulating error while the output is clamped; at clearance that stored error drives overshoot and oscillation in P, Q and voltage, and can trip the plant or disturb neighbours.
4. **14.4 kA.** I = 10,000 / (1.732 × 400) = 14.4 kA.

C4 · Power system studies and how a PM manages them

1. **D.** PLL instability is a fast control phenomenon on instantaneous waveforms; RMS phasor models cannot represent it reliably.
2. **About 91 %.** S = 1,500 / 0.95 = 1,579 MVA on the one remaining 1,732 MVA circuit: 1,579 / 1,732 = 91.2 %.
3. Change control: the vendor notifies MEWRE before the change, supplies updated RMS and EMT models for the new version, the OE assesses whether re-study is needed, and affected compliance tests are repeated before or after the update as agreed.
4. Protection settings approved by MEWRE and tested (then SCADA point-to-point checks, telecoms, metering sealed, switching programme approved).

D1 · From feasibility to handover

1. **B.** A single, dated gap with no fatal flaw is what "go with conditions" is for: proceed, with the report and its design consequences closed by a named date.
2. **0.82.** Final yield 2,665 / 50 = 53.3 kWh/kWp; reference yield 65 / 1 = 65 h; PR = 53.3 / 65 = 0.82.
3. **C.** The Performance Certificate comes within 28 days after the last DNP ends; the TOC transfers care and risk, stops delay damages and starts the DNP.
4. **72.8 MWh.** 100 × (800/1000) × [1 − 0.003 × (55 − 25)] × 1 h = 80 × 0.91 = 72.8 MWh.

D2 · Planning, control and reporting

1. **KWD 250 m.** CPI = 60 / 75 = 0.80; EAC = 200 / 0.80 = 250, i.e. KWD 50 m over budget if nothing changes.
2. **B.** SPI = 40 / 50 = 0.80 (behind); CPI = 40 / 38 ≈ 1.05 (under cost for the work done). AC below PV only reflects less work done.
3. **4 weeks.** Delay = max(0, 20 − 16) = 4; this is an Employer-caused delay, so the contractor or IPP is likely entitled to time relief.
4. **Mean 9.0 weeks, σ 1.67 weeks.** t_e = (6 + 4 × 8 + 16) / 6 = 54 / 6 = 9.0; σ = (16 − 6) / 6 = 1.67.

E1 · Construction contracts (FIDIC 2017)

1. **C.** Silver puts Unforeseeable Difficulties, including ground (4.12), on the Contractor; Red and Yellow leave it with the Employer.
2. **18 days.** EOT = max(0, 30 − 12) = 18; the first 12 days are absorbed by float.
3. **KWD 2 m.** 130 × 0.02 = 2.6, capped at 0.10 × 20 = 2.0.
4. **C.** Concurrent delay gives time but not money unless the contract says otherwise.

E2 · IPPs, PPAs and project finance

1. **1.25×.** 50 / 40 = 1.25.
2. **KWD 200,000.** 25,000 × 1,000 kWh × 0.008 KWD/kWh.
3. **C.** Offtaker default pays debt, breakage and equity with a return; MEWRE controlled the cause.
4. **B.** 1.183× is below the 1.20× lock-up (cash trapped) but above the 1.05× default level.

E3 · Time value of money, LCOE and comparing bids

1. **About 548.** 1,000,000 / 1.062^{10} = 547,968 KWD.
2. **2.0 US¢/kWh.** (100 × 0.08 + 2) / 500 = 0.02 USD/kWh.
3. **B.** A higher rate shrinks the weight of the late, higher payments (LT of 3.90 +1.5 %/yr: 4.67 at 4 %, 4.43 at 10 %).
4. **5.43 fils/kWh.** 1.764 × 10 / 3.25 = 5.43.

E4 · Procurement in Kuwait and working in the ministry

1. **KD 250,000.** 1,000,000 − (300,000 + 450,000); above that, CAPT applies (and Audit Bureau review may still apply).
2. **6 weeks.** (3 + 4 × 5 + 13)/6 = 36/6 = 6; σ = 10/6 = 1.67.
3. **B.** Unchallenged minutes become evidence against MEWRE, and minutes do not replace contractual notices.
4. **B.** Bidder offers are confidential; removing names does not anonymise them, and the committee, not a tool, decides.